

Inference of Last-Mile Network Characteristics

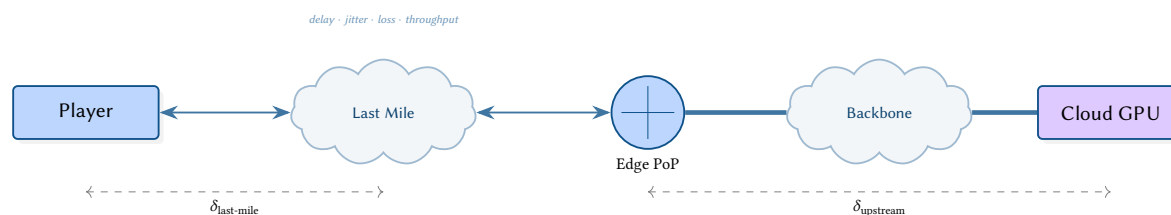
Adaptive Mitigation in Cloud Gaming

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Abstract. Cloud gaming shifts rendering from local hardware to remote GPU servers, creating an interactive control loop that cannot tolerate buffering. Unlike video streaming, a late frame is useless: it directly degrades Quality of Experience (QoE), which makes traditional Layer 4 congestion control alone insufficient. This *position paper* argues that cloud gaming platforms should treat last-mile inference as a first-order design concern. We propose a framework in which four telemetry signals—delay, jitter, loss, and throughput—measured at per-frame granularity expose structural fingerprints of eleven common access technologies. A two-tier decision architecture separates macro-level classification (10–30 s windows, motif matching, and cross-user correlation) from micro-level algorithm execution (per-frame congestion control, FEC, and codec adaptation). A session lifecycle state machine and persistent feature store enable progressively faster classification across sessions. We detail the mitigation toolkit—forward error correction, congestion-control selection, packetization, and jitter buffering—and map each access technology to its recommended strategy. Throughout, we rely on illustrative examples and architectural diagrams rather than empirical evaluation; a companion simulation report (LM-SIM-2026-01) covers the quantitative hypothesis testing. The goal is to define the problem space clearly and argue that fingerprint-driven adaptive mitigation is both feasible and necessary, not to present a finished production system.

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FOUNDATIONS

1 Introduction: Why Buffering Is Not an Option

The defining constraint of cloud gaming is that buffering—the universal solution to network variability in conventional video streaming—is fundamentally incompatible with interactive play. A video-on-demand service like Netflix routinely maintains 5–30 seconds of playout buffer, absorbing jitter, loss, and throughput fluctuations invisibly. A cloud gaming service has no such luxury: every frame must traverse the full network path from GPU render to player display within a strict latency budget, typically under 100 ms for casual genres and under 80 ms for competitive titles [0].

This report concerns not latency measurement per se—that is straightforward—but *inferring the characteristics of the last-mile network* from observable telemetry signals. Rather than prescribing a single algorithm, it develops the framework for using that inference to drive *adaptive mitigation* of impairments that cannot be eliminated.

Insight:
The “buffer” that solves streaming video is precisely the latency that destroys interactive gaming. This single constraint transforms network quality from a background concern into the primary engineering challenge.

1.1 Scope and Positioning

This is a *position paper*: it lays out a conceptual framework, identifies design dimensions, and argues for the feasibility of fingerprint-driven adaptive mitigation. It does not present a finished implementation or empirical evaluation. Telemetry plots and numerical examples throughout are *illustrative*—chosen to convey the characteristic shape and scale of each access technology’s fingerprint rather than to report measured data from a specific deployment. Quantitative hypothesis testing is deferred to a companion simulation report (LM-SIM-2026-01). The contribution of this paper is the *problem structuring*: decomposing the design space into access technology fingerprints, inference mechanisms, and mitigation strategies, and arguing that these three elements must be co-designed.

1.2 Problem Statement

At each decision epoch t , the cloud gaming server observes a telemetry window $\mathbf{x}_{t-W:t} = \{d_i, j_i, \ell_i, \tau_i\}_{i=t-W}^t$ comprising per-frame delay (d), jitter (j), loss (ℓ), and throughput (τ) measurements over a sliding window of W frames. The inference problem is:

1. **Classify** the current channel state $c_t \in C$, where C is a finite set of access technology fingerprints (e.g., DOCSIS 3.0 bufferbloat, Wi-Fi 5 contention, FTTH stationary), potentially including compound states for multi-hop paths.
2. **Select** a mitigation strategy $s_t \in S(c_t, g)$, where S is the set of feasible strategy configurations (congestion control algorithm, FEC mode, codec profile, jitter buffer depth) conditioned on the inferred channel class c_t and the game genre g .

The objective is to minimise expected QoE degradation over the session:

$$\min_{s_t} \mathbb{E} \left[\sum_t \mathcal{L}_{\text{QoE}}(\mathbf{x}_t, s_t, g) \right] \quad \text{subject to } s_t \in S(c_t, g), \quad (1)$$

where $\mathcal{L}_{\text{QoE}}$ captures frame delivery failure (late or lost frames), visual quality degradation (from bitrate reduction), and strategy transition costs (perceptible quality switches). The two-tier architecture (fig. 11) decomposes this: the macro tier solves the classification and strategy selection at 10–30 s intervals, while the micro tier executes the selected algorithms autonomously at per-frame rates within the chosen s_t .

Table 1. Notation used throughout this report.

Symbol	Description
t	Decision epoch (macro-tier cycle)
$\mathbf{x}_{t-W:t}$	Telemetry window: per-frame delay, jitter, loss, throughput
W	Sliding window length (frames)
$c_t \in C$	Inferred channel class (access technology fingerprint)
g	Game genre
$s_t \in S(c_t, g)$	Selected mitigation strategy (CC, FEC, codec, jitter buffer)
$\mathcal{L}_{\text{QoE}}$	QoE loss function (late frames, quality drops, switch costs)
θ	Confidence threshold for strategy switching
L_{MTP}	Motion-to-Photon latency (eq. (2))

1.3 Core Hypotheses

Two hypotheses underpin this framework:

Hypothesis 1: Layer 7 Adaptation over Layer 4 Control

Traditional TCP congestion control is mismatched for cloud gaming. Bulk transfer treats latency as secondary to guaranteed delivery; cloud gaming treats a late packet as useless—it directly degrades QoE. The goal is therefore to observe repeating network *motifs* and select the right combination of *Layer 7 strategies*—congestion control algorithm, FEC scheme, codec profile—to match the physical channel. This operates as a **two-tier** system: the selected algorithms (e.g., BBR, Raptor FEC, NVENC settings) run autonomously at per-frame rates, while the inference engine selects *which* algorithms are active on a slower timescale (10–30 s), switching only after accumulating sufficient confidence.

Hypothesis 2: Emergent Patterns in Large Datasets

Last-mile networks are not black boxes of random noise; their physical constraints produce structural, repeating fingerprints (stationarity, periodicity, linear bufferbloat ramps). A corpus of high-frequency time series across millions of sessions lets tools like Matrix Profile detect these patterns. Correlating signatures across users sharing an ISP routing prefix distinguishes local Wi-Fi issues from shared upstream congestion, enabling targeted mitigation.

1.4 Static vs. Interactive Streaming

Figure 1 illustrates the fundamental difference between buffered video delivery and interactive cloud gaming.

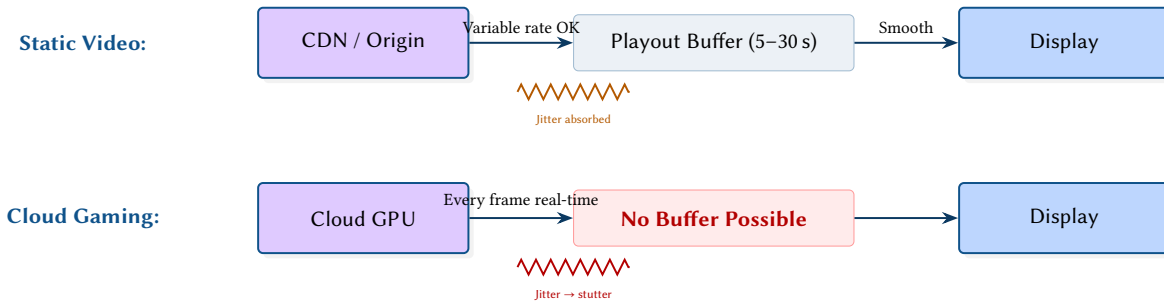


Figure 1. Static video streaming absorbs network variability in a playout buffer. Cloud gaming has no equivalent mechanism—every network impairment is directly visible to the player.

1.5 The Asymmetric Path

The cloud gaming network path is fundamentally asymmetric:

- **Upstream (player → server):** Small packets carrying controller input, key presses, and mouse movements. Typically < 100 bytes per input event at 60–120 updates/second. Low bandwidth but *critically latency-sensitive*—a single lost or delayed input packet directly affects gameplay responsiveness. Voice chat, when present, is typically carried on a separate channel.
- **Downstream (server → player):** Encoded video frames (H.264/H.265/VP9/AV1) at 5–45 Mbit/s, plus audio. High bandwidth, tolerant of individual packet loss (with Forward Error Correction (FEC)), but sensitive to sustained throughput drops and jitter.

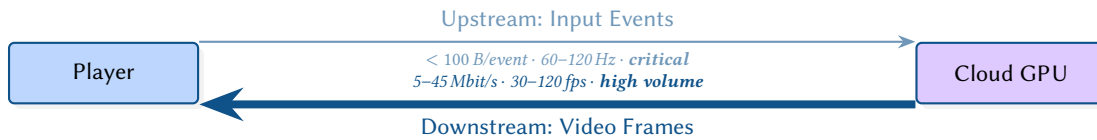
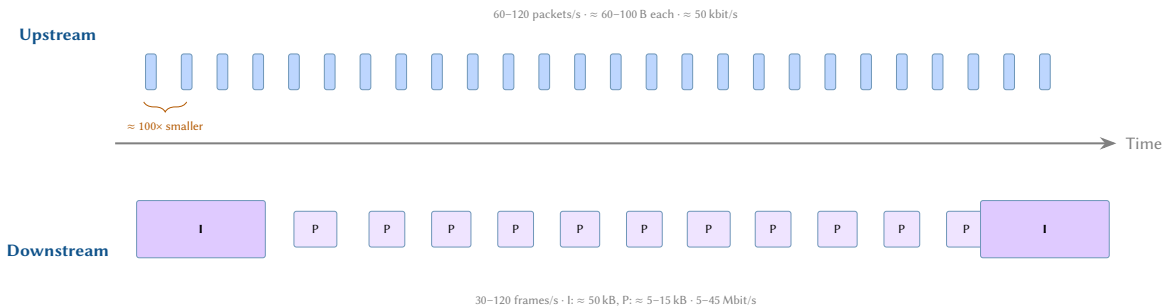


Figure 2. Asymmetric traffic profile of cloud gaming. Upstream input is tiny but loss-critical; downstream video is bandwidth-heavy but partially loss-tolerant with FEC.



Insight:
The asymmetry means that mitigation strategies must be different for each direction. Upstream: minimise loss and delay for small critical packets. Downstream: balance FEC redundancy, compression, and frame rate against available throughput.

Figure 3. Packet structure comparison between upstream (uniform tiny input packets) and downstream (bursty I/P-frame pattern). The bandwidth ratio is approximately 100:1 downstream-to-upstream.

1.6 System Overview

Before examining each component in detail, fig. 4 presents the end-to-end system architecture. The remainder of this report expands each block: Parts I–II describe the telemetry signals and access technology fingerprints that feed the inference engine; Part III covers the inference and classification logic; Part IV details the mitigation toolkit; and Parts V–VI address deployment, validation, and open questions.

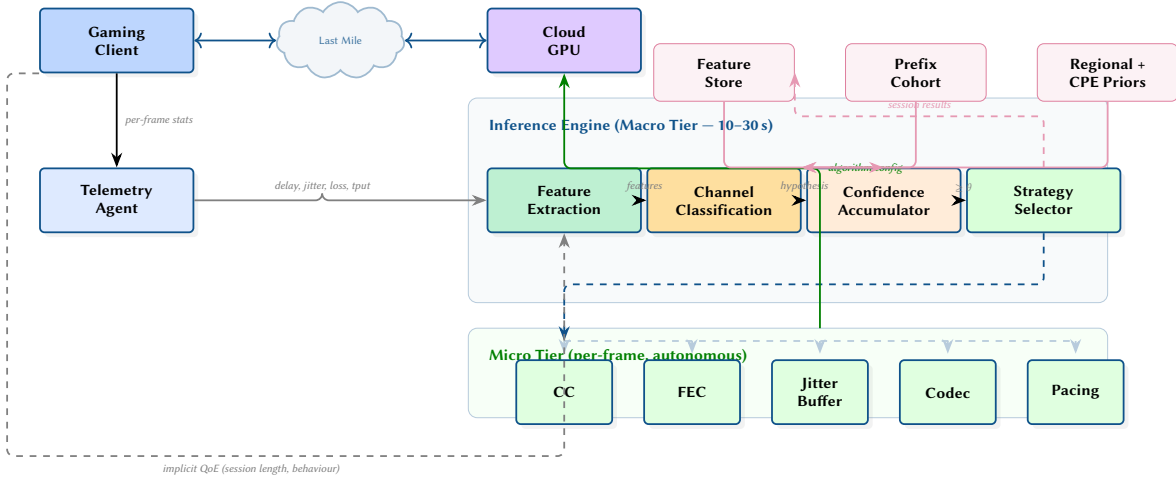


Figure 4. End-to-end system architecture. The client’s telemetry agent feeds per-frame measurements to the server-side inference engine. The macro tier classifies the channel using features, cohort signals, and stored priors, then selects which micro-tier algorithms to activate. Session results feed back into the feature store, improving future classification. Each numbered section of this report expands one or more blocks in this diagram.

2 Related Work

This section positions the present work relative to five areas of existing literature. As a position paper, our contribution is *integrating* these threads into a unified framework rather than advancing any single area.

2.1 Cloud Gaming Measurement

Foundational latency measurements by Chen et al. [0] and Lampe et al. [0] established the MTP budget as the central constraint. More recent platform-specific studies have dissected commercial deployments: Diomedee et al. [0] compared Stadia, GeForce NOW, and PSNow; Carrascosa and Bellalta [0] analysed Stadia’s traffic patterns; and Lyu et al. [0] provided the most detailed anatomy of GeForce NOW’s network behaviour to date. Barman et al. [0] studied platform behaviour under both synthetic and real cellular conditions. These studies characterise *what* cloud gaming traffic looks like; our framework addresses *what to do with that knowledge* at the server side.

2.2 QoE Modelling for Interactive Media

Metzger et al. [0] provide a comprehensive survey of QoS/QoE factors for online games. Sabet et al. [0] showed that delay sensitivity varies by game genre—a finding we formalise in our genre-specific latency budgets (table 2). Wang et al. [0] demonstrated that games are not equal in their QoE sensitivity, motivating our game-type-aware calibration. Where these studies model QoE as a function of network conditions, we go further by proposing that the inferred network *class* should drive strategy selection, not just the raw metrics.

2.3 Adaptive FEC and Loss Recovery

Static FEC for cloud gaming has been studied extensively: Wu et al. [0] applied source-FEC coding over TCP, while An et al. [0] demonstrated fine-grained per-frame FEC adjustment at NSDI ’25. Chen et al. [0] applied reinforcement learning to tune FEC parameters, and Meng et al. [0] proposed edge-based recovery at NSDI ’24. Shokrollahi [0] and the RaptorQ specification [0] provide the coding-theoretic foundation. Luby [0] recently proposed adaptive receiver-side scheduling using fountain codes for interactive delivery. Our framework builds on these by arguing that the *choice* between FEC modes (and their parameterisation) should be driven by access technology classification rather than reactive loss statistics alone.

2.4 Congestion Control for Real-Time Media

Copa [0] targets minimum delay, GCC [0] is designed for WebRTC, and SCReAM [0] provides self-clocked rate adaptation. Xu and Claypool [0] measured cloud gaming system response to competing TCP flows. On the AQM side, CoDel [0], PIE [0], and the L4S architecture [0] address bufferbloat at the network layer. Our contribution is to treat CC algorithm *selection*—not just parameterisation—as a decision variable that should change based on the inferred channel class.

2.5 Network Fingerprinting and Classification

Jiang et al. [0] showed that video quality problems cluster by ISP prefix—a finding that directly motivates our prefix cohort profiling. Time series classification methods including Matrix Profile [0] and ROCKET/MiniRocket [0] provide the algorithmic tools for motif discovery and rapid classification. Broadband measurement platforms (FCC MBA [0], Ofcom [0], RIPE Atlas [0], BISmark [0]) provide aggregate performance data by access technol-

ogy, but not at the per-frame granularity that cloud gaming requires. Our framework proposes that cloud gaming platforms themselves can serve as a large-scale measurement infrastructure, building fingerprint corpora from their own telemetry.

3 The Latency Pipeline

The total Motion-to-Photon (MTP) latency experienced by a cloud gaming player is the sum of eight stages (fig. 5), first formalised by Chen et al. [0] and refined by Lampe et al. [0].

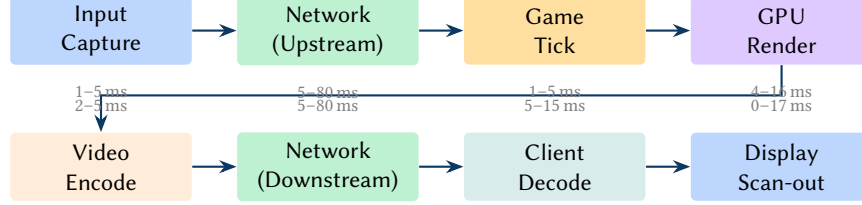


Figure 5. The eight-stage cloud gaming latency pipeline with typical per-stage contributions. The network stages are where inference and mitigation operate.

$$L_{MTP} = t_{\text{input}} + \underbrace{t_{\text{net}}^{\uparrow}}_{\text{upstream}} + t_{\text{tick}} + t_{\text{render}} + t_{\text{encode}} + \underbrace{t_{\text{net}}^{\downarrow}}_{\text{downstream}} + t_{\text{decode}} + t_{\text{display}} \quad (2)$$

The network terms can be further decomposed into last-mile and backbone contributions:

$$t_{\text{net}} = t_{\text{last-mile}}(a) + t_{\text{backbone}}(d) + t_{\text{queuing}} \quad (3)$$

where a is the access technology and d is the distance to the serving Point of Presence (PoP). Of these, $t_{\text{last-mile}}$ is the most variable and least controllable from the server’s perspective—making its inference the central challenge of this report.

Figure 6 shows how these stages map onto a concrete timeline for a single frame, illustrating where the latency budget is consumed.

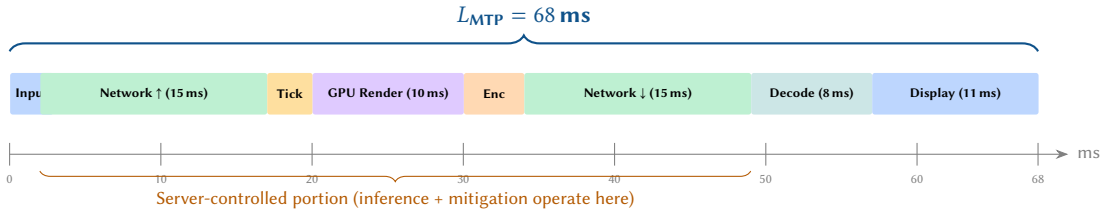


Figure 6. Lifecycle of a single frame on a 15 ms one-way path. The 68 ms total MTP leaves only 12 ms of margin before the 80 ms competitive gaming threshold.

Figure 7 disaggregates the MTP budget across pipeline stages for seven representative access technologies, using simulated P95 latencies from the companion Monte Carlo study (LM-SIM-2026-01). The network segments (upstream + downstream) consume the majority of the budget for wireless technologies; satellite exceeds the 80 ms competitive threshold from network transit alone.

3.1 Genre-Specific Latency Budgets

The 80 ms competitive threshold cited above is only one point on a spectrum (fig. 8). Different game genres tolerate substantially different MTP latencies, and—critically—the *allocation* of that budget across pipeline stages shifts with the genre.

Table 2. MTP budget allocation by game genre. All values in milliseconds. The “Network headroom” column shows the maximum tolerable one-way network delay after all other stages are accounted for.

Genre	MTP target	Input + Tick	Render + Encode	Decode + Display	Network ↓	Headroom
Competitive FPS / Fighting	60–80	4–6	12–16	10–14	10–20	<10
Action / Racing	80–100	4–6	14–18	12–16	15–25	10–20
Casual / Platformer	100–130	5–8	16–22	14–18	20–35	15–30
Turn-based / Strategy	150–200	8–12	20–30	16–22	30–50	30–60
VR / AR (per eye)	20–40	2–3	5–8	4–6	4–8	<5

Insight:
The genre determines not just the total latency budget but how aggressively the inference engine must act. A competitive FPS player on Wi-Fi 5 with 15 ms jitter spikes has already exhausted their headroom; the same

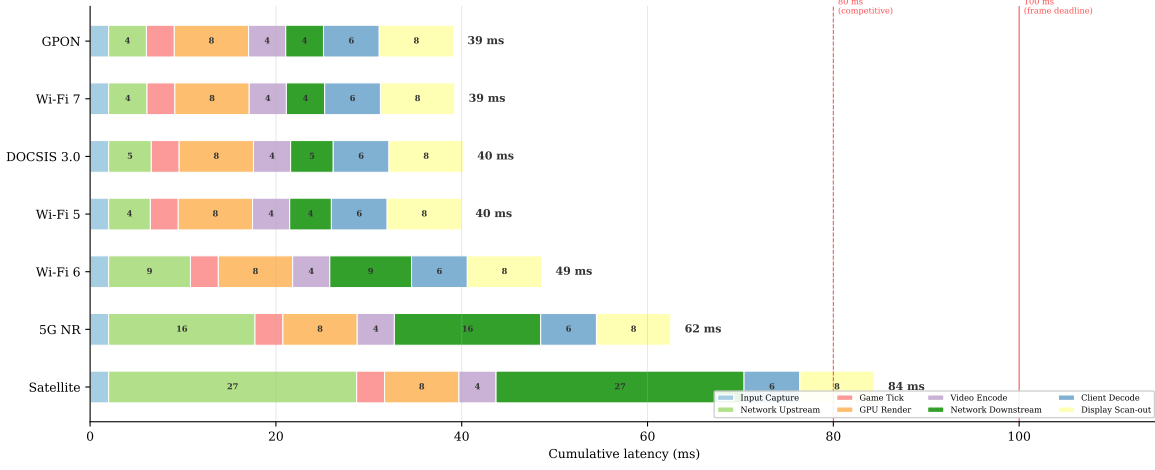


Figure 7. MTP latency budget breakdown by pipeline stage for seven access technologies. The two vertical lines mark the 80 ms competitive gaming and 100 ms frame deadline thresholds. Network segments use simulated P95 one-way latencies.

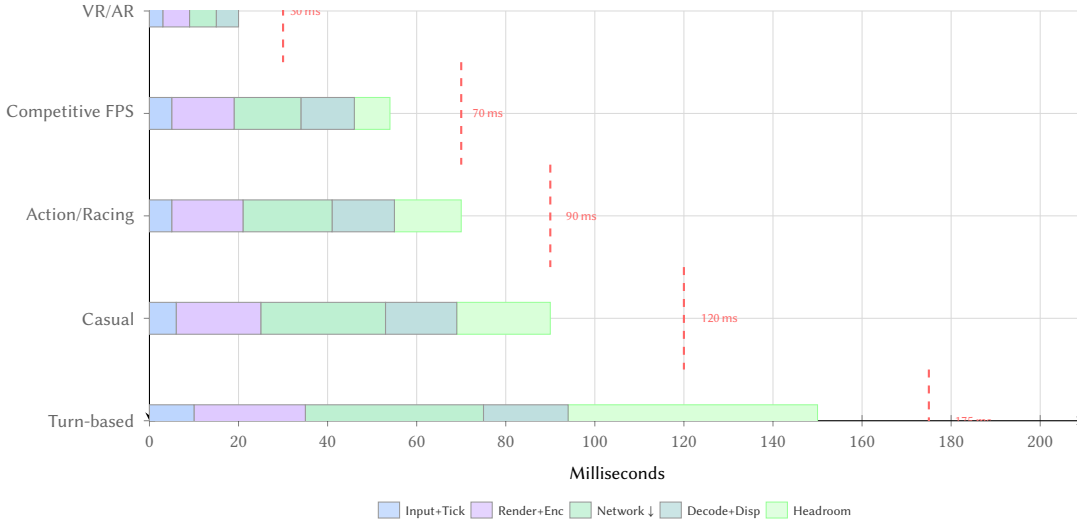


Figure 8. Stacked MTP budget by genre. The dashed red lines indicate typical MTP targets. VR/AR has essentially no network headroom, demanding edge deployment. Turn-based genres offer generous margins that tolerate even satellite paths.

3.2 Client-Side Decode Budget and Device Variability

The network is not the only variable in the MTP equation. Client-side decode time (t_{decode}) varies substantially with device hardware: a desktop PC with a dedicated hardware decoder may decode a 1080p frame in 2–3 ms, while a low-end mobile phone may require 8–12 ms. This variability directly affects the *available network budget*—the time remaining for upstream and downstream transit after all non-network stages are accounted for.

For a competitive gaming target of $L_{\text{MTP}} \leq 80$ ms, the available network budget is:

$$t_{\text{net}}^{\text{budget}} = 80 - t_{\text{input}} - t_{\text{tick}} - t_{\text{render}} - t_{\text{encode}} - t_{\text{decode}}(d) - t_{\text{display}} \quad (4)$$

where $t_{\text{decode}}(d)$ is device-dependent. A powerful desktop client may have $t_{\text{net}}^{\text{budget}} \approx 55$ ms, while a budget smartphone may have only ≈ 45 ms. The inference engine should factor in the client's reported decode capability (or infer it from the device class) when setting mitigation aggressiveness thresholds: a client with less network headroom requires earlier and more aggressive intervention.

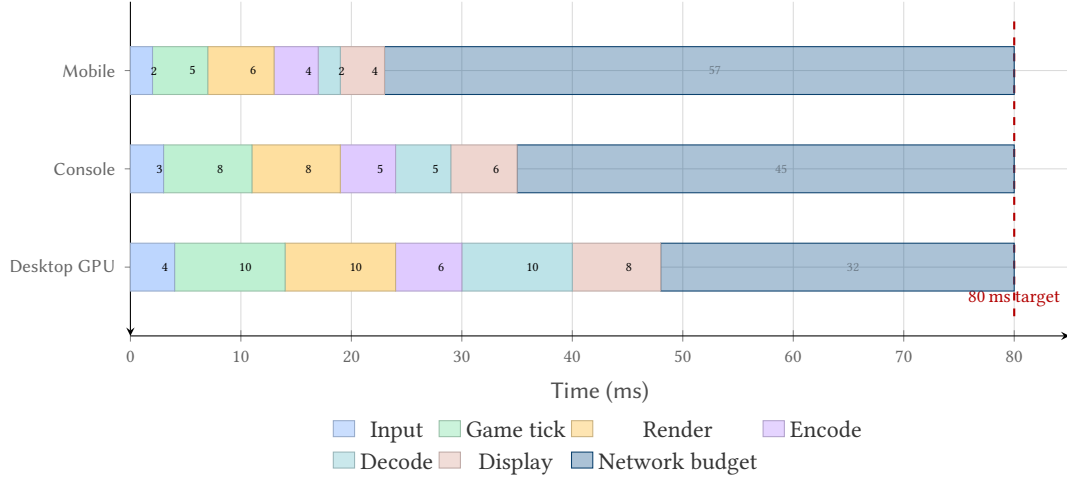


Figure 9. Client-side MTP budget waterfall for three device classes. Each stacked bar decomposes the fixed processing costs (input, game tick, render, encode, decode, display); the remaining segment (coloured) is the available network budget within an 80 ms competitive gaming target. A mobile client has roughly half the network headroom of a desktop GPU client.

4 The Four Telemetry Signals

Effective last-mile inference requires continuous measurement of four network telemetry signals at per-frame granularity (fig. 10). Together, these signals characterise the instantaneous and statistical properties of the network path.

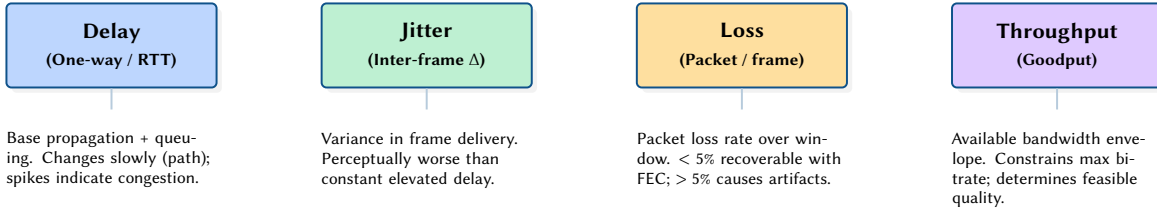


Figure 10. The four telemetry signals measured at per-frame granularity. Each captures a distinct aspect of last-mile network behaviour.

4.1 Delay

One-way delay (or the observable Round-Trip Time (RTT)) captures the base propagation time plus any queuing along the path. For cloud gaming, delay is typically measured via application-level timestamps embedded in the game’s own protocol packets, providing more accurate results than ICMP ping [0]. The relationship between network RTT and perceived MTP latency is:

$$L_{\text{MTP}} \approx \text{RTT} + \underbrace{t_{\text{encode}} + t_{\text{decode}} + t_{\text{display}}}_{\approx 30-80 \text{ ms overhead}} \quad (5)$$

4.2 Jitter

Inter-frame delay variation is often more perceptually disruptive than constant elevated latency [0]. Even brief spikes in frame delivery interval cause visible stuttering.

Telecommunications systems traditionally measure Packet Delay Variation (PDV) per RFC 3393 or use the smoothed RTP jitter formulation. Cloud gaming can simplify this: recording arrival times of fully reconstructed video frames with a high-resolution timer yields an approximation of the jitter the player actually perceives. This frame-level inter-arrival variance captures the underlying channel characteristics (e.g., Wi-Fi contention spikes) without tracking every constituent UDP packet. The metric is typically quantified as the standard deviation of frame inter-arrival times over a sliding window, or more robustly as the $P_{95}-P_{50}$ spread.

4.3 Loss

Packet loss rate measured over a sliding window. Cloud gaming platforms using FEC can typically recover from up to $\approx 5\%$ packet loss without visible degradation [0]. Beyond this threshold, loss manifests as macroblocking artifacts (downstream) or missed inputs (upstream). Importantly, packet *reordering* in UDP streams can be functionally equivalent to loss when packets arrive after their playout deadline.

The perceptual impact of downstream loss is highly asymmetric depending on the type of frame affected. Losing an I-frame (Intra-coded frame) is devastating: because it serves as the reference anchor for subsequent frames, its loss breaks the decoding chain, causing severe, sustained macroblocking or a frozen screen until the next I-frame arrives (often a full second or more later). Conversely, losing packets from a P-frame (Predicted frame) typically results in transient, localized visual artifacts that are quickly resolved by subsequent updates.

4.4 Throughput (Goodput)

The available end-to-end throughput constrains the maximum feasible video bitrate. Unlike the other three signals, throughput is a *capacity* metric rather than a *quality* metric—it determines the envelope within which the streaming server must operate. Throughput estimation is complicated by the bursty nature of video frame delivery (large I-frames followed by smaller P-frames).

Design Rule 1: High-Frequency Sampling vs. Strategy Reaction Time

While capturing these four signals at high frequency (e.g., per-frame, every 8–33 ms) is highly valuable for exposing transient impairments that 5-second averages obscure, it is critical to recognise that *the inference engine cannot and should not react at that rate*.

The system operates on two distinct timescales. At the **micro level**, the currently selected algorithms—congestion control (e.g., BBR), FEC coding (e.g., Raptor), and codec rate control—operate autonomously at sub-second or per-frame granularity, reacting to individual packet events as they are designed to do. At the **macro level**, the inference engine observes telemetry over a much longer window (typically 5–30 seconds) to build sufficient confidence before switching *which* algorithms or strategy profiles are active. This dampening is essential: the engine needs seconds of observation to distinguish a genuine regime change from a transient spike, and switching strategies prematurely—before convergence—risks compounding the problem rather than resolving it (section 10).

The core hypothesis is therefore that there are likely significant benefits to be gained from dynamically adapting *which* algorithms and mitigation strategies are applied over time, even though each individual algorithm operates at its own native timescale.

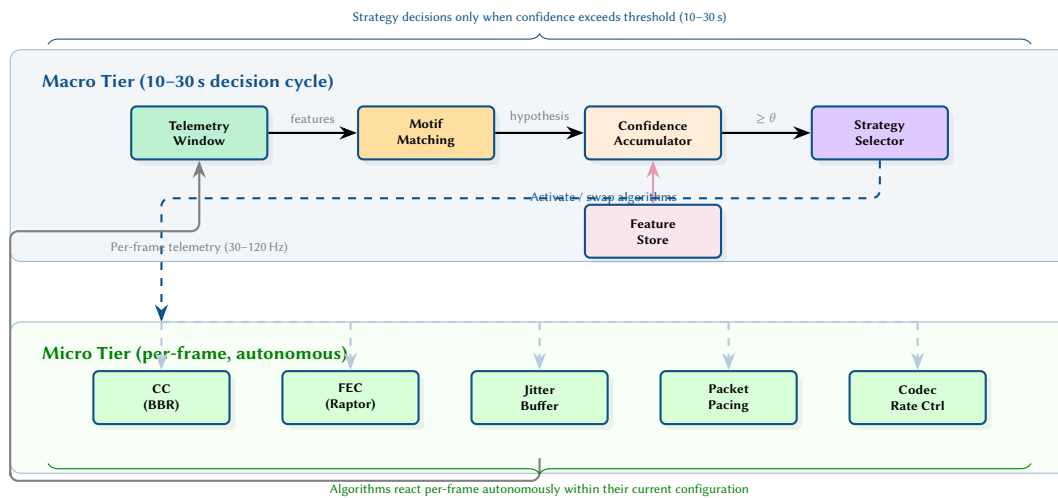


Figure 11. Two-tier decision architecture. The macro tier observes telemetry over 10–30 s windows, matches motifs, accumulates confidence, and selects which algorithm set to deploy. The micro tier runs the selected algorithms autonomously at per-frame rates. The macro tier never adjusts individual algorithm parameters—it only chooses *which* algorithms are active.

These transient events carry asymmetric weight. On the upstream path, a single missed key press can cascade into an unplayable state; even a brief period of perceived unresponsiveness can ruin a session. Baena et al. [0] and Wan and Jamieson [0] demonstrate frameworks for extracting high-frequency quality indicators without requiring perfect measurement.

Figure 13 illustrates how the four signals co-evolve during a real gaming session, showing the correlated signatures of a congestion event.

ACCESS TECHNOLOGY LANDSCAPE

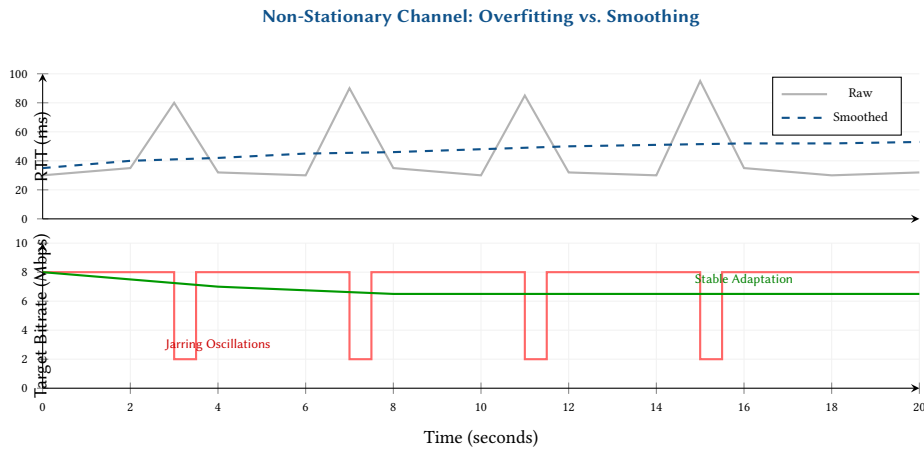


Figure 12. The danger of overfitting to per-frame telemetry. In a non-stationary channel with transient spikes (top, grey), reacting instantly causes severe quality oscillations (bottom, red). Using the data to build a converged profile (top, dashed) allows for a stable, proportional mitigation response (bottom, green).

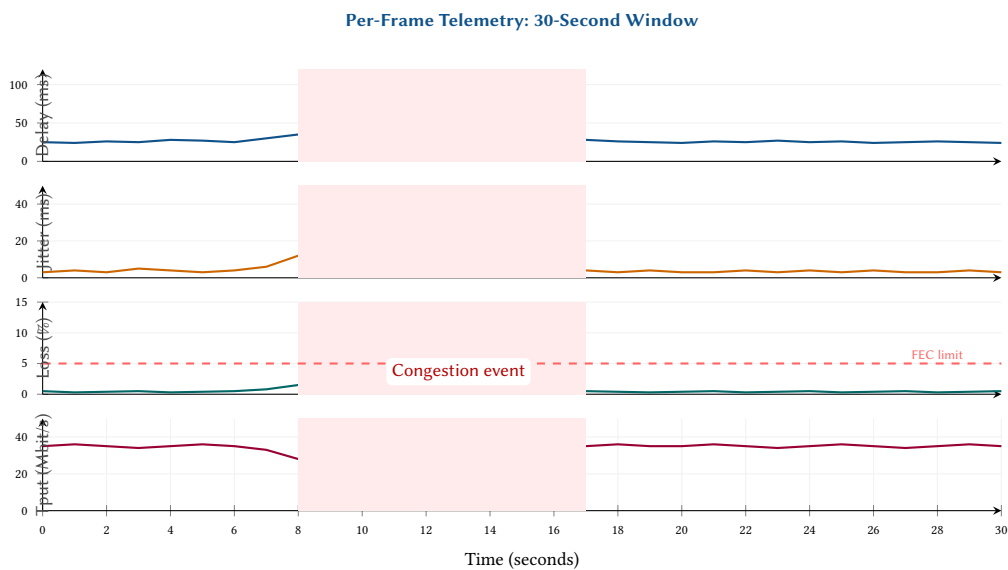


Figure 13. Four telemetry signals during a 30-second gaming session. A congestion event (shaded) at $t = 8\text{--}17\text{ s}$ produces correlated signatures: rising delay, spiking jitter, loss exceeding the FEC recovery threshold, and collapsing throughput.

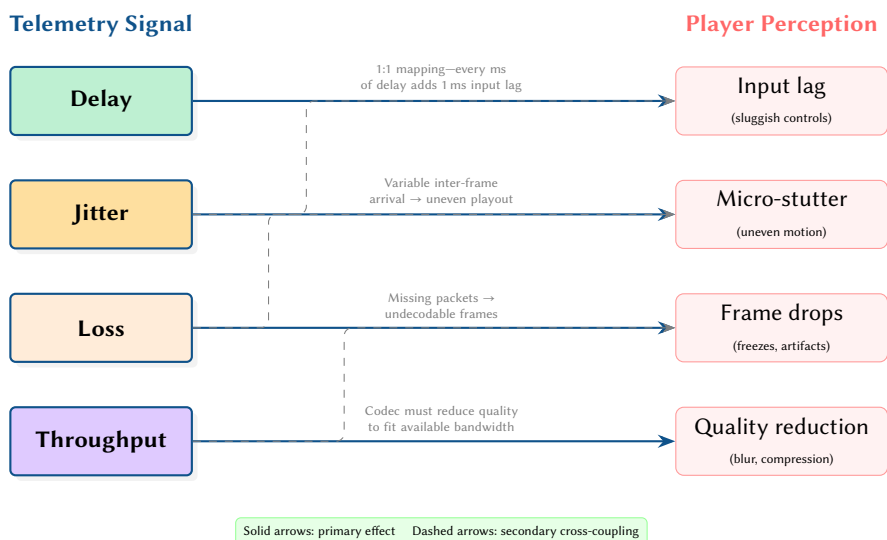


Figure 14. Telemetry signals mapped to player-perceived QoE effects. Each signal has a primary perceptual consequence (solid arrows) and secondary cross-coupling effects (dashed arrows). High jitter, for example, primarily causes micro-stutter but also contributes to perceived input lag through variable frame delivery timing.

5 Last-Mile Access Technology Characteristics

Last-mile networks produce structural, repeating *motifs*—distinct fingerprints in delay, jitter, and loss—imposed by their underlying physical constraints. Because cloud gaming cannot buffer, identifying these fingerprints is the only way to apply the correct Layer 7 mitigation. A strategy that cures one access technology will often cripple another.

5.1 Wi-Fi Generational Signatures

While the external access network defines the macro-characteristics of the path, the in-home Wi-Fi network frequently dominates the high-frequency jitter profile.

5.1.1 Wi-Fi 5 (802.11ac): Contention and Retries

Physical layer: Wi-Fi 5 operates in the 5 GHz band using up to 80 MHz (optionally 160 MHz) OFDM channels. The 5 GHz spectrum is shared with DFS radar and neighbouring networks. Multipath propagation—reflections from walls, furniture, and people—causes frequency-selective fading, where certain subcarriers experience deep nulls. When fading or co-channel interference degrades the Signal-to-Noise Ratio (SNR), the rate adaptation algorithm drops the Modulation and Coding Scheme (MCS), reducing throughput but maintaining connectivity.

Link layer (MAC): Wi-Fi 5 is a half-duplex, shared medium relying on Carrier-Sense Multiple Access with Collision Avoidance (CSMA/CA). Only one station transmits at a time; all others must wait. When a gaming device contends with a device performing a bulk download, the gaming packets are queued behind the bulk transmission. If a frame is corrupted on the air interface, the MAC layer retransmits it (up to a retry limit, typically 7 for short frames). To the application layer, successful retransmissions do not appear as packet loss but as sudden, unpredictable jitter spikes—the signature visible in fig. 16. When retries are exhausted, the MAC silently drops the frame, producing the burst-loss events visible in fig. 17.

Telemetry fingerprint (fig. 17): Stable throughput capacity with erratic, high-amplitude delay spikes (4–120 ms above baseline) and occasional correlated burst loss. Critically, throughput remains largely unaffected—the bottleneck is contention time, not channel capacity.

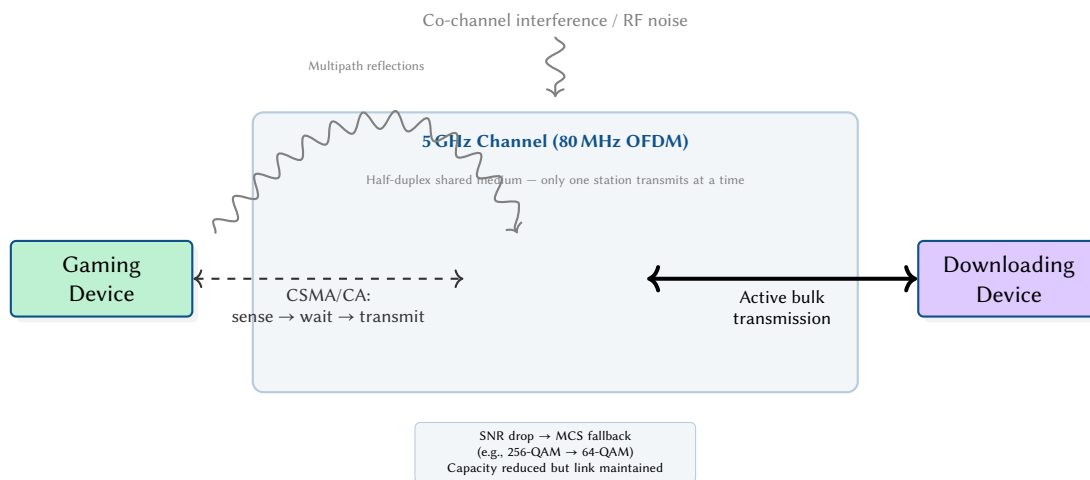


Figure 15. Wi-Fi 5 (802.11ac) physical layer. The 5 GHz OFDM channel is a half-duplex shared medium. Multipath reflections cause frequency-selective fading; co-channel interference degrades SNR, triggering MCS fallback.

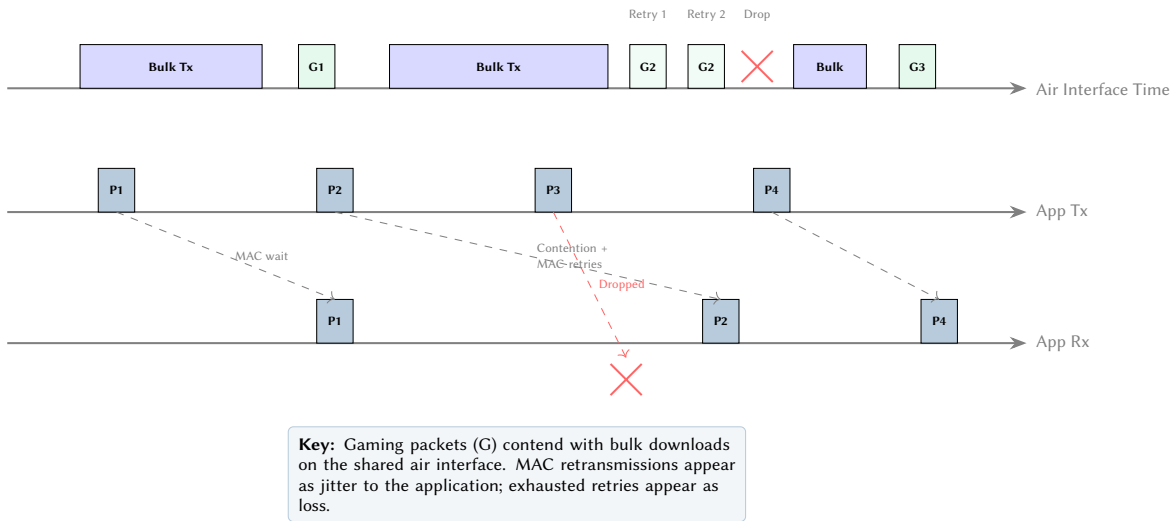


Figure 16. Wi-Fi 5 MAC-layer behaviour. The gaming device’s packets must wait for bulk transmissions to clear the shared medium (CSMA/CA). Failed air-interface transmissions trigger MAC retries, inflating delay; exhausted retries cause silent packet drops.

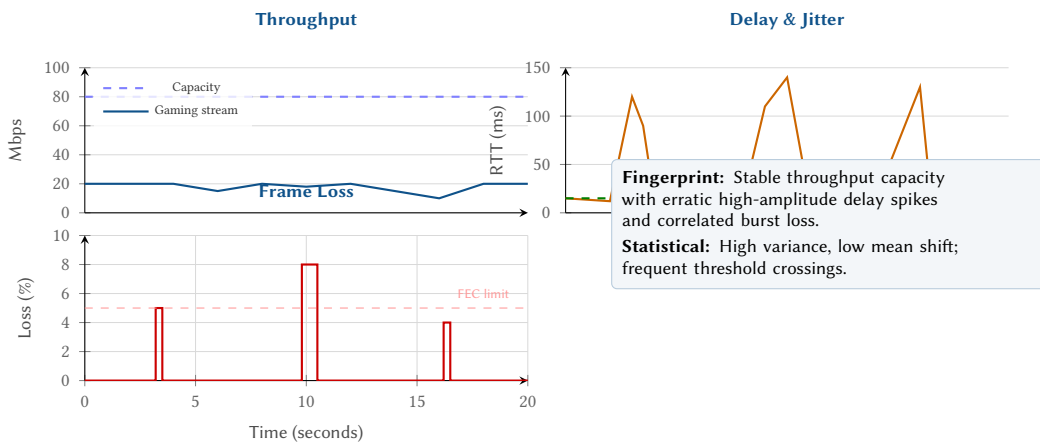


Figure 17. Wi-Fi 5 telemetry fingerprint over a 20-second window. Throughput remains largely stable (capacity is not the bottleneck), but delay exhibits violent spikes from MAC-layer contention, with correlated burst loss when retries are exhausted.

5.1.2 Wi-Fi 6 (802.11ax): OFDMA

Physical layer: Wi-Fi 6 retains the 5 GHz band but replaces single-user OFDM with Orthogonal Frequency-Division Multiple Access (OFDMA). The channel is subdivided into Resource Units (RUs)—groups of subcarriers that can be independently assigned to different clients within a single transmission opportunity. An 80 MHz channel can be divided into RUs as small as 2 MHz, providing fine-grained scheduling. Target Wake Time (TWT) further reduces contention by scheduling client wake periods.

Link layer (MAC): The fundamental change from Wi-Fi 5 is the elimination of per-packet contention for scheduled traffic. The Access Point (AP) acts as a centralised scheduler, assigning RUs to clients in each downlink OFDMA frame. The gaming device receives its own dedicated RU (fig. 18), transmitting in parallel with bulk downloaders rather than waiting behind them. This transforms the medium access from a random-access lottery into a deterministic schedule, drastically reducing jitter (fig. 19).

Telemetry fingerprint (fig. 20): Consistently low delay with reduced jitter amplitude compared to Wi-Fi 5. Some residual jitter spikes remain from uplink contention (which still uses CSMA/CA-based OFDMA random access) and from environments where the AP does not fully utilise OFDMA scheduling.

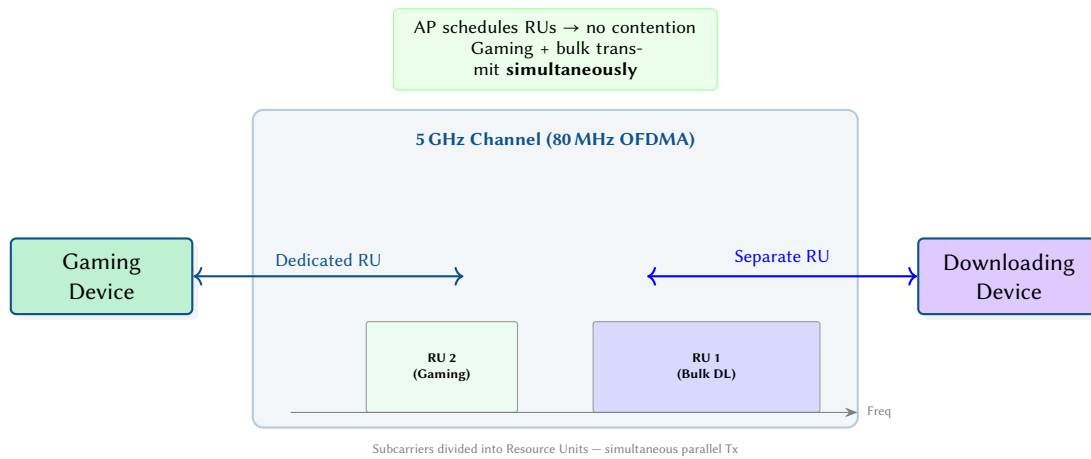


Figure 18. Wi-Fi 6 (802.11ax) physical layer. OFDMA divides the 80 MHz channel into Resource Units assigned to individual clients. The gaming device and bulk downloader transmit simultaneously on separate RUs, eliminating CSMA/CA contention.

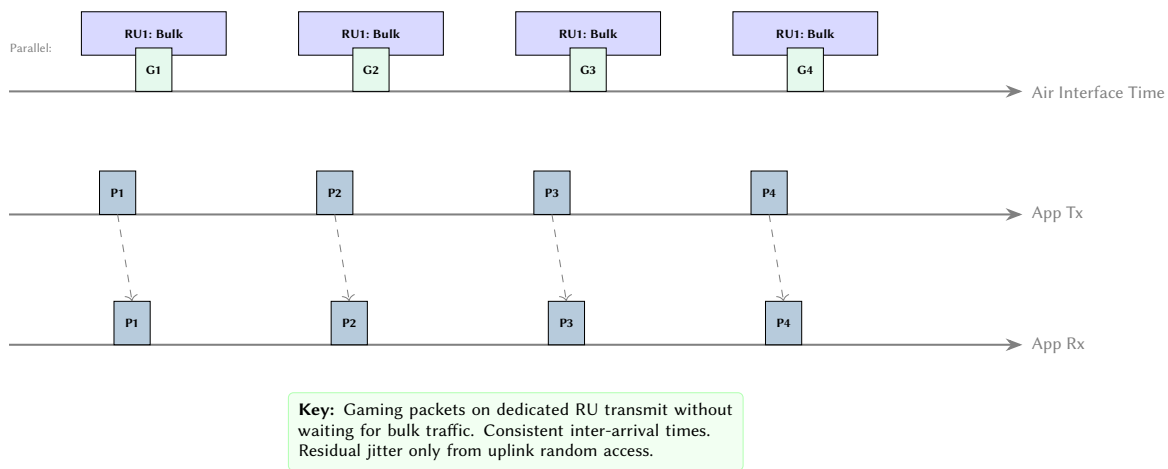


Figure 19. Wi-Fi 6 MAC-layer behaviour. The AP schedules gaming packets on a dedicated Resource Unit, transmitting in parallel with bulk downloads. Packet inter-arrival times are consistent, with minimal contention delay.

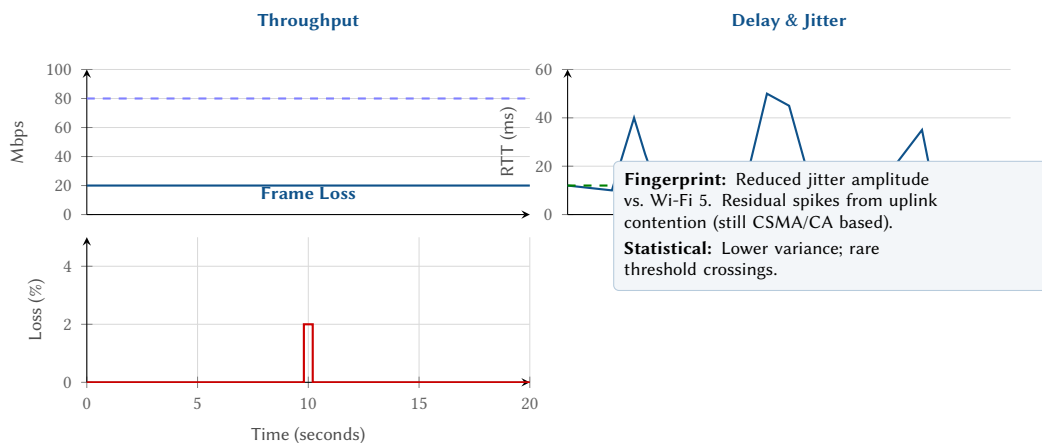


Figure 20. Wi-Fi 6 telemetry fingerprint. OFDMA scheduling dramatically reduces jitter amplitude compared to Wi-Fi 5. Residual spikes from uplink random access remain but are lower in magnitude and frequency.

5.1.3 Wi-Fi 6E (802.11ax in 6 GHz): Greenfield Spectrum

Physical layer: Wi-Fi 6E extends the 802.11ax standard into the 6 GHz band, adding up to 1200 MHz of new spectrum (5.925–7.125 GHz). This is *greenfield* spectrum: no legacy 802.11a/n/ac devices operate here, there are no DFS radar restrictions, and neighbouring networks are far fewer than on the crowded 5 GHz band. Channels up to 160 MHz are available without fragmentation. The PHY is identical to Wi-Fi 6 (OFDMA, 1024-QAM), but the clean spectral environment eliminates many of the interference sources that degrade Wi-Fi 6 on 5 GHz.

Link layer (MAC): Because every device on 6 GHz supports 802.11ax natively, there is no legacy protection overhead—no RTS/CTS for hidden 802.11a nodes, no legacy preambles, and no fallback to single-user OFDM. The AP’s OFDMA scheduler operates without contention from legacy devices, producing more consistent Resource Unit allocation (fig. 22). Mandatory WPA3 and Protected Management Frames (PMF) are required for 6 GHz operation. The key limitation compared to Wi-Fi 7 is the absence of Multi-Link Operation—if the 6 GHz channel experiences a deep fade or sudden interference event, there is no instantaneous failover to an alternative band.

Telemetry fingerprint (fig. 23): Very low jitter with a stable baseline, substantially better than Wi-Fi 6 on 5 GHz but not quite matching Wi-Fi 7’s near-wired stationarity. Residual uplink contention spikes are present but less frequent and lower in amplitude than on 5 GHz. The inference engine should recognise this as a low-overhead environment requiring minimal mitigation, while remaining alert for the occasional fade event that MLO would otherwise mask.

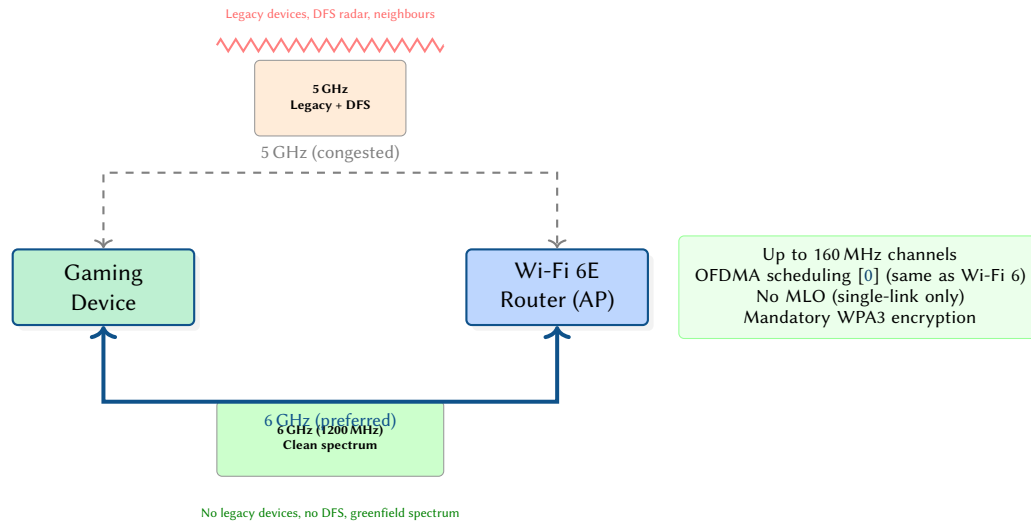


Figure 21. Wi-Fi 6E physical layer [0]. The 6 GHz band provides 1200 MHz of greenfield spectrum free from legacy device interference and DFS restrictions. While using the same OFDMA scheduling as Wi-Fi 6, the clean spectrum dramatically reduces contention and co-channel interference.

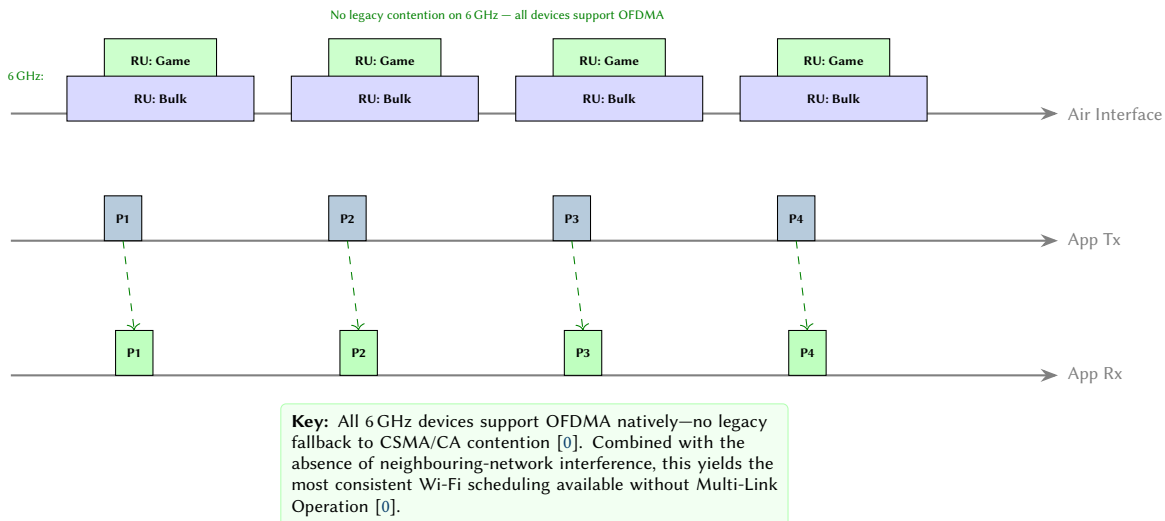


Figure 22. Wi-Fi 6E MAC-layer behaviour [0]. Because 6 GHz is a greenfield band, every device supports OFDMA scheduling—there is no legacy CSMA/CA fallback [0]. The absence of DFS interruptions and neighbouring legacy networks produces more consistent scheduling than Wi-Fi 6 operating on 5 GHz.

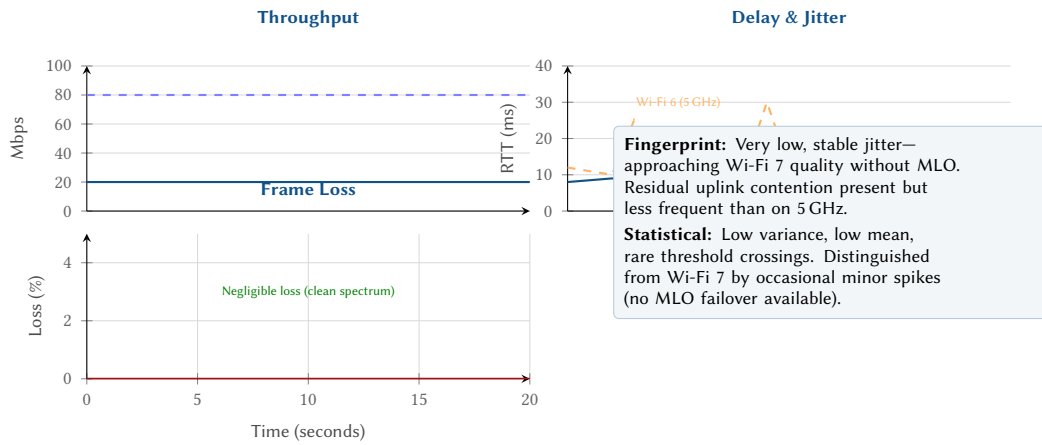


Figure 23. Wi-Fi 6E telemetry fingerprint [0]. The 6 GHz greenfield spectrum produces substantially lower jitter than Wi-Fi 6 on 5 GHz (shown dashed for comparison). Without MLO [0], occasional minor spikes from uplink contention remain, distinguishing this from Wi-Fi 7’s near-wired stationarity.

5.1.4 Wi-Fi 7 (802.11be): Multi-Link Operation

Physical layer: Wi-Fi 7 operates across three bands simultaneously: 2.4 GHz, 5 GHz, and the new 6 GHz band (up to 320 MHz channels). The 6 GHz band is particularly significant—it is largely free of legacy device interference and DFS restrictions, providing a “clean” spectrum. Channel bonding up to 320 MHz and 4096-QAM modulation deliver peak throughputs exceeding 40 Gbit/s, though these peaks are rarely relevant for cloud gaming’s modest bandwidth needs.

Link layer (MAC): The transformative feature is Multi-Link Operation (MLO). The client maintains simultaneous associations across multiple bands. At the MAC layer, each packet can be independently routed to whichever link currently offers the best conditions (fig. 25). If the 5 GHz link experiences a transient deep fade or interference event, the packet is instantly transmitted on 6 GHz with no MAC-layer backoff, no retry timeout, and no application-visible delay. This provides frequency diversity at the packet level, analogous to how MIMO provides spatial diversity at the symbol level.

Telemetry fingerprint (fig. 26): Near-wired stationarity. Delay variance approaches that of FTTH, with occasional minor fluctuations as MLO switches links. The inference engine should recognise this fingerprint to avoid applying unnecessary mitigation overhead.

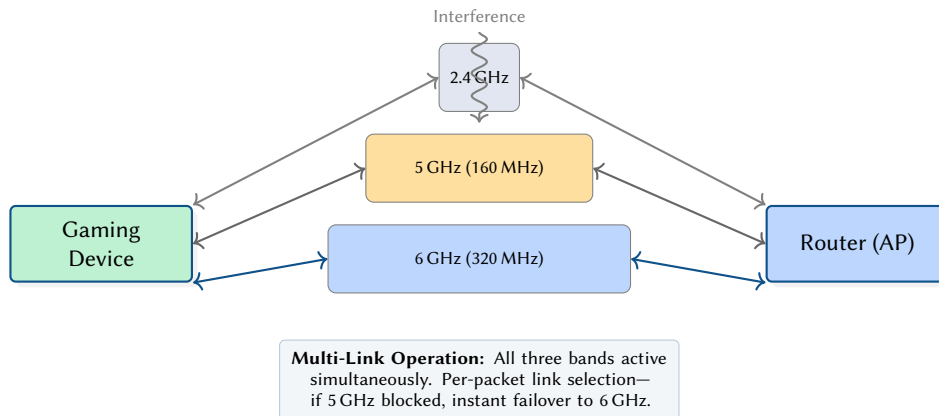


Figure 24. Wi-Fi 7 (802.11be) physical layer. Multi-Link Operation (MLO) maintains simultaneous associations across 2.4 GHz, 5 GHz, and 6 GHz. Interference on one band is bypassed by routing packets to another, providing frequency diversity at the packet level.

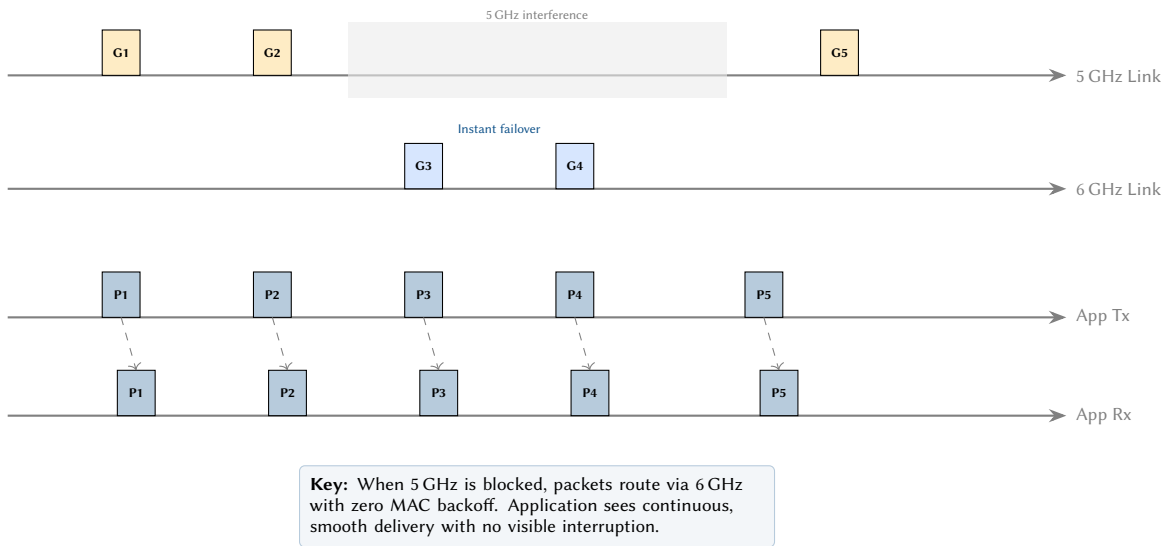


Figure 25. Wi-Fi 7 MLO link-layer behaviour. During 5 GHz interference, packets fail over instantly to the 6 GHz link without MAC-layer timeouts. The application perceives continuous smooth delivery.

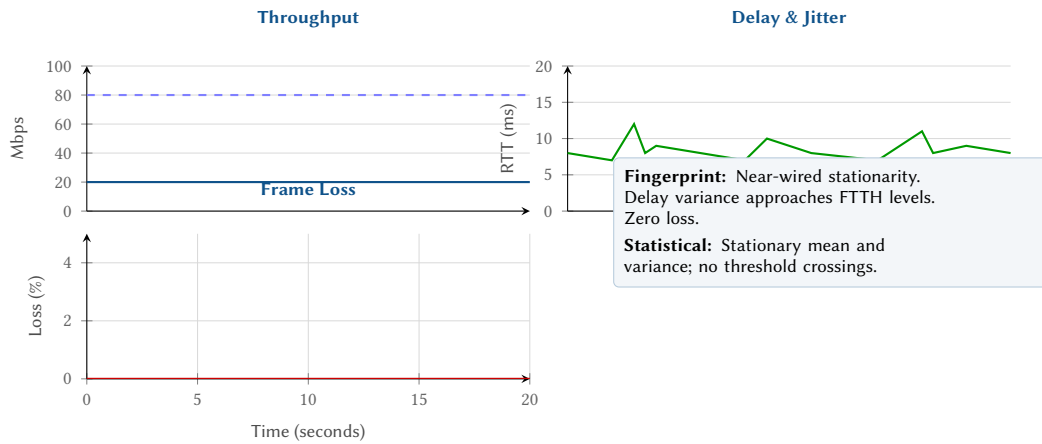


Figure 26. Wi-Fi 7 telemetry fingerprint. MLO frequency diversity produces near-wired stationarity. The inference engine should recognise this profile to avoid applying unnecessary mitigation overhead.

5.1.5 Wi-Fi Band Steering: Mid-Session Fingerprint Changes

Modern Wi-Fi access points use *band steering* to silently move clients between frequency bands based on signal quality, load balancing, or vendor-specific heuristics. When a gaming device is steered mid-session—for example, from a clean 5 GHz channel to the congested 2.4 GHz band because a signal quality threshold was crossed—the telemetry fingerprint changes abruptly.

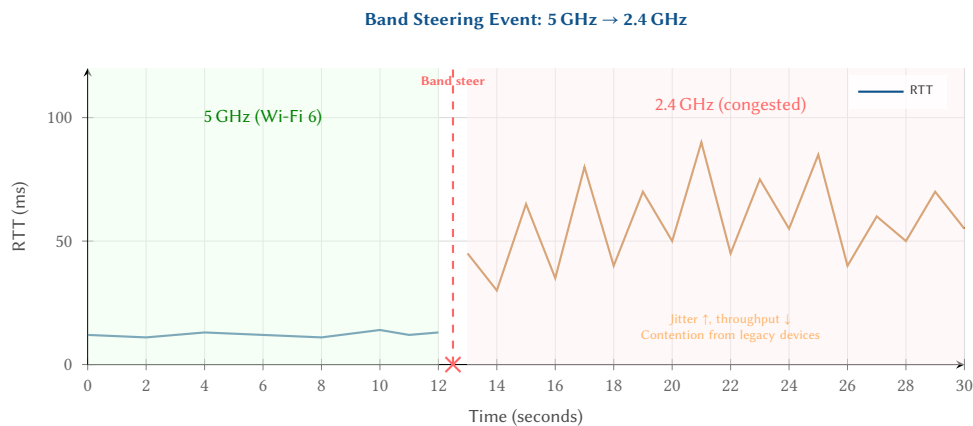


Figure 27. Band steering event mid-session. The AP moves the gaming device from 5 GHz to 2.4 GHz at $t = 12.5$ s. The telemetry shows an abrupt regime change: baseline delay jumps from ≈ 12 ms to ≈ 50 ms with substantially higher jitter from 2.4 GHz contention with legacy devices.

The inference engine must recognise band steering as a *regime change* rather than a gradual degradation. The hallmarks are: (1) an abrupt step change in both baseline delay and jitter variance, (2) a brief connectivity gap (10–500 ms) during the reassociation, and (3) the new fingerprint matching a different Wi-Fi generation’s characteristics. The appropriate response is to immediately trigger re-classification rather than attempting to mitigate the old fingerprint with increased intensity.

5.2 Fixed-Line Access Networks

5.2.1 FTTH / GPON

Physical layer: Fibre-to-the-Home networks transmit data as modulated light over single-mode glass fibre. The optical channel is immune to electromagnetic interference, multipath fading, and atmospheric effects—the phenomena that dominate wireless last-mile impairments. GPON (Gigabit Passive Optical Network) uses a passive optical splitter (typically 1:32 or 1:64) to fan out a single fibre from the Optical Line Terminal (OLT) at the ISP to multiple Optical Network Units (ONUs) at subscriber premises. Downstream, the OLT broadcasts to all ONUs at 2.488 Gbit/s using Time-Division Multiplexing (TDM); each ONU filters its assigned time slots.

Link layer: Upstream, each ONU transmits in a scheduled time slot assigned by the OLT via Dynamic Bandwidth Allocation (DBA). Because transmission windows are pre-scheduled, there is no contention—packets experience only propagation delay (approximately $5 \mu\text{s}/\text{km}$) plus minimal OLT processing. The shared downstream medium has sufficient capacity that queuing delay is negligible for cloud gaming traffic.

Telemetry fingerprint (fig. 30): The ideal stationary channel. Near-constant delay ($\approx 5\text{--}7\text{ ms}$ RTT for typical 10–20 km distances), negligible jitter (sub-millisecond variance), zero loss, and throughput far exceeding cloud gaming requirements. This fingerprint serves as the reference baseline against which all other access technologies are compared.

Insight:
Band steering is invisible to the application layer—the AP makes the decision unilaterally. The inference engine can only detect it retroactively from the telemetry discontinuity. On tri-band (Wi-Fi 7) routers with MLO, band steering is replaced by simultaneous multi-link operation, eliminating this class of regime change entirely.

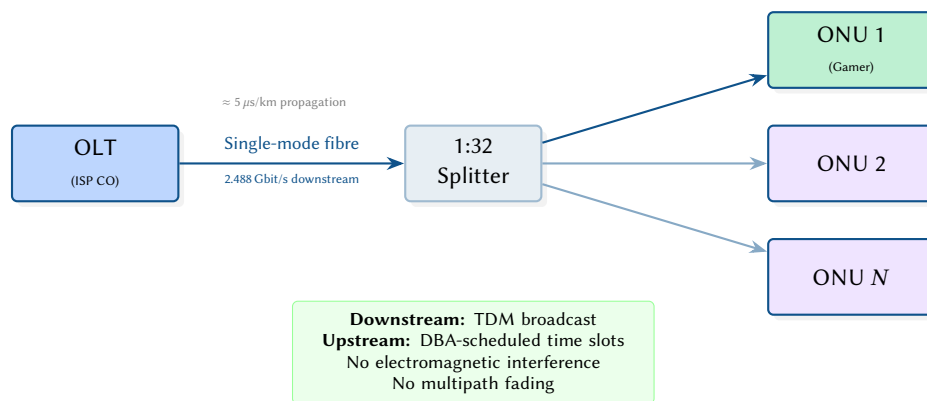


Figure 28. FTTH/GPON physical layer. The Optical Line Terminal (OLT) broadcasts downstream via TDM through a passive splitter. The optical medium is immune to electromagnetic interference and multipath fading, providing an inherently stationary channel.

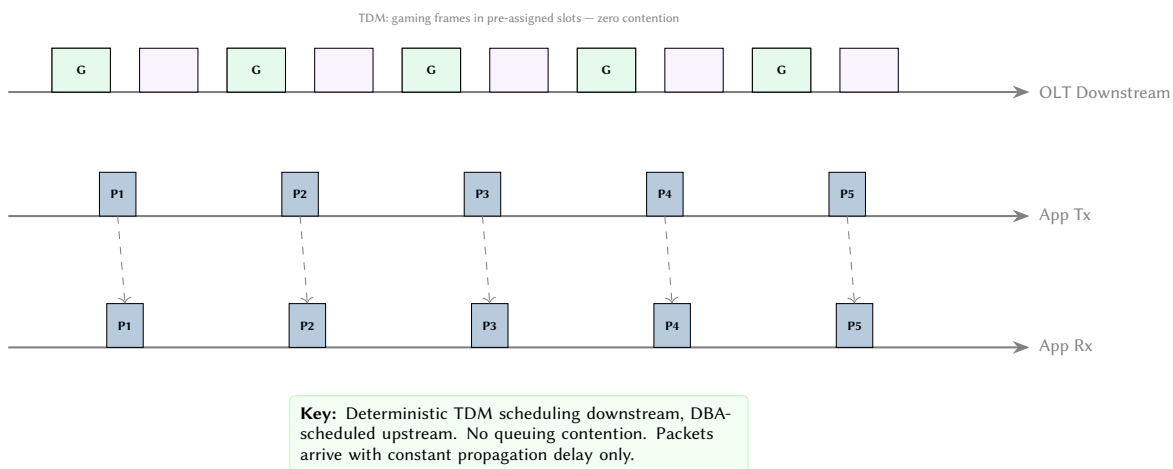


Figure 29. FTTH link-layer behaviour. Deterministic scheduling eliminates queuing contention. Packets experience only propagation delay, producing perfectly uniform inter-arrival times.

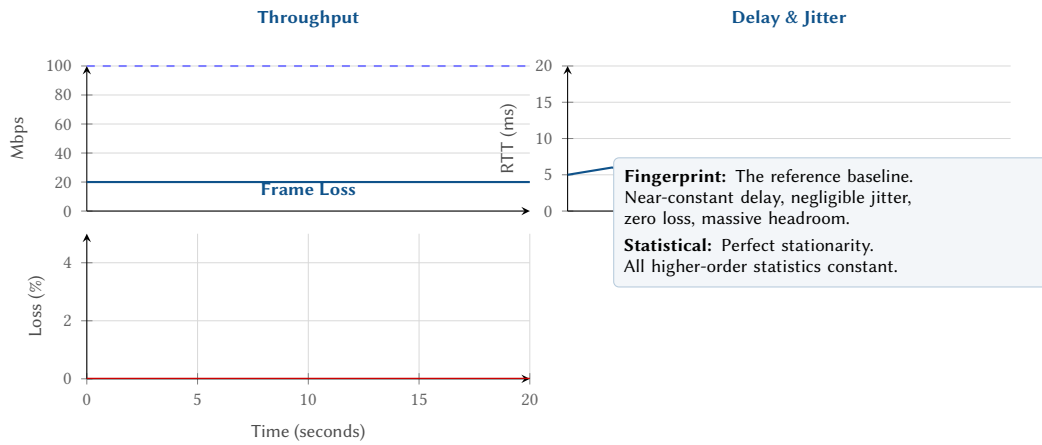


Figure 30. FTTH telemetry fingerprint—the ideal stationary channel. This profile serves as the reference baseline against which all other access technologies are compared.

5.2.2 DSL / VDSL: Copper Last Mile

Physical layer: Digital Subscriber Line (DSL) technologies transmit data over the existing twisted copper telephone pairs between the subscriber’s premises and the local exchange (or street cabinet for VDSL/G.fast). Unlike DOCSIS cable, each subscriber has a *dedicated* copper pair—there is no shared medium at the physical layer. However, copper pairs are bundled together in cables, and signals on adjacent pairs interfere through electromagnetic *crosstalk* (both Near-End and Far-End). The capacity of DSL is fundamentally constrained by loop length: VDSL2 achieves ≈ 100 Mbit/s at short distances (< 300 m) but degrades to < 10 Mbit/s at 1.5 km. Higher-frequency variants (G.fast at 106–212 MHz) achieve gigabit speeds but only over very short loops (< 100 m).

Link layer: The critical configuration for cloud gaming is the *interleaver*. DSL modems can operate in *fast path* mode (minimal latency, ≈ 1 –2 ms contribution) or *interleaved* mode (adds 8–16 ms of fixed delay but spreads data across time, allowing recovery from impulse noise bursts). Many ISPs default to interleaved mode for stability, which adds a constant delay penalty visible in the telemetry baseline. On fast path, impulse noise from electrical appliances (motors, switches, power adapters) produces brief burst loss events that are unrecoverable at the DSL layer (fig. 32). Vectoring (coordinated transmission across pairs) mitigates crosstalk in modern VDSL2 deployments but requires all lines in the bundle to be managed by the same DSLAM.

Telemetry fingerprint (fig. 33): An elevated but stable delay baseline (reflecting the interleaver contribution and copper propagation), with low-amplitude jitter from crosstalk variations. Throughput is stable but may be constrained (particularly on long loops or ADSL2+). On fast-path configurations, occasional brief burst loss events from impulse noise are the primary impairment. The key distinction from FTTH is the higher baseline delay; from DOCSIS, the stable throughput under load (no shared medium congestion).

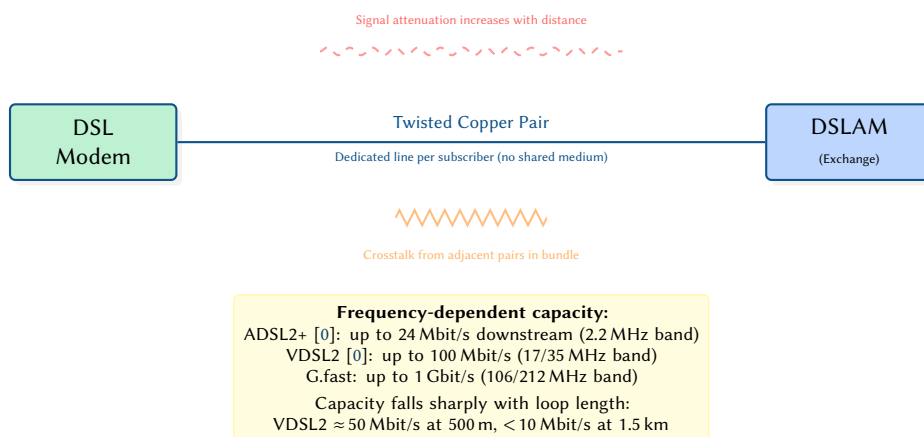


Figure 31. DSL/VDSL physical layer [0]. Each subscriber has a dedicated copper pair to the exchange (no shared medium), but signal attenuation increases with distance and frequency. Crosstalk from adjacent pairs in the cable bundle introduces noise [0] that varies with the activity of neighbours.

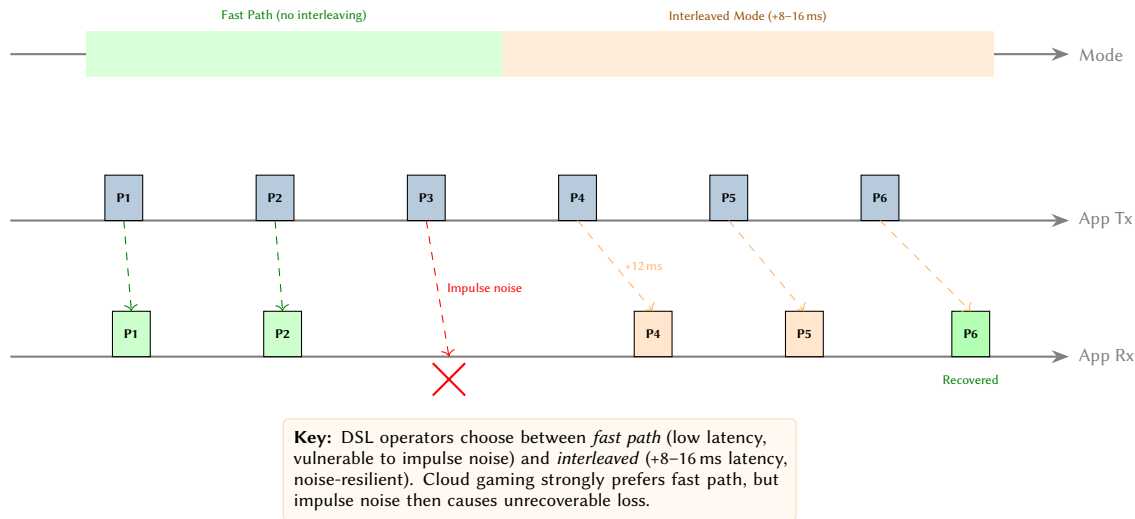


Figure 32. DSL link-layer interleaver tradeoff [0]. Fast path minimises latency but leaves packets vulnerable to impulse noise bursts (e.g., from electrical appliances). Interleaved mode adds 8–16 ms of fixed delay but can correct noise-induced errors through temporal spreading.

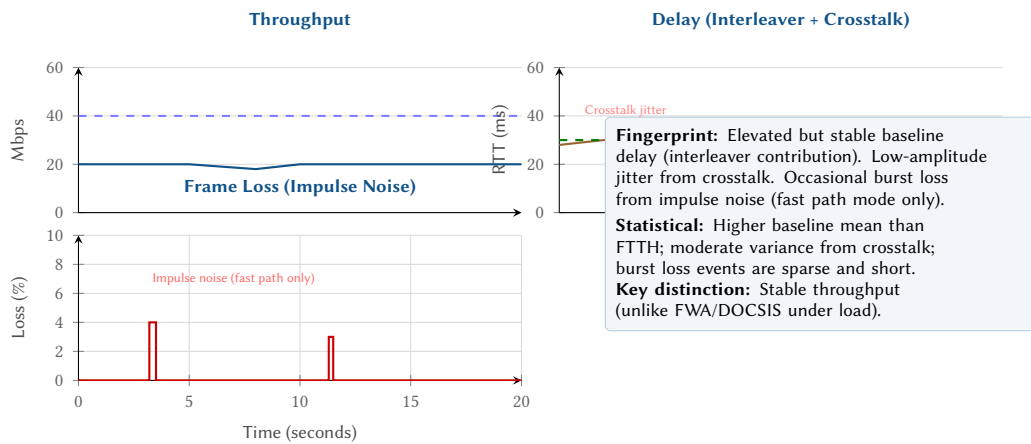


Figure 33. DSL/VDSL telemetry fingerprint [0]. The elevated but stable delay baseline reflects the interleaver contribution (if enabled) and copper propagation. Crosstalk from neighbouring lines [0] adds low-amplitude jitter. On fast-path configurations, impulse noise produces brief burst loss events.

5.2.3 DOCSIS 3.0 (Cable): Classic Bufferbloat

DOCSIS (Data Over Cable Service Interface Specification) transmits data over hybrid fibre-coaxial (HFC) infrastructure. The distinction between DOCSIS generations is critical for cloud gaming—the difference in queuing behaviour is the single largest determinant of last-mile quality for cable subscribers.

Physical layer: DOCSIS 3.0 uses SC-QAM (Single-Carrier Quadrature Amplitude Modulation) with channel bonding across 4–32 downstream and 1–8 upstream channels, each 6 MHz wide. Upstream, cable modems transmit in scheduled TDMA time slots. The coaxial segment is shared among all active modems on the node, typically serving 100–500 homes. While the downstream capacity is adequate (up to ≈ 300 Mbit/s with 24-channel bonding), the upstream is often the bottleneck at 10–30 Mbit/s.

Link layer: The critical impairment is the modem’s transmit buffer. DOCSIS 3.0 modems were designed with deep, unmanaged FIFO queues (100–300 ms worth of data) to absorb bursty TCP traffic and maximise throughput. There is no Active Queue Management—packets are served strictly in arrival order, and overflow is handled by indiscriminate tail-drop (fig. 35). When a household’s upstream traffic approaches the link capacity, the queue fills linearly, and gaming packets entering it experience progressively increasing delay—the classic “bufferbloat” signature. This is perhaps the single most impactful and most detectable last-mile impairment for cloud gaming.

Telemetry fingerprint (fig. 36): A highly characteristic linear ramp in delay correlated with sustained throughput, culminating in abrupt tail-drop loss. Jitter remains low during the ramp (packets are delayed but arrive in order). A monotonically increasing mean delay with constant variance is the strongest single diagnostic signal for DOCSIS 3.0 bufferbloat.

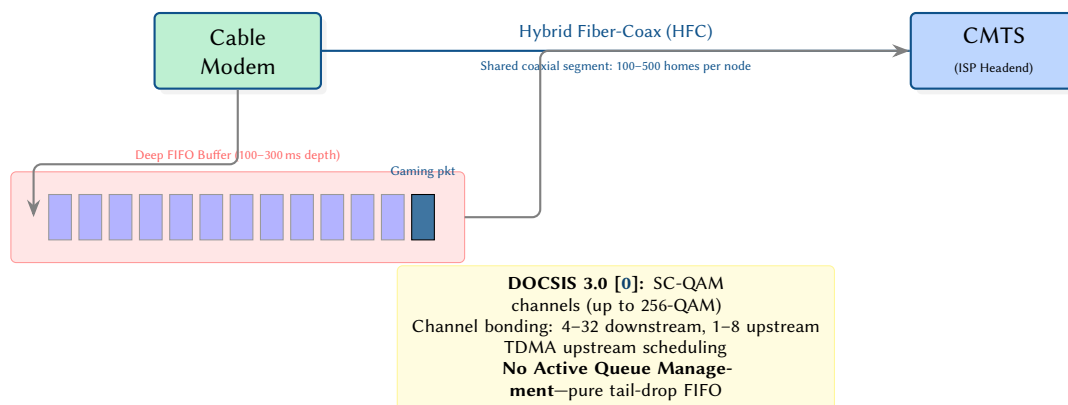


Figure 34. DOCSIS 3.0 physical layer [0]. The upstream transmit buffer is a deep, unmanaged FIFO queue—the root cause of bufferbloat [0]. When bulk TCP traffic fills the queue, gaming packets experience progressively increasing queuing delay with no prioritisation mechanism.

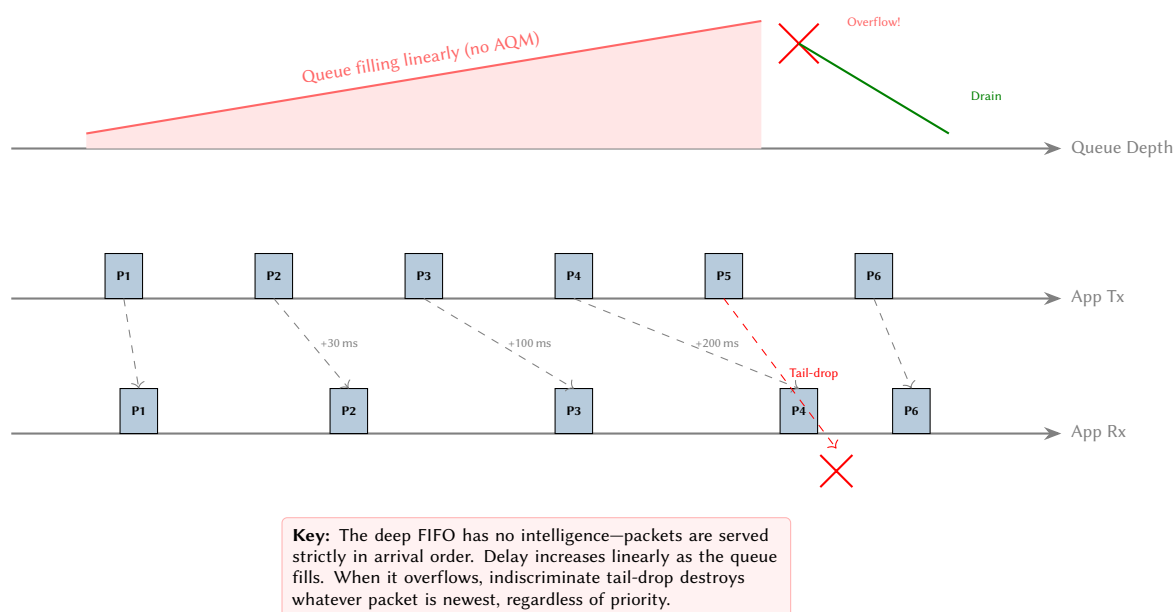


Figure 35. DOCSIS 3.0 link-layer behaviour (classic bufferbloat [0]). Gaming packets experience monotonically increasing delay as they queue behind bulk TCP traffic in the unmanaged FIFO. Tail-drop loss occurs abruptly when the buffer overflows [0].

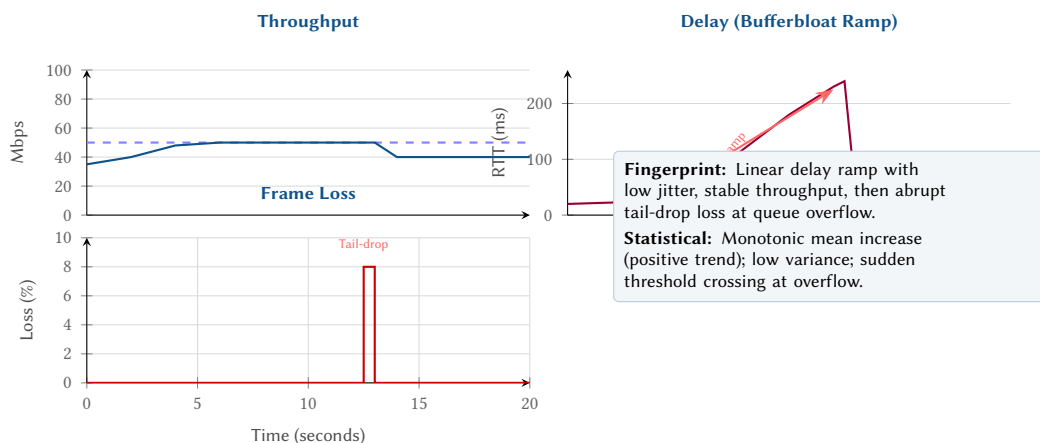


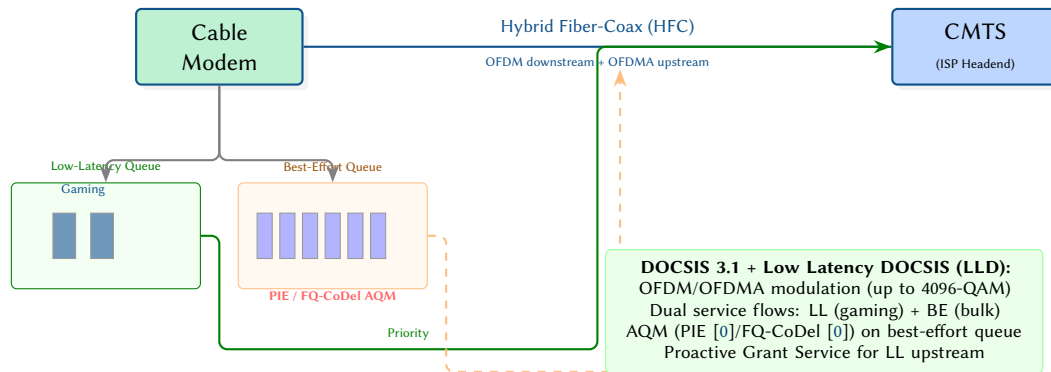
Figure 36. DOCSIS 3.0 telemetry fingerprint (classic bufferbloat [0]). The characteristic linear delay ramp with low jitter and abrupt tail-drop is one of the most reliably detectable motifs—a monotonically increasing mean with constant variance.

5.2.4 DOCSIS 3.1 + Low Latency DOCSIS (LLD): AQM-Managed

Physical layer: DOCSIS 3.1 replaces SC-QAM with OFDM downstream and OFDMA upstream, enabling much wider channel bonding (up to 192 MHz downstream) and higher-order modulation (up to 4096-QAM). This increases downstream capacity to >1 Gbit/s and upstream to >200 Mbit/s. The wider channels and OFDMA scheduling provide more granular upstream resource allocation than DOCSIS 3.0's fixed TDMA slots.

Link layer: The transformative feature for cloud gaming is Low Latency DOCSIS (LLD), which introduces two key mechanisms. First, *dual service flows* separate traffic into a low-latency queue (for gaming, VoIP, and interactive traffic) and a best-effort queue (for bulk downloads). Gaming packets are classified into the LL flow and bypass the bulk queue entirely (fig. 37). Second, *Active Queue Management (AQM)* using PIE or FQ-CoDel is applied to the best-effort queue, proactively dropping or ECN-marking packets to keep queue depth near a target (≈ 5 ms). This prevents the FIFO from filling and eliminates the bufferbloat ramp. Proactive Grant Service (PGS) further reduces upstream latency by pre-allocating transmission opportunities for the LL flow.

Telemetry fingerprint (fig. 39): With LLD enabled, the bufferbloat signature disappears. Delay is near-stationary and approaches FTTH-like quality, with no linear ramp and no tail-drop loss. The inference engine must distinguish this from FTTH (which has even lower baseline delay and no shared-segment congestion risk) and from DOCSIS 3.0 (where the ramp motif is absent simply because no competing traffic is present). The key differentiator is behaviour *under load*—DOCSIS 3.1+LLD maintains low delay even when household throughput is saturated, while DOCSIS 3.0 will ramp.



Insight: The difference between DOCSIS 3.0 and 3.1+LLD is not incremental—it is a qualitative change in queuing architecture. The inference engine must distinguish between them because the optimal mitigation strategy differs fundamentally: DOCSIS 3.0 requires aggressive congestion control (BBR/Copa) and pre-emptive bitrate reduction, while DOCSIS 3.1+LLD needs only light-touch monitoring similar to FTTH.

Figure 37. DOCSIS 3.1 with Low Latency DOCSIS (LLD) physical layer [0]. Dual service flows separate low-latency gaming traffic from best-effort bulk traffic. AQM on the best-effort queue [0] prevents bufferbloat [0] from affecting the low-latency path.

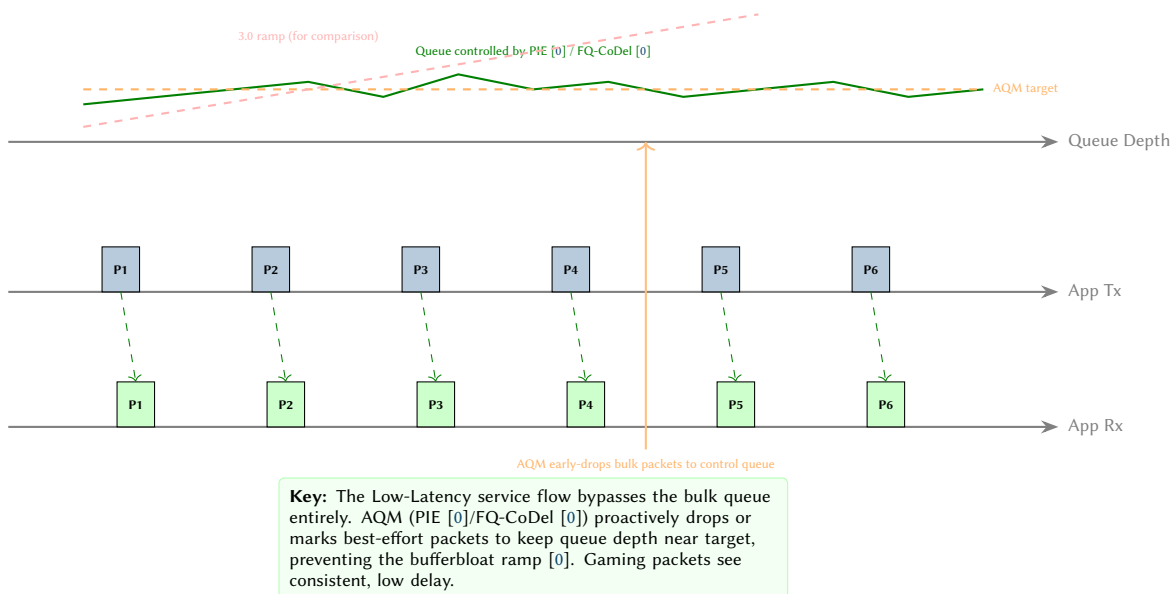


Figure 38. DOCSIS 3.1+LLD link-layer behaviour [0]. Active Queue Management holds the queue near a target depth by proactively dropping best-effort packets [0]. The low-latency service flow provides gaming traffic with consistent delay, eliminating the bufferbloat ramp.

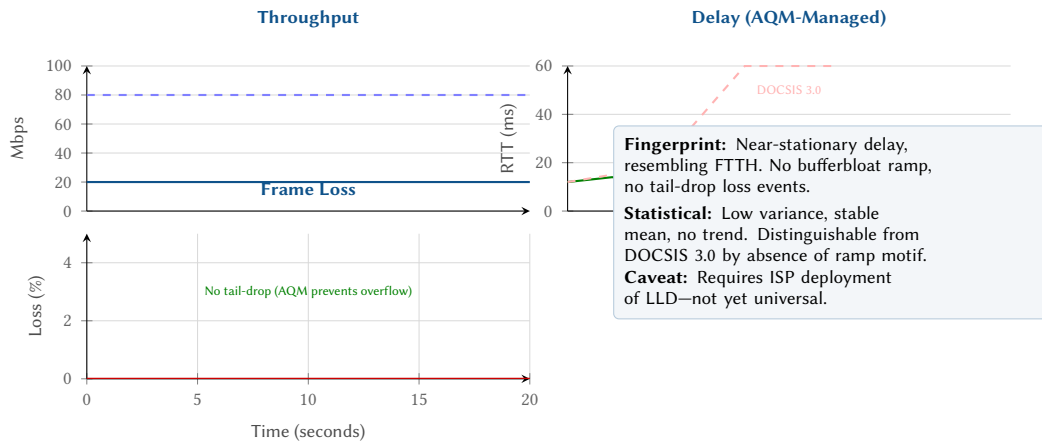


Figure 39. DOCSIS 3.1+LLD telemetry fingerprint [0]. With Active Queue Management [0] and low-latency service flows enabled, the bufferbloat signature disappears entirely. The fingerprint approaches FTTH-like stationarity, though the inference engine must still account for shared-segment congestion at the CMTS level.

5.3 Wireless and Non-Stationary Access

5.3.1 5G NR: Scheduling and Handovers

Physical layer: 5G New Radio (NR) operates in two frequency ranges: FR1 (sub-6 GHz, typically 3.5 GHz) and FR2 (mmWave, 24–52 GHz). FR1 provides broad coverage with moderate bandwidth; FR2 offers multi-gigabit throughput but limited range and penetration. The radio interface uses OFDMA with flexible numerology—subcarrier spacings of 15–240 kHz allow the scheduler to trade latency against spectral efficiency. At the physical layer, the gNB (base station) scheduler allocates resource blocks to each UE (User Equipment) every slot (0.5–1 ms), providing inherently low-jitter transmission when the device is stationary and connected to a single cell.

Link layer: The Packet Data Convergence Protocol (PDCP) layer provides in-order delivery and handles handover continuity. During a handover between gNBs, the radio link to the source cell drops while the target cell is being configured. Packets destined for the UE are buffered in the PDCP layer at the network side (fig. 41). When the target cell link is established (typically 20–100 ms later), the buffered packets are released as a massive burst—visible as simultaneous delay spikes and reordering. The UE perceives this as a brief gap followed by a flood.

Telemetry fingerprint (fig. 42): A stable, low-jitter baseline ($\approx 15\text{--}25$ ms RTT) interrupted by violent, quasi-periodic latency spikes (50–200 ms) and correlated burst loss. Between handovers, the channel is near-stationary due to the deterministic scheduling. The periodicity of spikes correlates with vehicle speed and cell density, making them predictable.

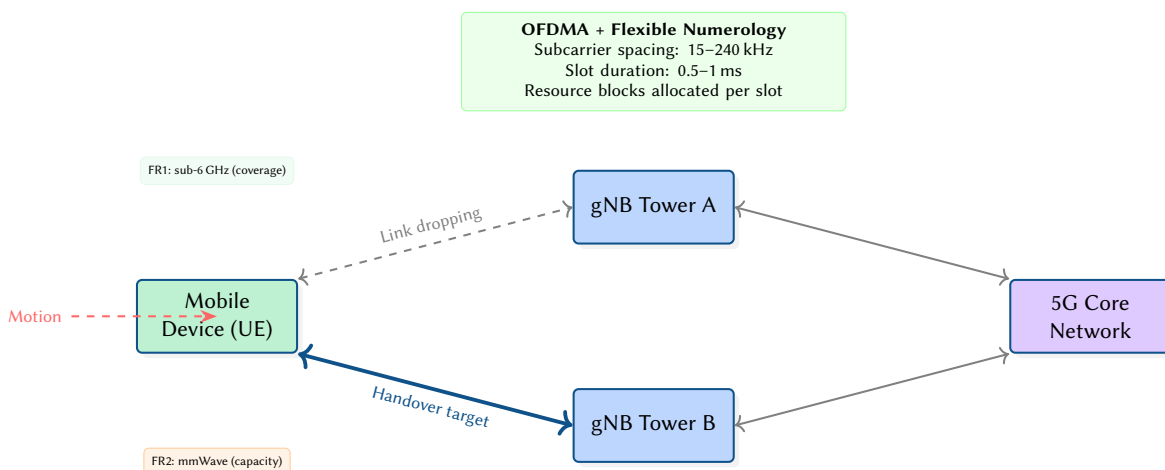


Figure 40. 5G NR physical layer. The gNB scheduler allocates OFDMA resource blocks per 0.5–1 ms slot, providing inherently low-jitter transmission. Device mobility forces periodic handovers between towers, during which the radio link briefly drops.

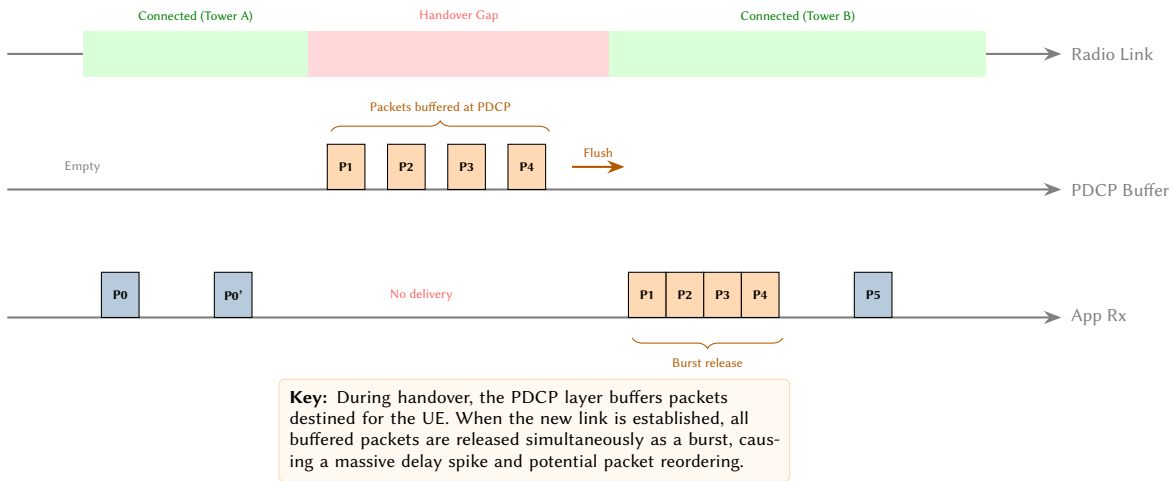


Figure 41. 5G PDCP-layer buffering during handover. The radio link gap (20–100 ms) causes packets to accumulate in the network-side PDCP buffer. Their burst release produces the characteristic handover spike.

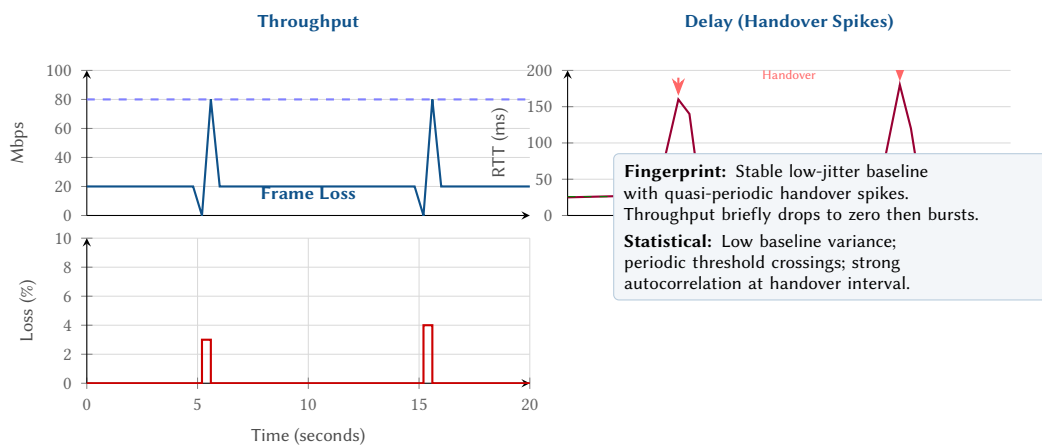


Figure 42. 5G NR telemetry fingerprint. Between handovers, the channel is near-stationary due to deterministic scheduling. Handover spikes are quasi-periodic and predictable from vehicle speed and cell density.

5.3.2 Fixed Wireless Access (WiMAX / Microwave)

Physical layer: Fixed Wireless Access (FWA) systems use point-to-point or point-to-multipoint microwave links, typically in the 5–60 GHz range. These high-frequency signals propagate as line-of-sight beams between a rooftop dish and an ISP tower. Under clear-sky conditions, the channel is stable and near-ideal. However, microwave propagation at these frequencies is subject to *rain fade*: water droplets in the propagation path absorb and scatter the signal, reducing the received SNR. Heavy rainfall at 28 GHz can attenuate the signal by 10–20 dB/km, forcing the radio to adapt.

Link layer: When the SNR degrades due to atmospheric fading, the radio’s adaptive modulation drops to a more robust but slower Modulation and Coding Scheme (MCS)—for example, from 256-QAM to QPSK. This trades spectral efficiency for reliability, but the consequence is that each packet requires physically more airtime to transmit (fig. 44). The available throughput drops proportionally, and delay inflates as packets wait for their extended transmission slots. If the fade deepens beyond the lowest MCS capability, the link drops entirely, producing sustained outages.

Telemetry fingerprint (fig. 45): A slow, correlated degradation across all metrics: throughput decreases gradually as MCS steps down, delay increases as airtime-per-packet grows, and sustained loss appears when the link approaches its fade margin. The key statistical indicator is a *slow ramp* in delay correlated with decreasing throughput—distinct from bufferbloat (where throughput remains stable during the delay ramp) and from Wi-Fi contention (where throughput is unaffected).

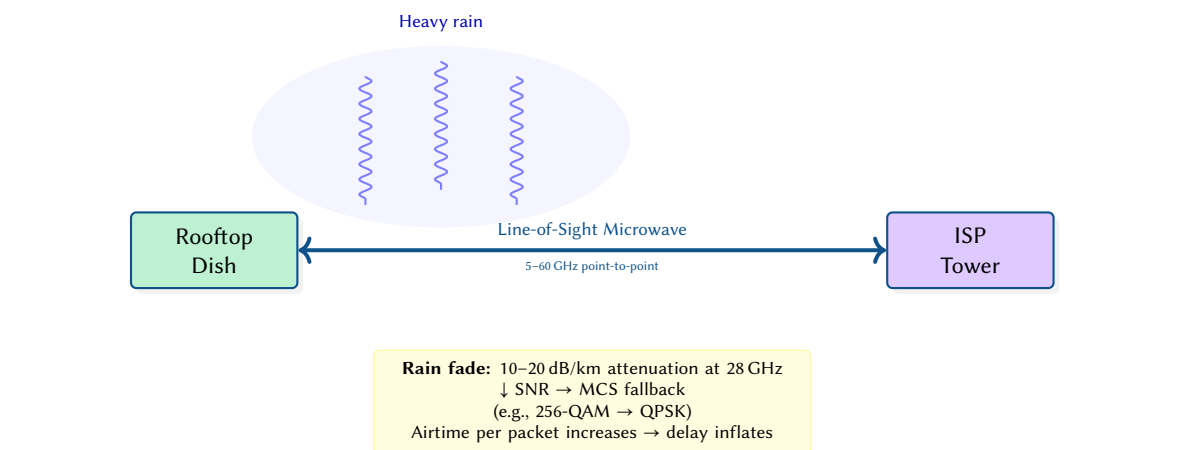


Figure 43. FWA physical layer. Line-of-sight microwave links are stable under clear conditions but subject to rain fade. As SNR degrades, the radio drops to a more robust MCS, increasing airtime per packet and inflating delay.

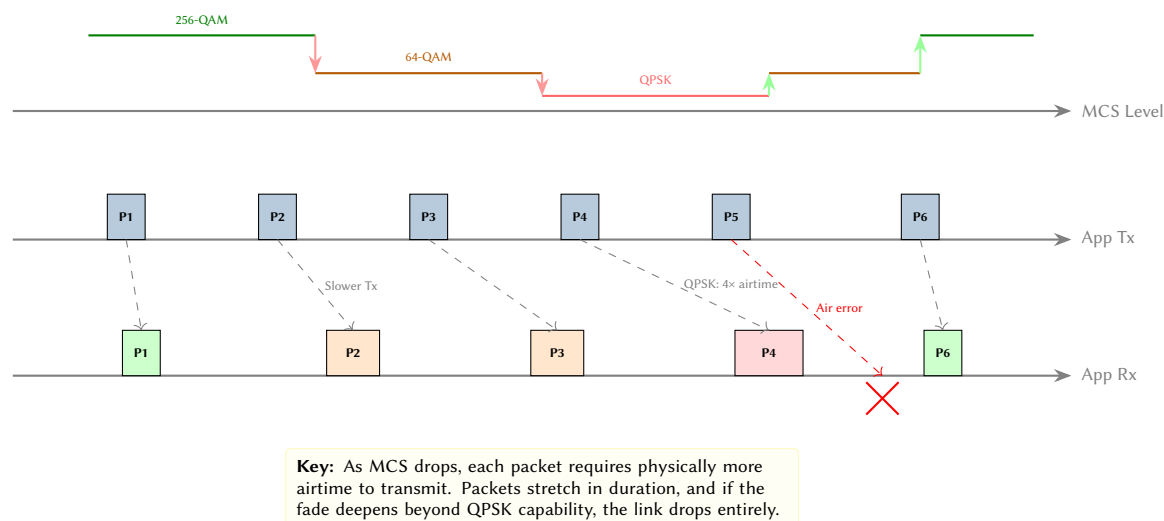


Figure 44. FWA link-layer behaviour during rain fade. The radio adapts its Modulation and Coding Scheme (MCS) as SNR degrades, progressively increasing per-packet airtime and inflating end-to-end delay.

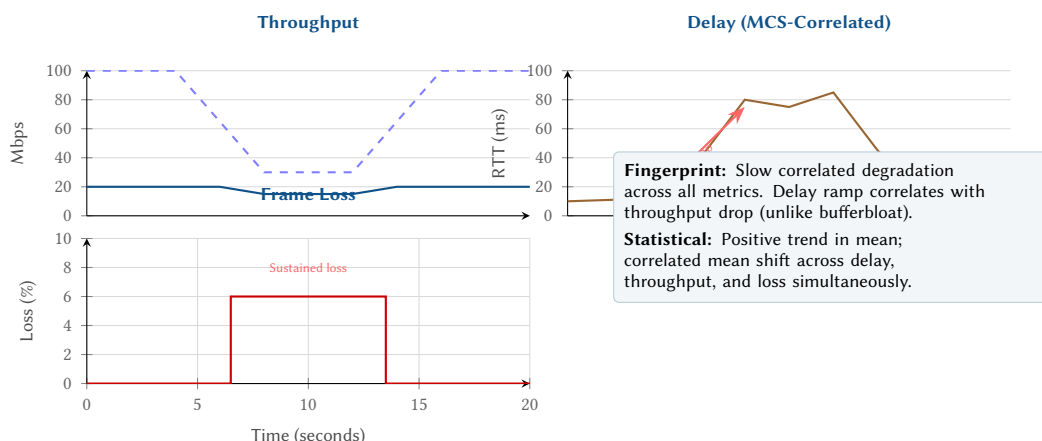


Figure 45. FWA telemetry fingerprint during rain fade. All metrics degrade in correlated fashion: throughput drops as MCS falls, delay increases with growing airtime, and sustained loss appears. The cross-metric correlation distinguishes this from bufferbloat or contention.

5.3.3 LEO and GEO Satellite

Physical layer: Satellite connections are governed by the speed of light and orbital mechanics. Geostationary (GEO) satellites orbit at 35 786 km altitude, imposing an inescapable ≈ 500 ms round-trip propagation delay—far beyond any cloud gaming latency budget. Low Earth Orbit (LEO) constellations (e.g., Starlink at ≈ 550 km) reduce this to 30–40 ms baseline RTT. The Ka-band (26.5–40 GHz) and Ku-band (12–18 GHz) links used by LEO systems are subject to rain fade (similar to FWA) and to *scintillation*—rapid signal fluctuations caused by

atmospheric turbulence. Additionally, the phased-array antenna on the user terminal must continuously steer its beam to track the rapidly moving satellite.

Link layer: Because LEO satellites traverse the sky in minutes, the user terminal must periodically switch from a setting satellite to a rising one. During this beam handover (fig. 46), the new path may have a significantly different propagation delay (the satellite is at a different range and the ground station backhaul route may differ). Packets in flight on the old path can arrive *after* packets sent on the new, shorter path (fig. 47)—a phenomenon the UDP decoder treats as functional loss. Inter-satellite links (where available) add further path diversity and potential reordering.

Telemetry fingerprint (fig. 48): An elevated but stable baseline RTT ($\approx 35\text{--}50\text{ ms}$) with quasi-periodic disruptions at 15–30 second intervals. During beam switches, delay spikes to 80–130 ms and packet reordering produces brief loss bursts. The strong autocorrelation structure makes this pattern highly predictable—the inference engine can detect the periodicity and pre-position FEC around predicted switch windows.

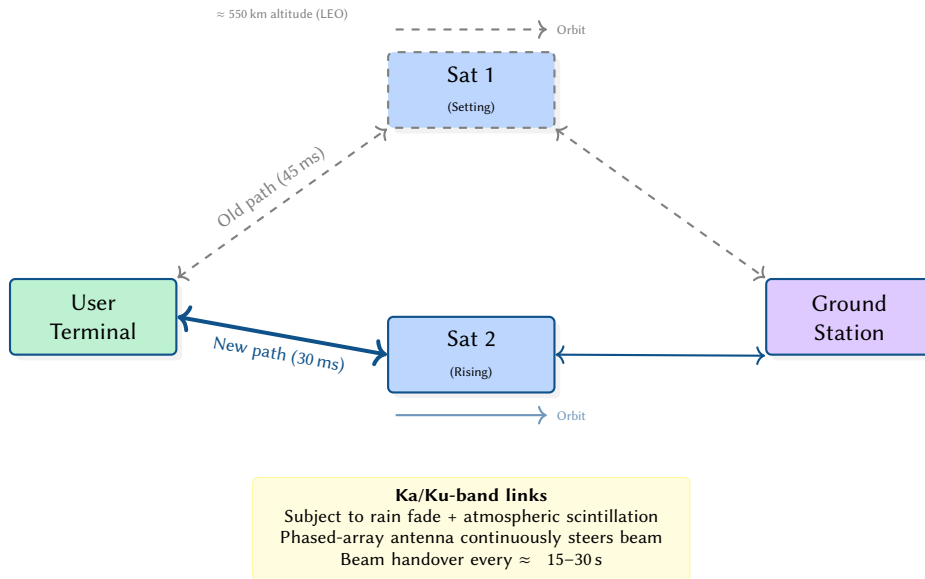


Figure 46. LEO satellite physical layer. Satellites in low orbit traverse the sky in minutes, requiring periodic beam handovers. The new path may have a significantly different propagation delay, causing packet reordering during the transition.

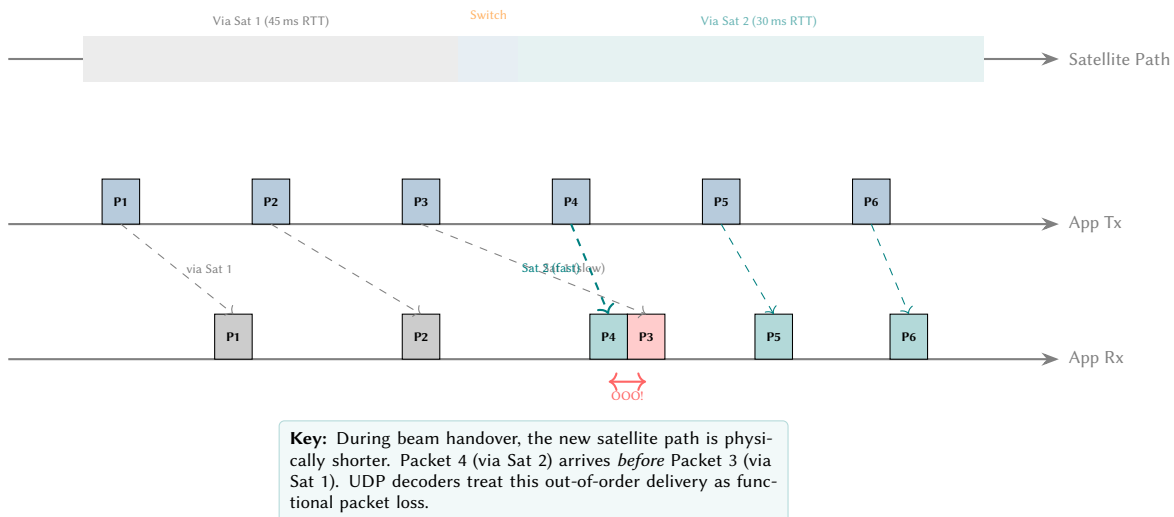


Figure 47. LEO satellite beam handover and packet reordering. The path-length difference between setting and rising satellites causes structural out-of-order delivery during the transition, which manifests as functional loss to the application.

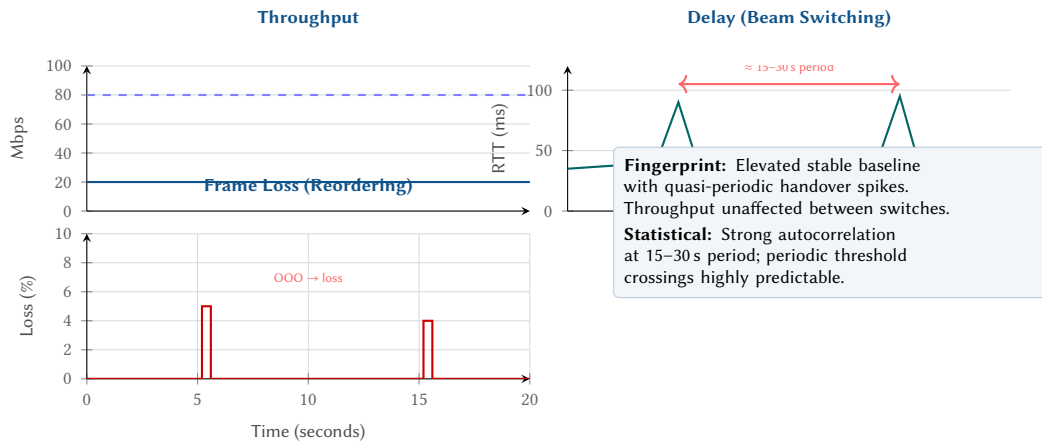


Figure 48. LEO satellite telemetry fingerprint. The elevated but stable baseline ($\approx 35\text{--}50\text{ ms}$) is interrupted by quasi-periodic beam handover spikes. The strong autocorrelation structure makes these highly predictable for pre-emptive mitigation.

5.4 Access Technology Summary

Table 3 provides a consolidated comparison of all access technologies and their characteristic telemetry signatures, serving as a quick reference for the inference engine’s prior expectations.

Table 3. Summary comparison of last-mile access technologies and their telemetry fingerprints. Baseline values assume typical deployment conditions (not worst-case).

Technology	Baseline RTT	Jitter	Loss Pattern	Shared?	Key Motif	Primary Mitigation
FTTH / GPON	5–7 ms	Negligible	None	No*	Stationarity	Baseline (minimal)
DSL / VDSL	15–30 ms†	Low (crosstalk)	Impulse bursts	No	Elevated stable baseline	FEC for impulse; fast-path if possible
DOCSIS 3.0	10–15 ms	Low during ramp	Tail-drop at overflow	Yes (coax)	Linear delay ramp	BBR/Copa CC; Raptor FEC
DOCSIS 3.1+LLD	10–15 ms	Low	None (AQM)	Yes (coax)	Near-stationarity	Baseline (light monitoring)
Wi-Fi 5 (ac)	8–15 ms	High (erratic)	Retry-exhaustion bursts	Yes (air)	High-freq. transients	QoS tagging; avoid bitrate cuts
Wi-Fi 6 (ax, 5 GHz)	8–12 ms	Reduced	Rare	Yes (air)	Reduced jitter	Light FEC; OFDMA scheduling
Wi-Fi 6E (ax, 6 GHz)	6–10 ms	Low	Negligible	No‡	Low stable jitter	Minimal; monitor for fade
Wi-Fi 7 (be, MLO)	5–8 ms	Negligible	Negligible	No‡	Near-wired stationarity	Baseline
5G NR (mobile)	15–25 ms	Low (between HO)	Burst at handover	No (sched.)	Quasi-periodic spikes	Pre-emptive FEC at HO
FWA (microwave)	8–15 ms	Low	Sustained (rain fade)	No	Correlated slow ramp	MCS-aware bitrate adaptation
LEO Satellite	35–50 ms	Low (between switch)	OOO at beam switch	Yes (beam)	Periodic beam-switch spikes	Pre-emptive interleaving

*Shared passive splitter, but scheduled TDM eliminates contention. †Includes interleaver delay if enabled. ‡Greenfield band: minimal neighbour contention.

Figure 49 presents a complementary radar-chart view of six representative technologies across five quality dimensions. The near-circular profile of GPON and Wi-Fi 7 reflects their uniformly strong performance, while the asymmetric profiles of 5G NR (low stability) and satellite (low latency score) highlight the dimensions where each technology is weakest—precisely the dimensions the inference engine should prioritise when selecting mitigations.

5.5 Regional Access Technology Prevalence

The prior probability of encountering each access technology varies significantly by geographic region, which the inference engine can exploit to narrow its hypothesis space. Table 4 summarises the key regional differences.

Knowing the region allows the inference engine to weight its prior expectations: a user connecting from suburban Australia is more likely on NBN FWA or HFC than FTTH, while a user in Seoul is almost certainly on FTTH with minimal last-mile impairment. This geographic prior reduces the time to confident classification by eliminating unlikely hypotheses early in the session.

5.6 Time-of-Day Congestion Patterns

Different access technologies exhibit distinct diurnal congestion profiles. Because cloud gaming sessions cluster during evening leisure hours—precisely when shared-medium technologies are most congested—understanding these patterns is critical for predictive pre-adaptation (section 14.8).

The diurnal patterns in fig. 50 suggest that a time-of-day prior could complement the regional and ISP priors: if a DOCSIS 3.0 user connects at 20:00, the inference engine can pre-load a bufferbloat-aware strategy before the first congestion ramp is observed.

5.7 ISP-Provided CPE and Its Inference Implications

A frequently overlooked but highly informative inference signal is the Customer Premises Equipment (CPE)—the modem, router, or gateway provided by the ISP or purchased by the subscriber. Many ISPs supply a

Insight:
Regional metadata is among the cheapest features available to the inference engine—derivable from IP geolocation or server region assignment with no telemetry required—yet it can halve the hypothesis space before the first frame is even received.

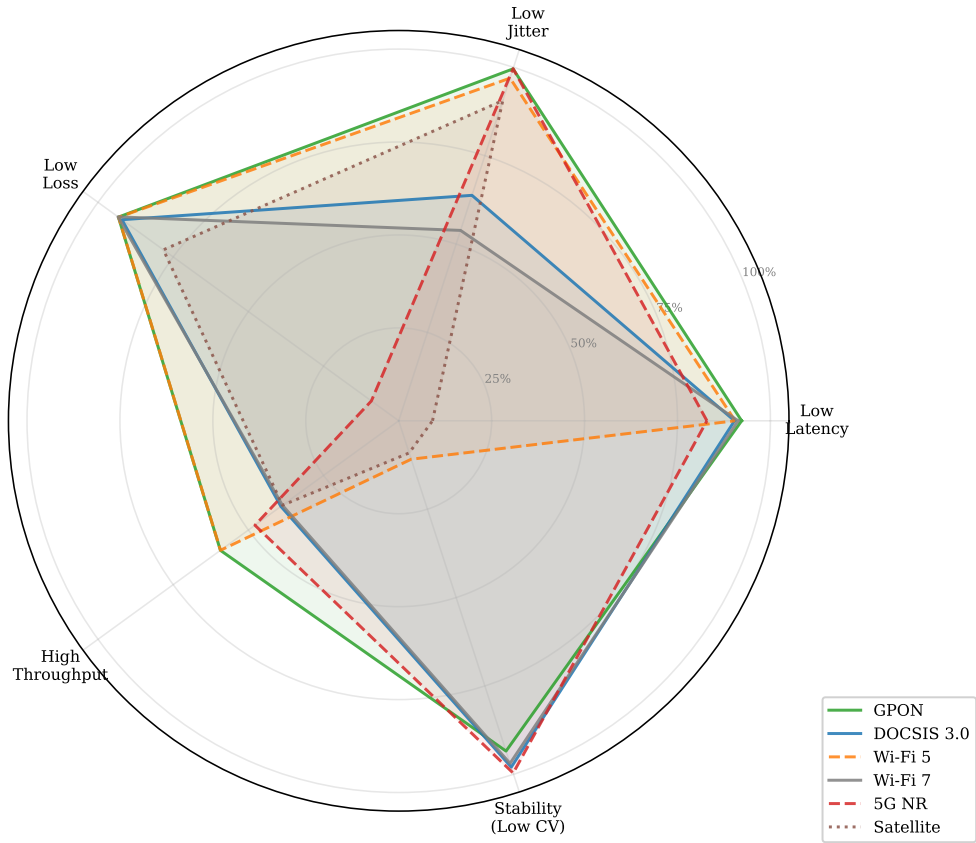


Figure 49. Multi-axis fingerprint comparison for six representative access technologies. Each axis is normalised so that the outer ring (100%) represents the best-performing technology on that dimension. Asymmetric profiles indicate targeted vulnerability.

Table 4. Regional variation in access technology prevalence and spectrum availability. The inference engine can use geographic metadata (IP geolocation, server region) to adjust prior probabilities for each technology, improving classification speed and accuracy.

Region	Dominant Fixed	Wi-Fi Spectrum	Mobile	Inference Prior Notes
North America	DOCSIS 3.0/3.1 dominant; FTTH growing	Full 6 GHz (1200 MHz)	5G NR, with mmWave in cities	Strong DOCSIS prior; Wi-Fi 6E/7 more likely in newer homes
Western Europe	DSL or VDSL dominant; FTTH expanding	Lower 6 GHz band (500 MHz)	5G sub-6 dominant	Strong DSL prior; DOCSIS rarer; 6E channels narrower
Nordics / Baltics	FTTH dominant (>70% penetration)	Full 6 GHz expected	Advanced 5G rollout	FTTH baseline likely; low last-mile impairment expected
Australia / NZ	Mixed NBN: FTTH, FWA, legacy HFC	6 GHz 500 MHz under review	5G FWA common regionally	FWA rain fade matters; NBN access mix shifts priors
East Asia (JP/KR)	FTTH near-universal	Full 6 GHz	Dense urban 5G	Very low last-mile impairment expected
Southeast Asia	DSL plus mobile-first access	6 GHz varies by country	4G or 5G primary for many users	Mobility and handover spikes dominate
Rural / Remote	LEO satellite or FWA	Limited Wi-Fi 6E adoption	Sparse 5G, often satellite backhaul	Satellite beam-switch signatures more likely

standardised CPE to all subscribers on a given plan, creating strong correlations between ISP identity and last-mile behaviour:

- **ISP-mandated CPE:** Large ISPs often deploy a single modem/router model across hundreds of thousands of subscribers. If that device has a known bufferbloat issue (e.g., a DOCSIS 3.0 modem with a deep, unmanaged FIFO and no AQM firmware update), then every subscriber on that ISP’s network will exhibit the same characteristic delay ramp under load. Conversely, an ISP that deploys DOCSIS 3.1+LLD-capable gateways with factory-enabled AQM will produce a qualitatively different telemetry profile.
- **Gateway Wi-Fi generation:** The ISP-provided gateway determines the Wi-Fi generation available to the subscriber. An ISP still deploying Wi-Fi 5 gateways constrains its entire subscriber base to CSMA/CA contention behaviour, regardless of whether the user’s client device supports Wi-Fi 6 or 7. The inference

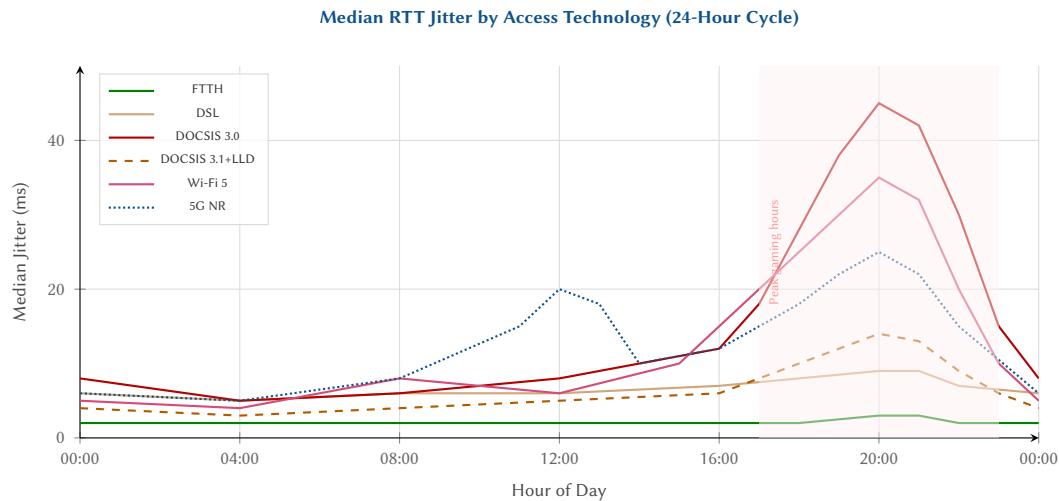


Figure 50. Illustrative diurnal jitter profiles by access technology. FTTH and DSL remain stable throughout the day. DOCSIS 3.0 shows a dramatic evening peak from shared-segment congestion and bufferbloat. Wi-Fi 5 degrades during family hours as household devices contend. 5G exhibits a secondary lunch-hour peak from urban cell congestion. Note: values are illustrative of characteristic patterns, not measured data.

engine can use the ISP identity (derivable from the IP prefix or AS number) as a proxy for the likely Wi-Fi generation.

- **Firmware and configuration:** CPE firmware determines whether features like SQM (Smart Queue Management), OFDMA scheduling, WMM prioritisation, and band steering are enabled by default. ISPs vary widely in their default configurations—some enable QoS features that significantly reduce gaming jitter, while others ship conservative defaults that prioritise stability over latency. Firmware updates deployed by the ISP can cause step-changes in the telemetry fingerprint of an entire subscriber base overnight.
- **All-in-one vs. bridge mode:** ISP-provided gateways often combine modem, router, Wi-Fi AP, and sometimes even a managed switch into a single device. Performance-conscious users may place the ISP gateway in bridge mode and use their own high-quality router behind it. The inference engine may observe a bimodal distribution within the same ISP prefix—one cluster with ISP-gateway Wi-Fi characteristics and another with substantially better performance from aftermarket equipment.

Over time, the feature store (fig. 58) accumulates CPE-correlated fingerprints at the ISP level. When a new user connects from a known ISP prefix, the system can immediately load the CPE-informed prior: “this ISP deploys model X gateways with known Wi-Fi 5 and DOCSIS 3.0 behaviour; expect contention spikes and bufferbloat ramps under load.” This CPE prior stacks with the regional prior, further narrowing the hypothesis space.

5.8 User Empowerment and Out-of-Band Mitigation

When the inference engine determines that an impairment is structural and persistent (e.g., severe bufferbloat on a DOCSIS 3.0 modem, or 2.4 GHz Wi-Fi multipath fading), in-band mitigations alone may be insufficient. The platform can use the inferred profile to advise the user directly:

- **Router Settings:** Prompt the user to log into their router to enable “Smart Queue Management” (FQ-CoDel).
- **Hardware Upgrades:** Advise that their specific latency signature indicates a Wi-Fi 4/5 bottleneck, recommending an upgrade to a Wi-Fi 6 router or an Ethernet connection.

5.9 Multi-Hop Last Miles and Compound Fingerprints

Real-world last-mile paths are frequently multi-hop: the gaming device connects via Wi-Fi to a router, which may be connected via a powerline adapter (Ethernet-over-mains) or MoCA (coax) bridge to a modem, which terminates on a DSL, DOCSIS, or FTTH access line. Each hop contributes its own impairment layer, and the telemetry observed at the cloud server reflects the *convolution* of all layers.

Common multi-hop combinations include:

- **Wi-Fi + DOCSIS 3.0:** The most problematic compound path. Wi-Fi contention spikes (high-frequency, short-duration) superimpose on DOCSIS bufferbloat ramps (low-frequency, long-duration). The inference

Insight:
The ISP–CPE correlation means that a single firmware update pushed to millions of gateways can shift the telemetry fingerprint of an entire subscriber population. The inference engine must detect and adapt to these fleet-wide regime changes, which manifest as a simultaneous mean shift across all users on the affected prefix.

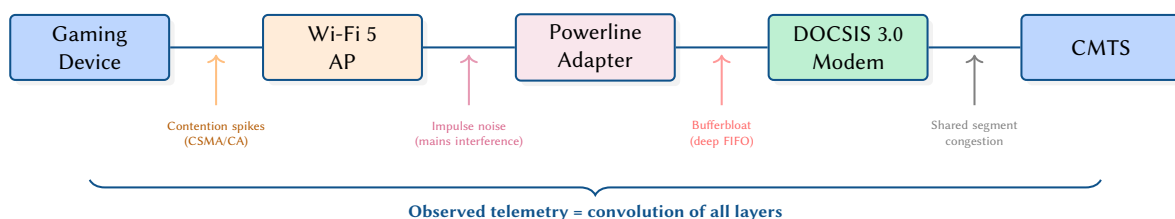


Figure 51. A compound multi-hop last mile. Each hop contributes a distinct impairment layer. The cloud server observes the convolution of all layers, making decomposition into root causes substantially harder than for single-hop paths.

engine can attempt decomposition by frequency: the slow monotonic ramp is characteristic of bufferbloat, while the high-frequency transients indicate Wi-Fi contention.

- **Wi-Fi + Powerline:** Powerline adapters introduce impairments correlated with electrical load on the mains circuit. These manifest as throughput drops and burst loss events that correlate with time-of-day but not with network congestion indicators—a useful disambiguation signal.
- **Wi-Fi + DSL (interleaved):** The interleaver adds a stable baseline delay offset ($\approx 8\text{--}16\text{ ms}$) that elevates the floor beneath Wi-Fi jitter. The inference engine observes a shifted distribution rather than a changed shape.
- **Carrier-grade NAT + any access technology:** As discussed in section 7.2, CGNAT confounds prefix-based cohort analysis by placing unrelated subscribers behind the same public IP.

The companion simulation report (LM-SIM-2026-01) validates compound path behaviour quantitatively. Figure 52 compares measured compound-path P95 latencies against the sum of their component single-link P95 values. The compound latency is typically *less* than the arithmetic sum of components, because the serialisation delays do not stack additively—the Wi-Fi and access-line stages process packets in pipeline fashion rather than sequentially per frame. This sub-additive property is practically important: it means compound paths are not as bad as worst-case analysis would suggest.

Insight: Compound paths are the norm, not the exception. The inference engine should default to a multi-layer hypothesis and use frequency decomposition, cross-user correlation, and time-of-day patterns to attribute impairments to individual hops.

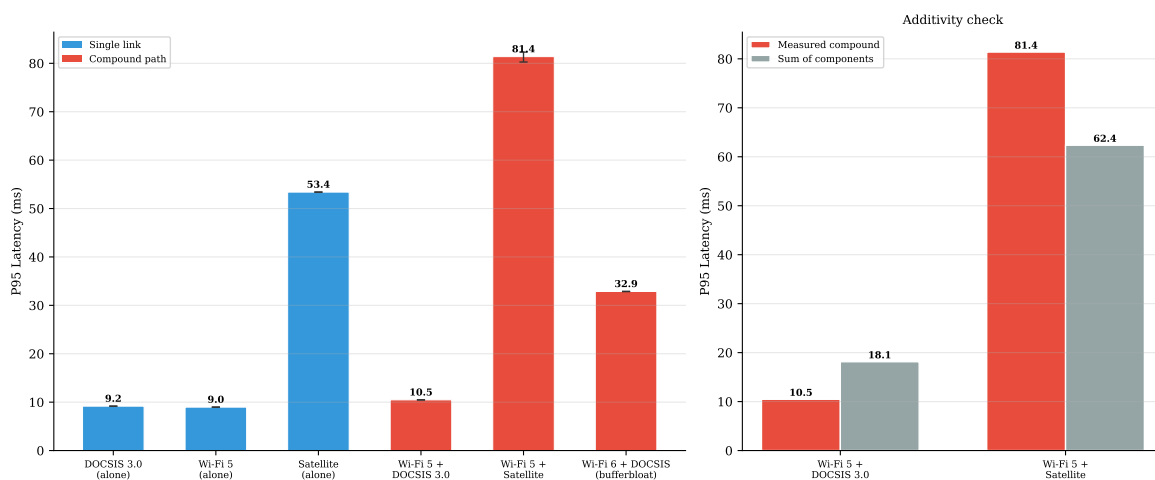


Figure 52. Compound path P95 latency comparison. Left: single-link baselines vs. compound paths. Right: measured compound latency vs. sum of component latencies, showing sub-additive composition.

6 Mobility, VPNs, and Non-Stationary Paths

Cloud gaming is particularly attractive on mobile devices—phones and tablets—precisely because offloading rendering eliminates the need for local GPU hardware. This means a disproportionate fraction of cloud gaming sessions occur on mobile or otherwise non-stationary network paths, which present unique inference challenges.

6.1 Cellular Handover Spikes

A common deployment scenario is a child playing on a tablet in a moving vehicle. The 5G/LTE path exhibits not only the baseline cellular characteristics (scheduling jitter, moderate loss) but also periodic latency spikes as the device hands over between cell towers. These handover events typically cause 50–200 ms latency spikes and occasional packet loss bursts.

Ni et al. [0] address the multipath cellular streaming problem with network coding across 5G/LTE links, providing redundancy that spans handover boundaries.

Insight: Handover spikes are quasi-periodic and predictable—their timing correlates with vehicle speed and cell density. An inference system that detects the onset of a handover pattern can pre-emptively increase FEC depth and input redundancy

Latency Profile: Mobile 5G with Handover Events

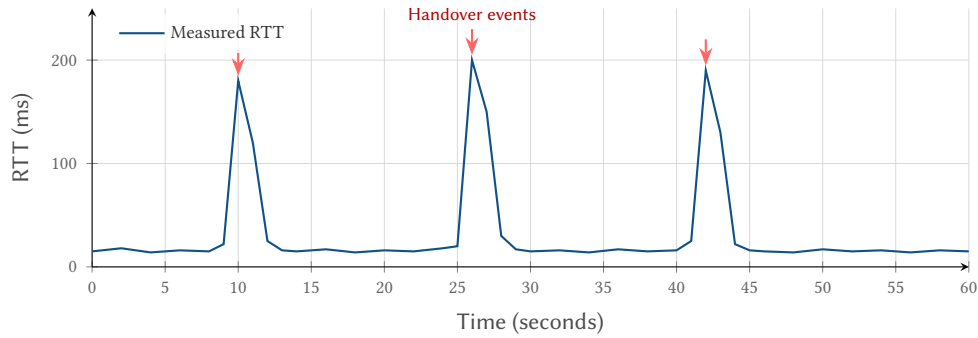


Figure 53. Mobile 5G latency profile showing periodic handover spikes (50–200 ms) against a low baseline (≈ 15 ms). The spikes are quasi-periodic, occurring roughly every 15–20 seconds at motorway speed.

6.2 Satellite (Starlink) Periodicity

Satellite internet connections, particularly LEO constellations like Starlink, exhibit a characteristic periodic latency signature caused by satellite-to-ground-station geometry changes and beam switching. These spikes are not truly periodic but are quasi-periodic with intervals of roughly 15–30 seconds, and the spike magnitude (20–100 ms above baseline) depends on the orbital mechanics of the serving satellite.

Latency Profile: LEO Satellite (e.g., Starlink)

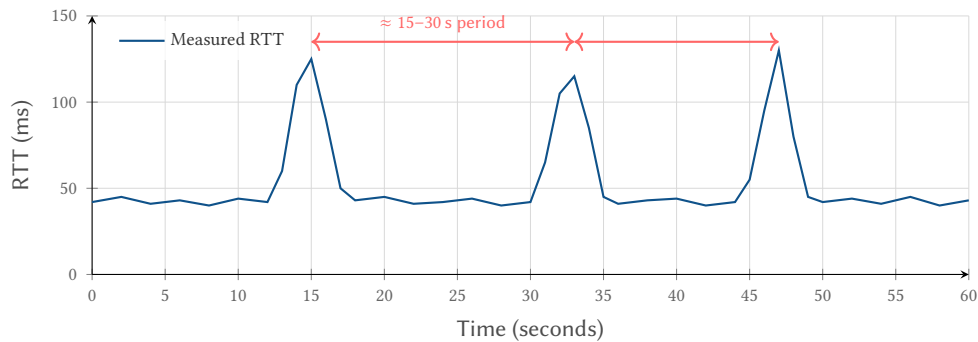


Figure 54. LEO Satellite latency profile. The baseline RTT (≈ 40 ms) is interrupted by quasi-periodic spikes (up to 130 ms) caused by beam switching and satellite handovers. The highly structured autocorrelation makes these spikes predictable.

The inference system can potentially identify and model this periodicity:

- **Detection:** Autocorrelation of the delay signal over a 60–120 second window reveals the quasi-periodic structure.
- **Prediction:** Once detected, the next spike can be anticipated within a time window, allowing pre-emptive FEC increase or frame rate adjustment.
- **Mitigation:** During predicted spike windows, increase downstream FEC and upstream input redundancy; between spikes, operate at reduced mitigation to preserve quality.

6.3 VPN and Overlay Networks

Users connecting via VPN or overlay networks add an additional routing hop that may increase both delay and jitter. The inference system should detect VPN-induced latency by comparing the observed delay against the expected delay for the user’s geographic location and prefix. A user in Sydney showing 200 ms RTT to an Australian edge PoP is likely transiting an overseas VPN endpoint.

6.4 Account Mobility

The same user account may appear from different network locations across sessions: home (FTTH), office (corporate LAN/VPN), commute (5G), or travel (hotel Wi-Fi, different city). Within-user profiling must therefore be keyed not just to the account but to the *(account, network-prefix)* tuple, maintaining separate profiles for each access context and selecting the appropriate profile at session start based on the observed network characteristics.

INFERENCE

7 Network Profile Inference

Raw telemetry signals become useful when aggregated into *network profiles* that characterise the behaviour of a user's last-mile connection and the upstream path. Profiles can be built at two levels: within-user (across sessions) and between-user (across a prefix cohort).

7.1 Within-User Profiling

By accumulating telemetry across multiple gaming sessions for the same user, the system can build a statistical model of the user's typical network behaviour:

- **Baseline characterisation:** Median delay, typical jitter envelope, steady-state loss rate, and available throughput.
- **Time-of-day patterns:** Many access networks exhibit predictable congestion patterns (evening peak, weekend load).
- **Session warm-up:** Some connections (particularly cellular) exhibit higher initial jitter that stabilises after the first few seconds.
- **Degradation signatures:** Characteristic patterns that precede quality drops (e.g. rising queuing delay indicating onset of congestion).

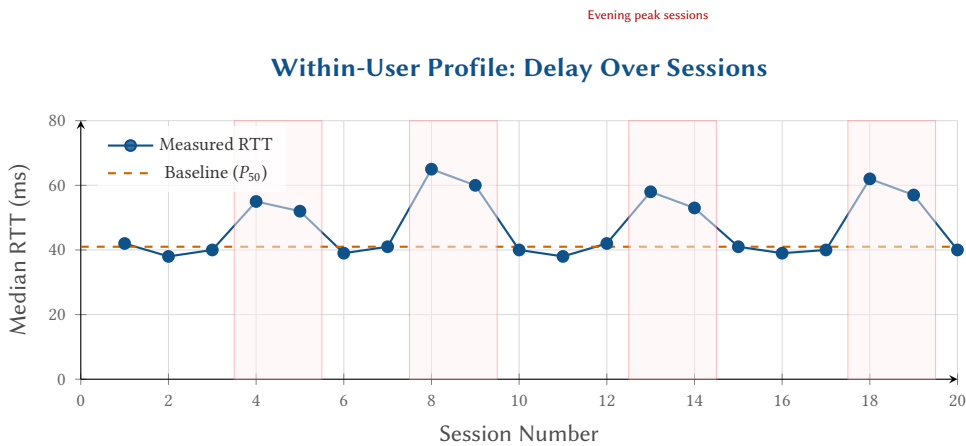


Figure 55. Within-user profiling across sessions reveals time-of-day patterns (shaded: evening peak periods with elevated delay) against a stable baseline.

7.2 Prefix Cohort Profiling

By correlating telemetry across users sharing the same network prefix (typically a /24 or /48 for IPv4/IPv6), the system can distinguish between (fig. 56):

1. **Individual impairment:** Only one user's metrics degrade → likely Wi-Fi contention, local congestion, or device issue.
2. **Prefix-correlated impairment:** Multiple users on the same prefix degrade simultaneously → shared access network congestion (CMTS overload, DSLAM saturation, cell tower congestion).
3. **Upstream impairment:** Users across *different* prefixes but sharing the same upstream path degrade → ISP backbone or peering congestion.

Insight:
When multiple users on the same IP prefix or within the same ISP/access network exhibit correlated degradation, the impairment is on a shared last-mile or upstream segment—not the individual user's local network.

Jiang et al. [0] demonstrated this approach for video streaming, showing that correlating quality metrics across users sharing ISP/AS/CDN prefix structure reveals systematic infrastructure problems invisible from individual sessions.

CGNAT complication: Carrier-Grade NAT (CGNAT), increasingly deployed by ISPs to conserve IPv4 address space, causes multiple unrelated households to share a single public IPv4 address. This confounds prefix cohort analysis: users behind the same CGNAT endpoint appear to share a prefix but may be on entirely different access segments. The inference engine must account for this by requiring stronger temporal correlation (simultaneous spikes from CGNAT-grouped users are more likely coincidental) or by preferring IPv6 addresses for cohort grouping where available. IPv6 deployment—which assigns unique /56 or /48 prefixes per household—eliminates this ambiguity entirely and should be the preferred cohort key when both address families are present.

7.3 Temporal Correlation and Anti-Correlation

The prefix cohort model becomes substantially more powerful when the inference engine tracks *temporal alignment* of impairment events across users within a cohort.

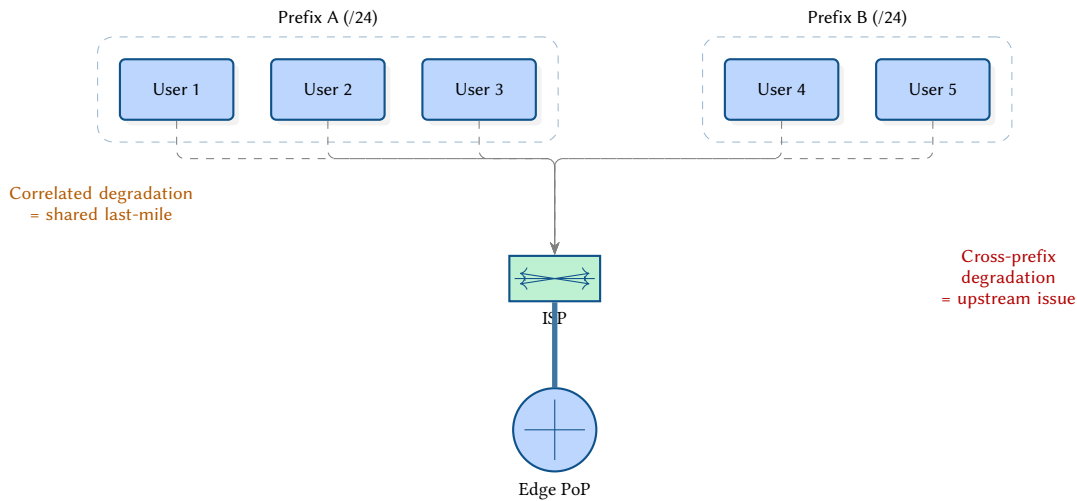


Figure 56. Prefix cohort analysis: correlating telemetry across users sharing a prefix distinguishes local from shared impairments.

7.3.1 Correlation: Shared Upstream Events

When multiple users on the same prefix or subnet exhibit a simultaneous degradation—a spike in delay, a step change in loss, or a coordinated throughput drop—the inference engine can attribute the event to a shared upstream cause with high confidence (fig. 57). The more users who exhibit the same signature at the same time, the stronger the evidence:

- **2–3 users:** Suggestive; may be coincidental Wi-Fi channel congestion in a shared building.
- **> 5 users across different households:** High confidence that the impairment is on the shared access segment (CMTS, DSLAM, cell tower, or ISP peering link).
- **Cross-prefix correlation:** If users on *different* prefixes but the same ISP degrade simultaneously, the bottleneck is upstream of the access network—likely a peering or backbone link.

7.3.2 Anti-Correlation: Isolating In-Home Events

The converse pattern is equally diagnostic. If five users are known to share a prefix cohort and four maintain stable telemetry while one exhibits a sudden degradation, the impairment is almost certainly local to that individual user’s premises—Wi-Fi contention with a household appliance, a competing download on their home network, or a device-level issue. Anti-correlation effectively rules out shared infrastructure as the cause and narrows the diagnosis to the in-home segment.

Over successive sessions, the system builds a per-user history of correlated versus anti-correlated events. A user who frequently experiences anti-correlated degradation is likely on a congested home network; conversely, a user whose impairments consistently align with the cohort is limited by the shared access segment.

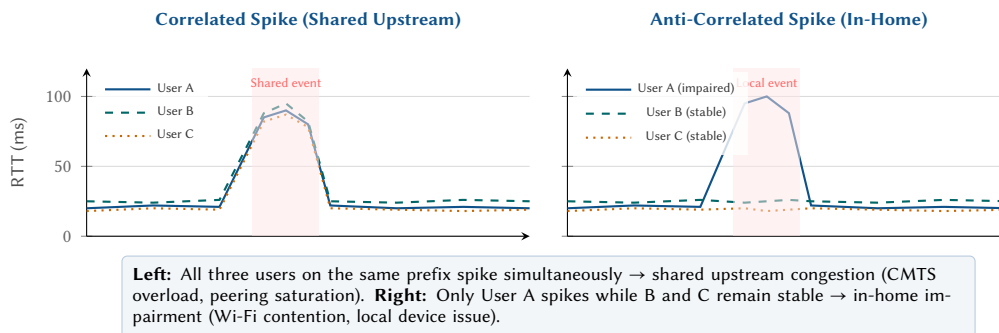


Figure 57. Temporal correlation vs. anti-correlation across a prefix cohort. Correlated degradation across multiple users (left) identifies shared infrastructure impairment. Anti-correlated degradation (right)—where one user degrades while the cohort remains stable—isolates the event to the individual user’s premises.

7.3.3 Building the Fingerprint Corpus and Feature Store

Over time, the accumulation of correlated and anti-correlated events across millions of sessions creates a rich *fingerprint corpus* (fig. 58). Each event is characterised by a feature vector combining:

- **Shape motifs:** Temporal pattern classifications (stationarity, bufferbloat ramp, periodic spikes, transient bursts) extracted by the motif discovery engine.

- **Statistical indicators:** Mean shift magnitude, variance change, threshold crossing frequency, trend slope, and burst statistics for the affected window.
- **Correlation metadata:** Whether the event was correlated (shared) or anti-correlated (local), the cohort size, and the fraction of the cohort affected.
- **Contextual features:** Time of day, day of week, inferred access technology, geographic region, and ISP identity.

This feature store serves multiple purposes. At the *session level*, it provides the inference engine with prior probabilities: a user on a DOCSIS prefix that has historically exhibited evening bufferbloat should have its inference engine pre-positioned for that signature as peak hours approach. At the *platform level*, it enables systematic identification of infrastructure degradation patterns—for example, discovering that a particular ISP’s peering link consistently saturates during major game launches, affecting all users on that path.

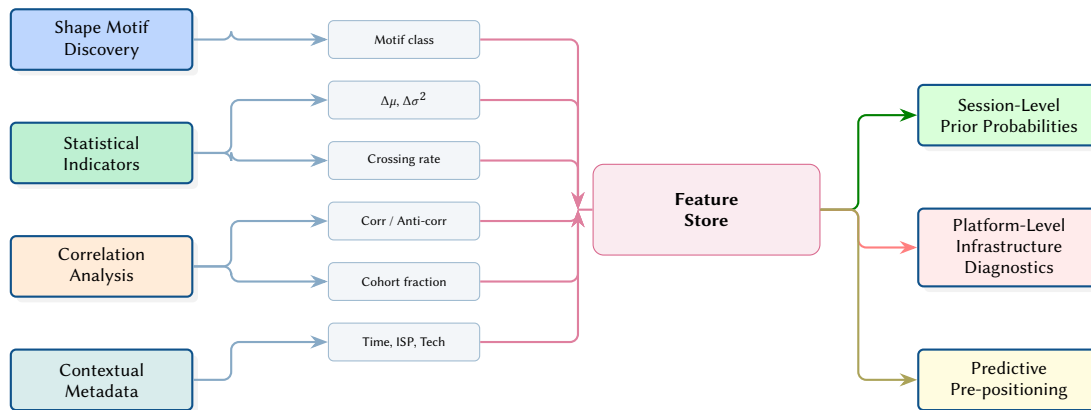


Figure 58. Feature store architecture. Shape motifs, statistical indicators, correlation metadata, and contextual features are extracted from each session and accumulated into a persistent feature store. This store feeds session-level priors (pre-positioning the inference engine), platform-level infrastructure diagnostics, and predictive models for anticipating degradation before it occurs.

7.4 Profile Evolution Over Time

Network profiles are not static. Access network characteristics change due to infrastructure upgrades (e.g. DOCSIS 3.0 → 3.1, LTE → 5G), ISP congestion management policy changes, and shifting usage patterns. Profiles should therefore be maintained with exponential decay weighting, where recent observations carry more weight than historical ones, while preserving long-term baseline statistics.

The inference system can classify the network into discrete states based on the telemetry signals, as shown in fig. 59. State transitions drive mitigation adjustments: entering the *degrading* state triggers pre-emptive FEC increase, while entering *handover* activates maximum input redundancy.

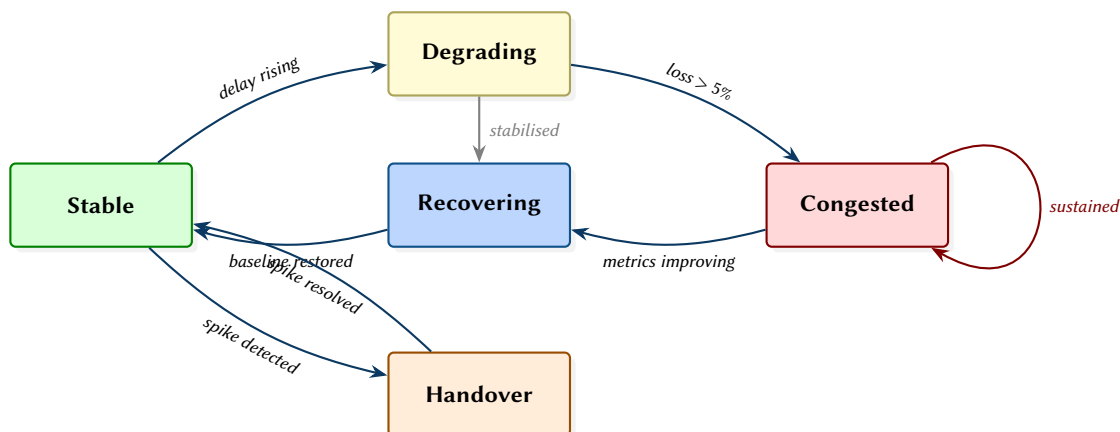


Figure 59. Network profile state machine. Transitions are driven by telemetry signal thresholds. Each state maps to a mitigation posture: *stable* → minimal correction; *degrading* → pre-emptive FEC; *congested* → aggressive adaptation; *handover* → input redundancy + FEC burst.

Insight:
The combination of temporal correlation across a prefix cohort and accumulated fingerprint features creates a feedback loop: each session both consumes prior knowledge from the feature store and contributes new observations back to it, progressively refining the system’s ability to distinguish shared from local impairments and to predict degradation before it manifests.

7.5 Sub-Second Motifs and Channel Signatures

The underlying conjecture is that last-mile network behaviour exhibits structural, repeating *motifs*—patterns of bandwidth, delay, jitter, and loss at the sub-second level that persist or evolve over 5–20 s windows.

Reliably detecting these motifs enables informed algorithmic selection: treating the channel as a composite of identifiable conditions rather than a single mean value, and matching mitigation strategies to the underlying channel physics.

Key shape-based motifs include:

- **Stationarity:** The channel mean and variance remain constant over the observation window (e.g., a clean FTTH Ethernet connection). Mitigation: Fixed, low-overhead FEC.
- **Linear Ramp-Up (Bufferbloat):** Jitter is low, but delay increases linearly as a function of sustained throughput. Loss remains zero until the queue overflows, at which point massive tail-drop occurs. Mitigation: Switch Congestion Control to BBR/Copa; slash video bitrate.
- **Quasi-Periodic Spikes:** Delay and jitter spike violently but at predictable intervals (e.g., 5G handovers every 15s, or satellite beam switching). Mitigation: Pre-emptive interleaving and burst-FEC just before the predicted spike window.
- **High-Frequency Transients (Contention):** Rapid, isolated jitter spikes (4–10 ms) without throughput degradation or loss, indicating Wi-Fi MAC layer retransmissions. Mitigation: QoS tagging (WMM/L4S) and slight decoder buffer expansion; avoid overreacting by cutting bitrate, as capacity is unaffected.

7.5.1 Statistical Indicators Beyond Shape Motifs

Channel characterisation extends beyond waveform shapes. The inference engine should also track *statistical properties* of the telemetry time series, whose evolution provides additional diagnostic power. Figure 60 illustrates these indicators.

- **Mean shift (regime change):** A sustained step change in the mean delay or loss rate—distinct from a transient spike—indicates a structural change in the path. This may signal a routing change, a new competing flow saturating a shared bottleneck, or an access technology transition (e.g., a mobile device moving from Wi-Fi to cellular). Detection via CUSUM or similar change-point algorithms triggers a full strategy re-evaluation.
- **Variance change:** An increase in variance (second-order statistic) without a corresponding mean shift indicates growing instability in the channel—for example, increasing Wi-Fi contention from neighbouring networks, or atmospheric scintillation on a microwave link. Conversely, a decrease in variance (the channel “settling”) may indicate that a previously applied mitigation is working and should be maintained.
- **Threshold crossings:** Counting how frequently a signal crosses a critical threshold (e.g., delay exceeding the frame playout deadline, or loss exceeding the FEC recovery limit) within a window provides a direct measure of impairment severity that is more actionable than the mean alone. A channel with low mean delay but frequent threshold crossings is perceptually worse than one with moderately elevated but stable delay.
- **Slow ramp (trend):** A gradual, monotonic increase in delay or decrease in throughput over tens of seconds may indicate an emerging congestion event or thermal throttling of a wireless radio. Unlike step changes, ramps give the inference engine advance warning to begin pre-positioning mitigation before the channel degrades severely.
- **Burst statistics:** Brief interference events (e.g., a microwave oven activating near a 2.4 GHz Wi-Fi link) typically manifest as a short burst of correlated packet loss with minimal impact on delay or throughput outside the burst window. Characterising the burst length, inter-burst interval, and loss density within bursts distinguishes this from sustained lossy channels and avoids over-provisioning FEC for the entire session.

These statistical indicators complement shape-based motifs and are often more robust to noise. A shape motif (e.g., a bufferbloat ramp) identifies a specific temporal pattern; a statistical indicator (e.g., monotonically increasing mean with constant variance) describes the same phenomenon in distributional terms. Tracking both gives the inference engine redundant confirmation before committing to a strategy change—directly supporting the macro tier’s confidence requirement.

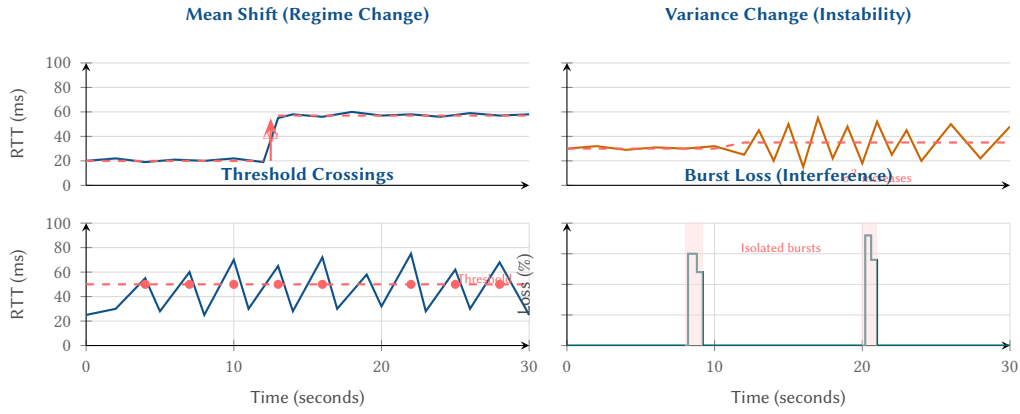


Figure 60. Statistical indicators beyond shape motifs. **Top-left:** A step change in mean delay indicates a regime change (e.g., routing shift). **Top-right:** Variance increases while mean remains stable, indicating growing channel instability (e.g., Wi-Fi contention). **Bottom-left:** Threshold crossing frequency—the channel mean is acceptable, but frequent exceedances of the playout deadline make the path unsuitable without mitigation. **Bottom-right:** Brief, isolated loss bursts characteristic of transient interference (e.g., microwave oven on 2.4 GHz), distinct from sustained lossy channels.

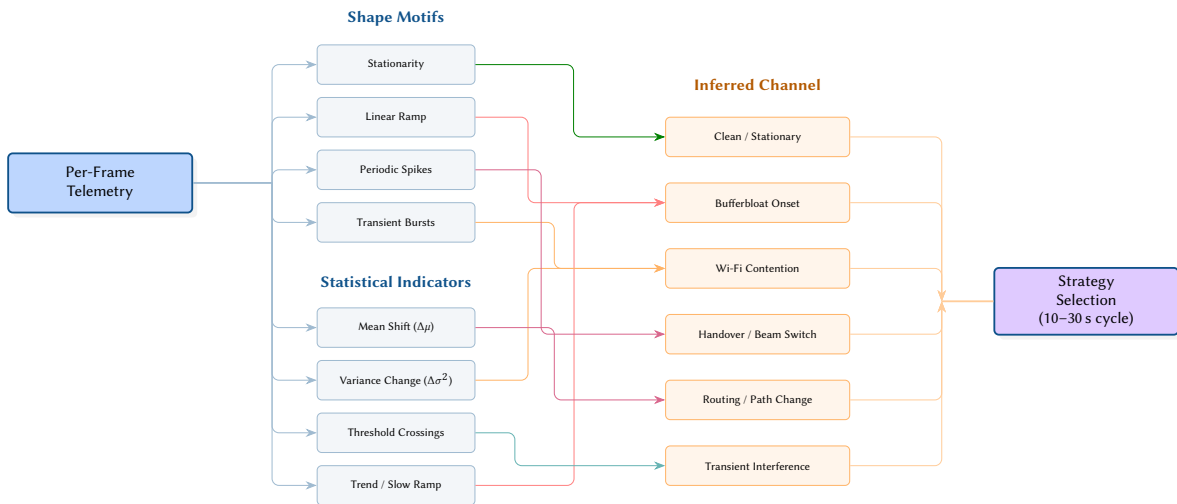


Figure 61. Inference pipeline: per-frame telemetry is analysed for both shape-based motifs (temporal patterns) and statistical indicators (distributional changes). Multiple indicators converging on the same channel diagnosis build the confidence required for macro-level strategy selection. For example, a bufferbloat diagnosis requires both a linear ramp motif *and* a positive delay trend—either alone is insufficient.

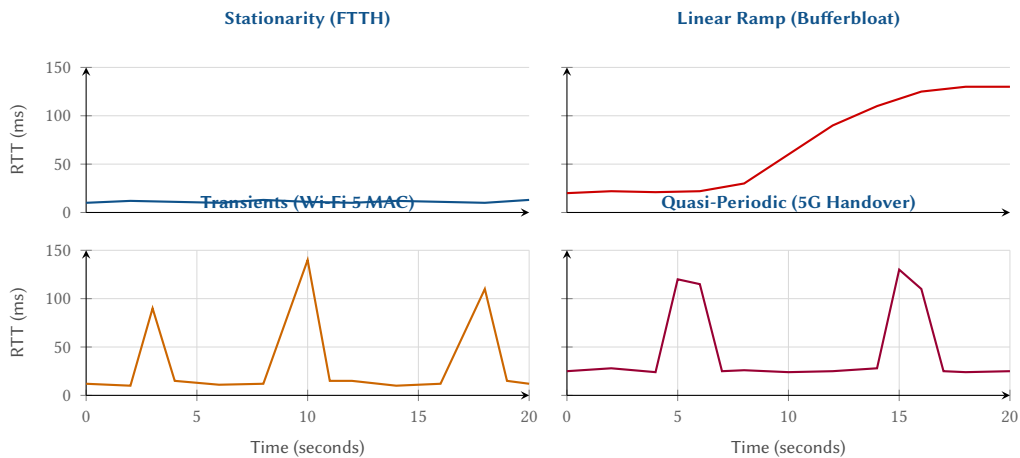


Figure 62. Sub-second motifs combining to show structural channel characteristics over a 20-second window. Each distinct signature requires a fundamentally different algorithmic mitigation strategy. Treating the bottom-left contention spikes as a bandwidth drop (top-right) would result in a disastrously incorrect mitigation (unnecessary video degradation).

8 Detection and Inference Methods

8.1 Traditional Measurement (Literature Review)

Network inference has historically relied on passive, active, and hardware-based methods. Passive protocol fingerprinting [0] identifies sessions at the ISP level. Active application-level probing embeds timing information within the protocol [0]. Hardware methods provide ground-truth MTP measurement (e.g., OpenLDAT [0] and G2GDelay [0]). More recent approaches, such as Telesa [0], feed 5G RAN telemetry (SINR, MCS) directly to the server. Each approach, however, is either too invasive, requires specialised hardware, or is bound to a single access technology.

8.2 Time Series Profiling and ML-Guided Fingerprinting

The core method is high-frequency, multivariate time series profiling. Network telemetry forms a continuous stream of delay, jitter, loss, and throughput data; detecting structural signatures—Wi-Fi contention spikes, bufferbloat ramps—requires pattern recognition beyond simple moving averages.

As the platform accumulates sessions, emergent patterns materialise. The system transitions from treating each session as an isolated packet sequence to viewing it as a composite of repeating mathematical *motifs* (stationarity, quasi-periodicity, linear ramps).

This inference operates across three distinct domains:

- **Offline Trace Analysis (Corpus Building):** By capturing millions of session traces over time, the system builds a comprehensive corpus of impairment signatures. Algorithms like the *Matrix Profile* [0] can efficiently compute all-pairs similarity joins across massive datasets to discover repeated motifs and discords (anomalies). This offline analysis distils raw telemetry into a library of distinct access network “fingerprints.”
- **Online / Real-Time Inference:** Crucially, tools like the Matrix Profile can also be applied *online* to streaming data. Paired with modern time-series classification transforms based on random convolutional kernels (such as ROCKET [0] and MiniRocket [0]), the server can deterministically classify the current network state within the strict time budget of a single frame.
- **Cross-User Prefix Correlation:** By aggregating telemetry at the subnet or ISP routing prefix level, the system can correlate impairments across different users. If multiple users on the same local ISP prefix exhibit identical bufferbloat ramps at 7:00 PM, the system can confidently infer that the congestion is upstream at the ISP peering link rather than inside the user’s home Wi-Fi. This allows the system to differentiate between local physical constraints and shared network congestion.

These ML-derived classifications form an orchestration layer operating on a 10–30 s decision timescale. The engine matches accumulated telemetry against the established motif corpus, requires sufficient confidence before acting, and then *selects and switches* the appropriate mechanisms (congestion control algorithm, FEC scheme, codec profile). The selected algorithms then operate autonomously at their native sub-second rates. This two-tier approach combines ML’s pattern-matching power at the macro level with classical network engineering’s predictability at the micro level, avoiding the instability of per-frame strategy switching.

Figure 63 illustrates the complete detection and inference pipeline from raw telemetry through to strategy deployment.

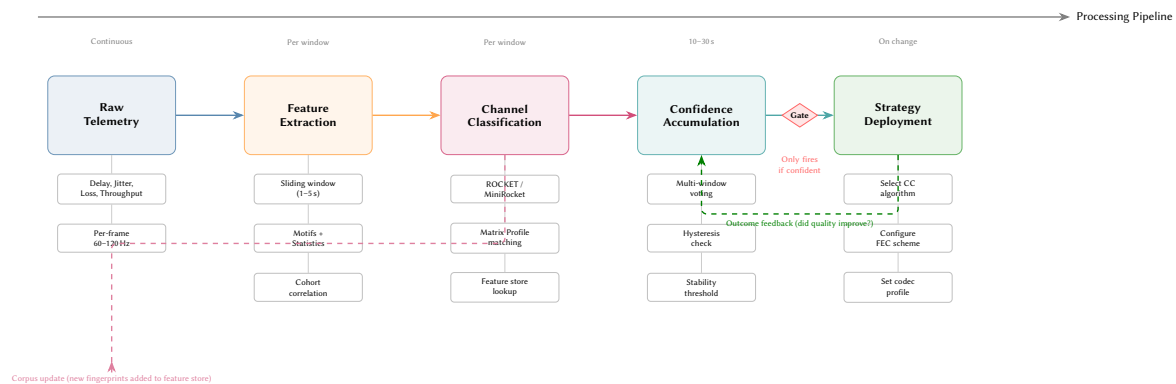


Figure 63. End-to-end detection and inference pipeline. Raw per-frame telemetry flows through windowed feature extraction (motifs, statistics, cohort correlation), ML-based channel classification (ROCKET/MiniRocket + Matrix Profile), and confidence accumulation over a 10–30 s window. A strategy change is deployed only when the confidence gate is satisfied and hysteresis constraints are met. Outcome feedback and corpus updates create a continuous learning loop.

8.3 Multi-Modal Disambiguation and Local Interference

A significant challenge for pure network-layer inference is distinguishing between network-induced jitter and local device impairments. High-frequency delay spikes in telemetry can be caused by last-mile contention, but also by local CPU/GPU thermal throttling or OS-level scheduler latency.

To resolve this ambiguity, the inference engine can ingest *multi-modal signals* from the client device:

- **Radio Resource Management (RRM) Metrics:** Direct access to the Wi-Fi or Cellular driver provides RSSI, SINR, and MCS values. A jitter spike correlated with a drop in RSSI confirms a wireless link issue; a spike with stable RSSI suggests an upstream or local bottleneck.
- **Device Telemetry:** Monitoring the client’s thermal state and CPU load allows the engine to differentiate between "network jitter" and "decode jitter." If the decoding time Δ_{decode} increases alongside network delay, the bottleneck is likely the local hardware, and the appropriate mitigation is a resolution/bitrate reduction rather than a network-layer change.
- **Display VSync Alignment:** As identified by Li et al. [0], misaligned display cycles can manifest as periodic latency spikes. By correlating telemetry with the client’s display refresh rate, the inference engine can subtract these "visual artifacts" from the true network delay profile.

8.4 Diagnostic Disambiguation

A key challenge for the inference engine is that several distinct channel conditions produce superficially similar telemetry signatures. Figure 64 maps the primary observable indicators to channel diagnoses and highlights the critical differentiators that resolve ambiguity.

	Delay Ramp	Delay Spike	High Jitter	Stable Delay	Throughput Drop	Burst Loss	Sustained Loss	Periodic	Correlated Across Users
DOCSIS 3.0 Bufferbloat	✓	—	Low	—	—	—	At overflow	—	If shared
DOCSIS 3.1+LLD (AQM Active)	—	—	Low	✓	—	—	—	—	—
Wi-Fi 5 Contention	—	✓	High	—	—	✓	—	—	—
Wi-Fi 6	—	Rare	Low	✓	—	—	—	—	—
5G Handover	—	✓	Low	Between	Brief	✓	—	✓	—
FWA Rain Fade	✓	—	Low	—	✓	—	✓	—	✓
Satellite Beam Switch	—	✓	Low	Between	—	OOO	—	✓	✓
DSL Impulse Noise	—	—	Crosstalk	✓	—	✓	—	—	—
Upstream ISP Congestion	✓	—	Moderate	—	✓	—	✓	—	✓

Key disambiguation pairs:

- **Bufferbloat vs. FWA rain fade:** Both show delay ramps, but bufferbloat has *stable* throughput during the ramp; FWA shows correlated throughput drop.
- **Wi-Fi 5 contention vs. 5G handover:** Both produce delay spikes, but 5G spikes are *periodic* with stable baseline between; Wi-Fi 5 spikes are random.
- **5G handover vs. satellite beam switch:** Both are periodic spikes, but satellite has elevated baseline ($\approx 35\text{--}50\text{ ms}$) vs. 5G ($\approx 15\text{--}25\text{ ms}$). Cross-user correlation confirms satellite.
- **Bufferbloat vs. ISP congestion:** Both show delay ramps, but ISP congestion shows throughput drop and is correlated across users on different prefixes.

Insight: Multi-modal inference transforms the network classifier into a system-wide diagnostic tool. By correlating network telemetry with local hardware state, the platform avoids the "false positive" of applying network mitigations to local compute bottlenecks.

Figure 64. Diagnostic disambiguation matrix. Each row represents a channel condition; columns show observable telemetry indicators. Green cells indicate a strong signal; yellow indicates sometimes present. The callout box highlights the most common confusion pairs and the differentiating features that resolve them.

8.5 Worked Example: Evening DOCSIS Session

To illustrate the complete inference pipeline in action, consider a concrete scenario: a cloud gaming session on a DOCSIS 3.0 connection during the evening peak (fig. 65).

Phase 1: Session Start ($t = 0\text{--}30\text{ s}$). The session begins during the early evening. Telemetry shows a clean, stationary channel: delay $\approx 12\text{ ms}$, negligible jitter, no loss. The inference engine classifies this as "clean" and applies the default strategy: CUBIC congestion control, modest Reed-Solomon FEC, standard codec profile.

Phase 2: Bufferbloat Onset ($t = 30\text{--}85\text{ s}$). Another household member starts a large file upload. The DOCSIS 3.0 modem’s unmanaged FIFO begins filling. The telemetry shows a characteristic linear ramp in delay with stable throughput and low jitter. After two consecutive 10 s windows confirm the ramp motif and positive mean trend, the inference engine classifies the channel as "DOCSIS 3.0 bufferbloat" with sufficient confidence.

Phase 3: Strategy Switch ($t = 45\text{ s}$). The macro tier fires: congestion control switches from CUBIC (which exacerbates bufferbloat by aggressively filling the queue) to BBR (which targets bandwidth without inflating queues). FEC switches from static Reed-Solomon to Raptor (anticipating potential tail-drop at overflow). Video bitrate is capped to reduce the gaming traffic’s contribution to queue depth. The switch takes effect within 2 s.

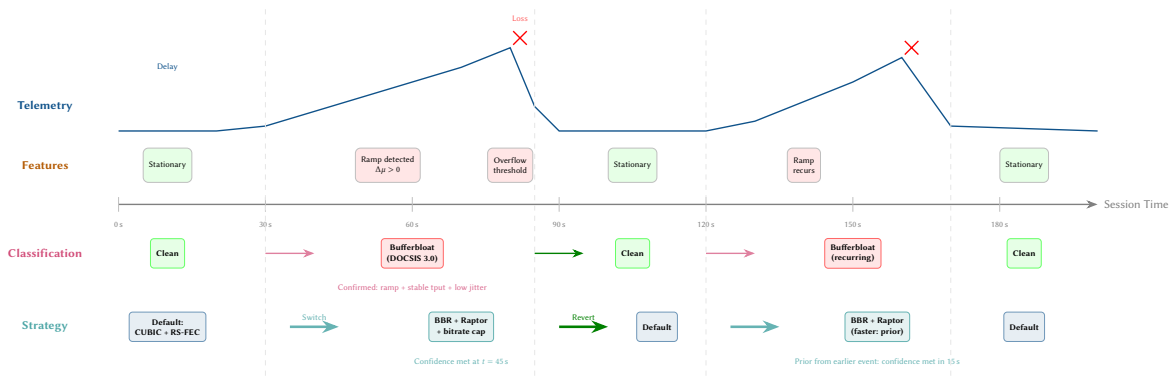


Figure 65. Worked example: a DOCSIS 3.0 session during the evening peak. **Top:** Raw delay telemetry showing a bufferbloat ramp followed by tail-drop, then recovery, then recurrence. **Middle:** Feature extraction detects the ramp motif and confirms bufferbloat through multi-indicator convergence. **Bottom:** Strategy switches from default (CUBIC + RS-FEC) to bufferbloat-optimised (BBR + Raptor + bitrate cap) after confidence is reached at $t = 45$ s. When the ramp recurs, the prior from the earlier event allows the second switch to occur in only 15 s.

Phase 4: Recovery and Recurrence ($t = 90$ – 180 s). The upload completes at $t = 85$ s. The queue drains, and delay returns to baseline. After 20 s of confirmed stationarity, the inference engine reverts to the default strategy. At $t = 120$ s, a second upload begins. The ramp recurs—but this time, the feature store contains a prior from the earlier event. Confidence accumulates faster (15 s instead of 30 s), and the switch to BBR + Raptor occurs before the queue reaches the depth that caused tail-drop in Phase 2.

8.6 Session Lifecycle

Every cloud gaming session passes through a predictable sequence of inference states (fig. 66). The transitions are driven by confidence thresholds, telemetry regime changes, and feature store lookups.

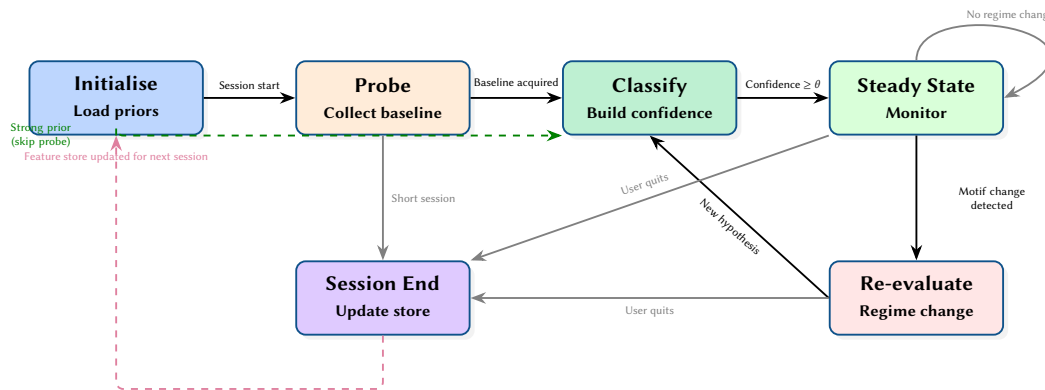


Figure 66. Session lifecycle state machine. A new session loads priors from the feature store, probes the channel to establish a baseline, builds classification confidence, then monitors in steady state. Regime changes trigger re-evaluation. On session end, accumulated fingerprints are written back to the feature store.

Initialise: The session begins by loading priors from the feature store—regional, ISP prefix, CPE, and any user-specific history from prior sessions. If the prior is strong (e.g., the same user connected from the same prefix yesterday with a confirmed DOCSIS 3.0 bufferbloat profile), the system may skip directly to classification with a high initial confidence.

Probe: For the first 5–15 seconds, the inference engine collects baseline telemetry without applying aggressive mitigation. This “cold start” period establishes the channel’s resting state—baseline delay, ambient jitter, and background loss rate. During probing, only conservative default mitigations are active.

Classify: As telemetry accumulates, the engine matches observed motifs against known fingerprints. Confidence grows with each consistent observation window. Once a classification threshold is reached, the macro tier selects the corresponding mitigation strategy.

Steady state: The engine monitors for regime changes—a sudden mean shift, new motif appearance, or correlation signal from the prefix cohort. If no change is detected, the current strategy persists.

Re-evaluate: When a regime change is detected (e.g., a bufferbloat ramp begins, or Wi-Fi band steering shifts the device to a different frequency), the engine re-enters classification with the new telemetry. Feature store priors from the current session accelerate re-classification.

Insight: The worked example illustrates two critical properties of the two-tier model: (1) the macro tier’s confidence requirement prevents false positives during the initial ramp (avoiding premature switching on a transient event), and (2) the feature store’s prior knowledge enables faster response on recurrence—the system learns from its own session history.

Session end: The accumulated fingerprint—including which strategies were applied and their measured QoE effect—is written back to the feature store, enriching priors for subsequent sessions.

MITIGATION

9 The Mitigation Toolkit

The upstream and downstream paths impose fundamentally different constraints on the mitigation toolkit (fig. 67). Upstream input packets are tiny but irreplaceable: a single delayed or lost packet directly affects playability. Downstream video frames are large and deadline-bound: if a frame misses its playout window, it is useless and must be discarded.

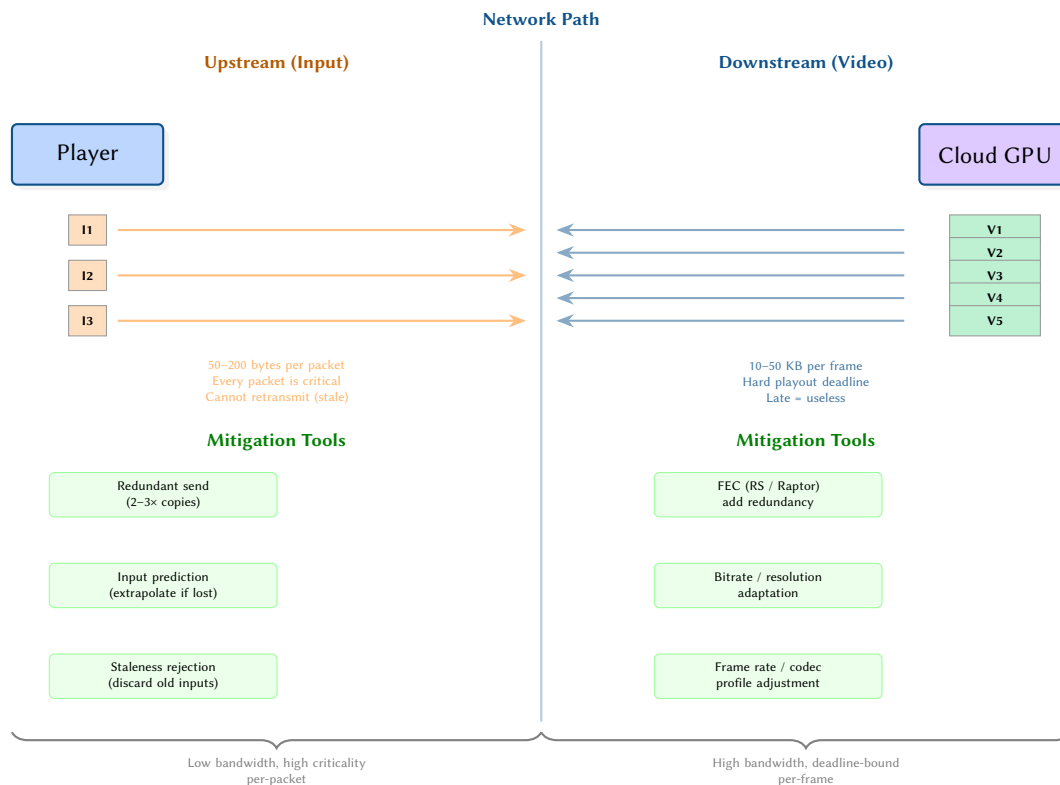


Figure 67. Upstream vs. downstream asymmetry. Upstream carries tiny, irreplaceable input packets; downstream carries large, deadline-bound video frames. The mitigation tools available for each direction are fundamentally different, reflecting the distinct failure modes.

Given the inferred network profile, the streaming server has several “knobs” it can adjust (fig. 69). These fall into two categories: *input factors* (what the server controls) and *output metrics* (what the player experiences).

9.1 Packetization and Interleaving

When configuring the downstream RTP transport, the server must decide how to split large video frames into MTU-sized packets. The optimal packetization strategy depends heavily on the inferred network conditions:

- **Pure Latency:** If the path exhibits high delay but no loss or jitter, standard sequential packetization is fine. The focus should be on reducing frame size and compression time.
- **High Jitter:** If packets arrive in bursts, the decoder buffer must be slightly deeper, and pacing algorithms (like SCReAM or BBR) become critical to avoid overwhelming the last-mile bottleneck.
- **High Loss:** If the path is lossy, simply adding FEC sequentially leaves the frame vulnerable to burst losses. *Interleaving* packets across multiple frames or using dispersive FEC spreads the risk, ensuring a brief burst of interference destroys a recoverable fraction of several frames rather than an entire unrecoverable I-frame.

9.2 Forward Error Correction: Block vs. Fountain Codes

FEC adds redundant data to the video stream so that lost packets can be recovered without server-side retransmission (which would invariably arrive too late). Figure 70 illustrates why.

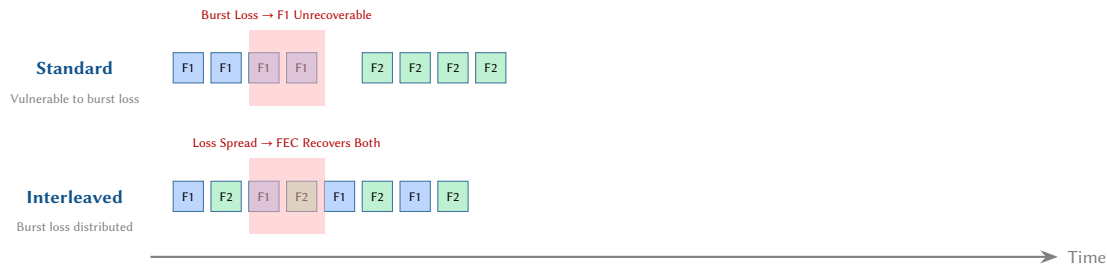


Figure 68. Impact of packetization strategy. Standard sequential transmission leaves individual frames vulnerable to burst loss. Interleaving distributes burst loss across multiple frames, keeping the loss per-frame below the FEC recovery threshold, albeit at the cost of slightly higher baseline latency.

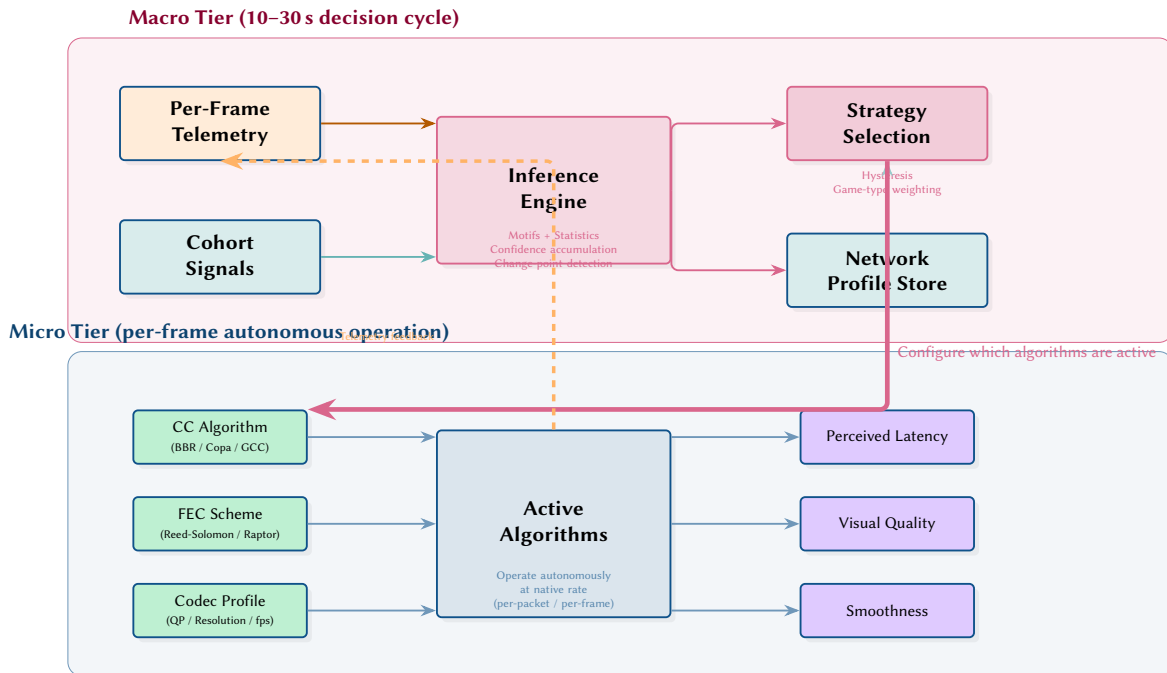


Figure 69. Two-tier adaptation engine architecture. The **macro tier** (10–30 s cycle) accumulates telemetry, builds confidence through motif and statistical analysis, and selects which algorithms to activate—subject to hysteresis and game-type weighting. The **micro tier** runs the selected algorithms autonomously at their native per-frame or per-packet rates. The heavy downward arrow represents the infrequent strategy configuration; the dashed upward arrow represents continuous telemetry feedback.

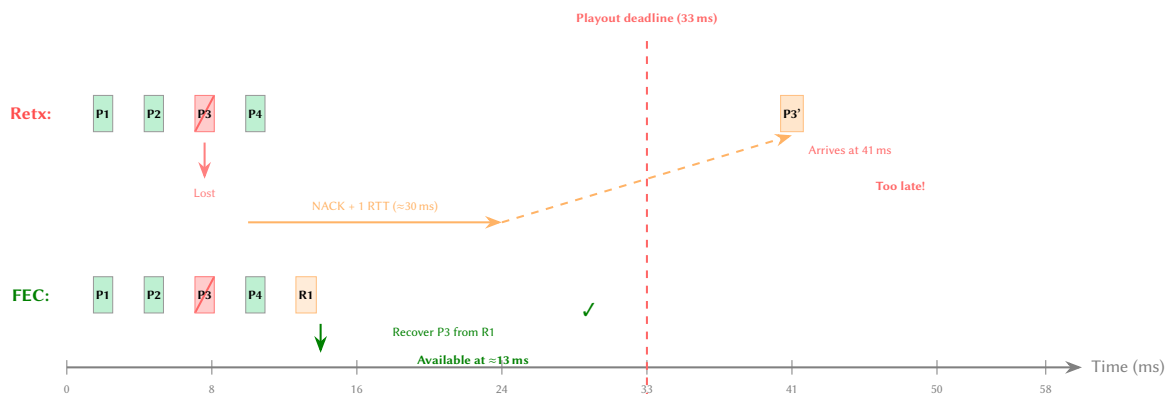


Figure 70. Retransmission vs. FEC recovery timing. With a 15 ms one-way delay, a retransmission request (NACK) and its response take ≈ 30 ms—arriving after the 33 ms playout deadline. FEC repair symbols, sent proactively alongside the original data, enable recovery within the same frame window at no additional round-trip cost.

Static Block Codes (Reed-Solomon): The standard approach relies on rigid Block Codes. However, standard Reed-Solomon is primarily designed for static data at rest (e.g., CD-ROMs or hard drives) and struggles with fluid, non-stationary channels. The primary trade-off with rigid Block FEC is:

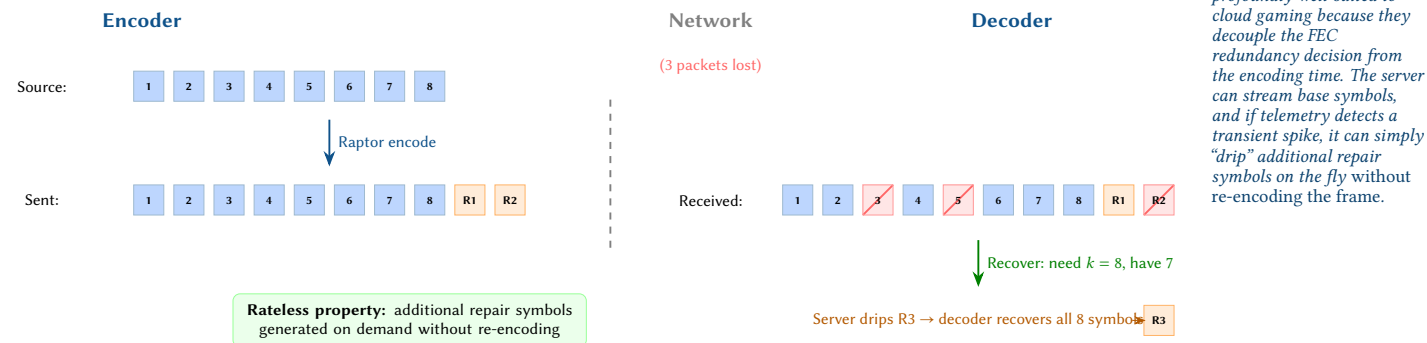
- **Too little FEC** (< loss rate): unrecoverable losses → visual artifacts.

- **Too much FEC** ($\gg$ loss rate): wasted bandwidth, consuming throughput that could carry higher-quality video.

An et al. [0] demonstrate fine-grained per-frame FEC adjustment specifically for cloud gaming, and Chen et al. [0] apply reinforcement learning to adaptively tune FEC parameters. Meng et al. [0] propose edge-based FEC recovery that reduces the round-trip penalty of server-side retransmission.

Dynamic Fountain Codes (Raptor): To solve the rigidity problem, modern cloud gaming platforms can utilise dynamic, rateless erasure codes, known as Fountain Codes. Raptor codes [0] (and their successor RaptorQ) are a highly efficient form of Fountain FEC. The encoder can generate an effectively unlimited number of encoding symbols from any input block, and the decoder can recover the original data from any sufficient subset of received symbols.

Luby [0] recently proposed adaptive receiver-side scheduling using fountain codes for interactive delivery, addressing the cloud gaming and XR use case directly [0].



Insight: Fountain codes are profoundly well-suited to cloud gaming because they decouple the FEC redundancy decision from the encoding time. The server can stream base symbols, and if telemetry detects a transient spike, it can simply “drip” additional repair symbols on the fly without re-encoding the frame.

Figure 71. Fountain Code (Raptor) encoding and rateless recovery. The encoder generates source and repair symbols; the decoder recovers from any sufficient subset. Unlike static Reed-Solomon block codes, additional repair symbols can be dynamically “dripped” based on ML telemetry without re-encoding the video frame.

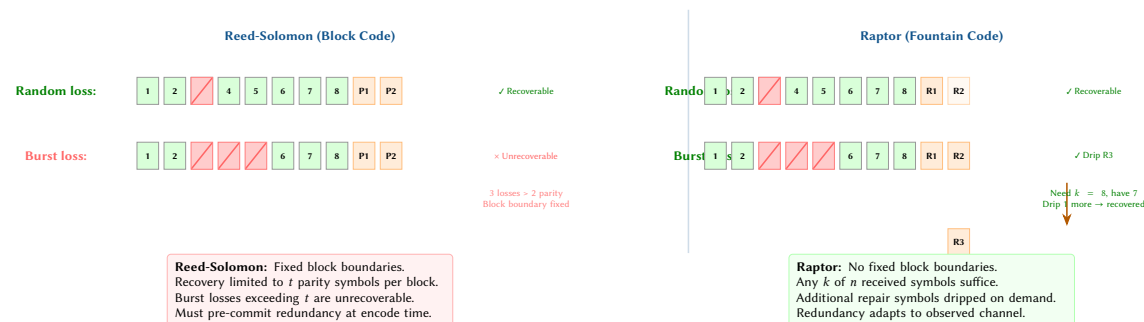


Figure 72. Reed-Solomon vs. Raptor FEC under burst loss. Reed-Solomon recovers from random losses within a block but fails when burst losses exceed the fixed parity count. Raptor codes have no block boundary constraint and can drip additional repair symbols on demand, making them inherently burst-tolerant.

Dynamic Telemetry Application

Consider a downstream video frame split into $k = 100$ source symbols. On a path experiencing a non-stationary fading event, the server initially sends $k + 5 = 105$ encoding symbols (5% overhead). At the **micro level**, the Raptor codec operates autonomously: if packet acknowledgements indicate loss, it can immediately drip additional repair symbols *without re-encoding the frame*—a property unique to rateless codes. At the **macro level**, the inference engine observes the telemetry over a 10–30 second window. If it builds sufficient confidence that the channel has shifted (e.g., from a stationary profile to a high-contention Wi-Fi profile), it adjusts the *baseline* FEC overhead from 5% to 15%, switches the congestion control algorithm, or both. The algorithms then continue to operate autonomously within their new configuration.

9.3 Frame Rate Adjustment

Reducing frame rate from 60 fps to 30 fps halves the downstream bandwidth requirement and doubles the per-frame encoding budget, but directly degrades perceived smoothness. This is most acceptable for slow-paced game genres (strategy, puzzle) and least acceptable for fast-action titles (FPS, racing) [0].

9.4 Adaptive Codec Selection: AV1 vs. H.264

Within a fixed frame rate, the server can adjust video quality via Layer 7 application-level controls:

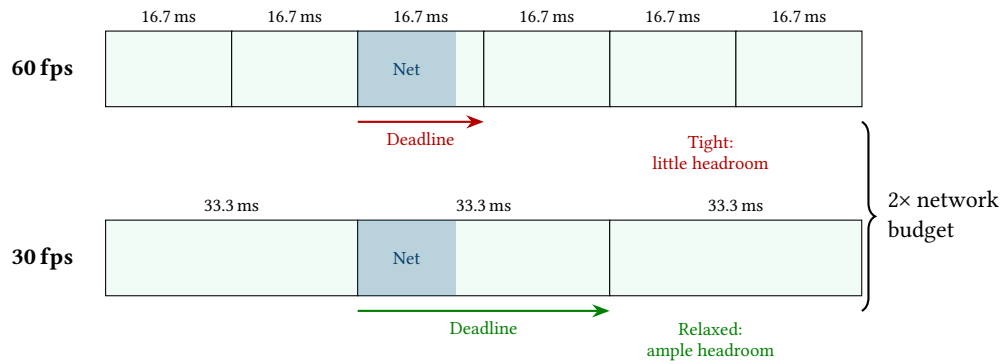


Figure 73. Frame rate determines the per-frame network budget. At 60 fps each frame must complete within 16.67 ms, leaving little headroom for network transit. Halving to 30 fps doubles the available budget per frame, providing substantially more tolerance for jitter and retransmission.

- **Resolution scaling:** Reducing from 1080p to 720p (or dynamic resolution) significantly reduces frame size.
- **Quantisation parameter (QP):** Increasing QP reduces bitrate at the cost of visual quality (blocking artifacts).
- **Codec selection:** Modern codecs like AOMedia Video 1 (AV1) and High Efficiency Video Coding (H.265) (HEVC) deliver 20–40% bitrate savings over Advanced Video Coding (H.264) at equivalent quality.

The choice between codecs is not merely one of compression efficiency but of *latency trade-offs*:

- **AV1 / H.265:** Superior for "Slow-Ramp FWA Rain Fade" or "Low-Throughput DSL" where the network bottleneck is the dominant constraint. Reducing the bitrate reduces the probability of queue-building.
- **H.264:** Often preferred for "High-Jitter Wi-Fi 5" on older client hardware. While less efficient, H.264's decoder is ubiquitous and typically hardware-accelerated with lower decoding latency than early AV1 software implementations.

Table 5 quantifies the relationship between resolution, frame rate, codec, and the resulting bitrate demand. The "Tier" column maps each configuration to the mitigation tiers used in table 7, enabling direct cross-referencing between the quality a user desires and the access technology required to sustain it.

Table 5. Cloud gaming bitrate ladder. Target bitrates assume a competitive action genre with moderate scene complexity. "Avg Frame" is the mean encoded frame size at the target bitrate. Tiers correspond to the Bitrate Strategy column in table 7.

Resolution	FPS	Codec	Target (Mbit/s)	Avg Frame (KB)	Tier
720p	30	H.264	3	12.5	Conservative
720p	60	H.264	5	10.4	Conservative
720p	60	H.265	3	6.3	Conservative
1080p	30	H.264	8	33.3	Moderate
1080p	60	H.264	15	31.3	High
1080p	60	H.265	10	20.8	Moderate
1080p	60	AV1	8	16.7	Moderate
1440p	60	H.265	20	41.7	High
4K	60	H.265	35	72.9	Premium
4K	60	AV1	25	52.1	Premium

Codec efficiency as a mitigation lever. Table 5 reveals that codec selection is not merely a quality-of-implementation detail—it is a first-class mitigation lever that determines which access technologies can sustain a given quality tier. H.265/HEVC and AV1 deliver 20–40% bitrate savings over H.264 at equivalent perceptual quality. The practical consequence is substantial: a DOCSIS 3.0 upstream capped at ≈ 10 Mbit/s can sustain 1080p/60 only with H.265 or AV1 (10 and 8 Mbit/s respectively), not with H.264 (15 Mbit/s). Similarly, constrained wireless links (Wi-Fi 5 under contention, FWA during rain fade) benefit disproportionately from modern codecs, effectively shifting their achievable tier upward without any network-layer change. The inference engine's output should therefore include a codec recommendation alongside its congestion control, FEC, and bitrate selections—treating the encoder as part of the adaptive mitigation stack rather than a fixed upstream dependency.

Insight:
Codec selection is a zero-cost mitigation lever: switching from H.264 to H.265/AV1 can shift a constrained link from an unplayable tier to a viable one, with no change to the network or transport layer.

Bitrate Ladder vs. Typical Upstream Capacity

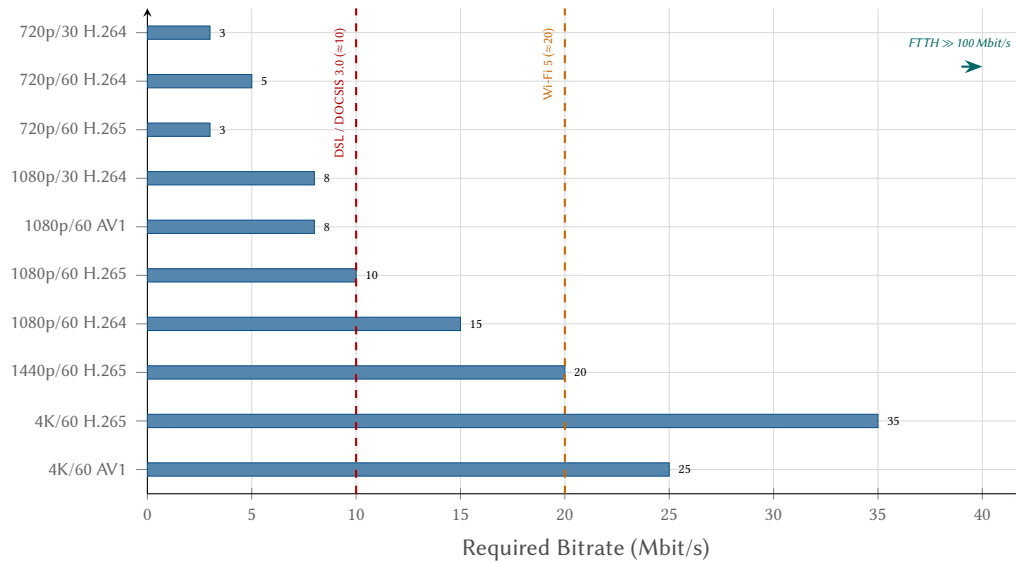


Figure 74. Bitrate ladder overlaid with typical upstream capacities. Bars extending past a technology’s dashed line cannot be sustained on that access type. FTTH capacity ($\gg 100$ Mbit/s) is far off-chart. Modern codecs (H.265, AV1) shift several tiers below the DOCSIS 3.0 / DSL threshold, enabling viable cloud gaming on constrained links.

Hysteresis and QoE: The system must employ *hysteresis* when making these adjustments. Frequent quality oscillations (e.g., rapidly toggling between 720p and 1080p) disrupt QoE more than a stable, slightly lower resolution. Users prefer consistency. The control loop should degrade quality quickly in response to congestion but wait for sustained improvement before upgrading. The broader implications of hysteresis—particularly for strategy-level switching—are discussed in section 10.

9.5 L4S and NQB Implementation: Network-Layer Participation

While Layer 7 mitigations are the primary toolkit, the inference engine can also signal its requirements directly to the network layer to take advantage of emerging Low Latency features in modern ISP infrastructure.

- **L4S (Low Latency, Low Loss, Scalable Throughput):** If the inference engine detects a path that is L4S-capable (e.g., DOCSIS 3.1 with AQM), it should mark packets with the ECN bit ‘ECT(1)’. This signals to the network that the stream is a “scalable” flow that can react to congestion signals more rapidly than traditional TCP, maintaining near-zero queuing delay.
- **NQB (Non-Queue-Building):** For the tiny but critical upstream input packets, the client should mark traffic with the NQB DSCP bit (‘45’ or ‘0x2D’). This signals to the home router and CMTS that these packets should be placed in a dedicated low-priority, non-queue-building queue, bypassing the bulk video traffic and avoiding self-induced bufferbloat.

9.6 Congestion Control and Layer 7 Bitrate Targeting

Cloud gaming congestion control differs fundamentally from traditional Layer 4 TCP. A standard algorithm like CUBIC manages a sliding window for guaranteed delivery, using loss to trigger retransmissions. In cloud gaming, a retransmitted frame invariably arrives after its playout deadline, rendering it useless.

Cloud gaming therefore uses CC algorithms as estimators of a *target Layer 7 video bitrate* (fig. 75). The CC algorithm monitors delay and loss to throttle the encoder’s output, ensuring the stream stays within bottleneck capacity. Several algorithms have been evaluated for cloud gaming [0]:

9.7 Mitigation Strategy by Access Technology

With the full mitigation toolkit established, table 7 maps each access technology to its recommended strategy. The table reflects the fingerprint-to-strategy logic that the inference engine implements: once a channel is classified, the corresponding row defines the initial mitigation configuration.

Figure 76 provides a complementary view: a heatmap of expected effectiveness for each mitigation–technology pair. Green cells indicate beneficial interventions; red cells indicate strategies that would waste bandwidth, add unnecessary latency, or actively degrade QoE. The “wrong-technique” risk is visually apparent—heavy FEC strongly helps satellite and DSL but actively hurts GPON and Wi-Fi 7, where the overhead degrades an already-clean path.

Insight:
No single Congestion Control (CC) algorithm is optimal for all last-mile conditions. A DOCSIS path with bufferbloat benefits from BBR or Copa; a lossy cellular path needs loss-tolerant algorithms that do not catastrophically back off. The inferred network profile should inform CC selection.

Table 6. Congestion control algorithms and their cloud gaming trade-offs.

Algorithm	Approach	Strength	Weakness
CUBIC [0]	Loss-based	High throughput	Fills buffers (bloat)
BBR	Model-based	Low queuing delay	Can over-estimate BW
Copa [0]	Delay-based	Low latency target	Under-utilises on lossy paths
Google Congestion Control (GCC) [0]	Delay + loss	Designed for WebRTC	Moderate throughput
SCReAM [0]	Self-clocked	Frame-aware pacing	Limited deployment

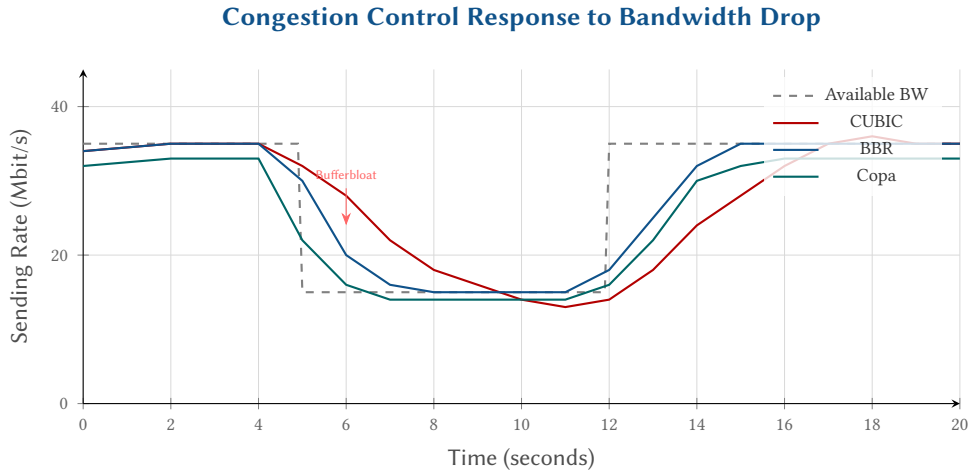


Figure 75. Congestion control response to a bandwidth drop at $t = 5$ s. CUBIC overshoots (filling buffers, adding latency), BBR adapts via bandwidth model, Copa targets minimum delay but slightly under-utilises.

Table 7. Recommended mitigation strategy by access technology. CC = congestion control, JB = jitter buffer, FEC mode = preferred forward error correction approach. Strategies assume a competitive action genre; casual genres permit more conservative settings.

Access Technology	CC Algorithm	FEC Mode	JB Depth	Bitrate Strategy	Key Rationale
FTTH / GPON	CUBIC / BBR	Light RS	Minimal (1–2 ms)	Max quality (up to 4K/AV1)	Near-ideal path; minimal intervention
DSL (fast path)	BBR	Raptor (burst)	3–5 ms	Moderate	Impulse noise needs burst FEC; avoid interleaver
DSL (interleaved)	CUBIC	Light RS	Minimal	Moderate	Interleaver handles noise; accept baseline delay
DOCSIS 3.0	BBR / Copa	Raptor (adaptive)	5–10 ms	Capped at Moderate tier (720p–1080p H.265)	Avoid filling queue; Raptor for tail-drop recovery
DOCSIS 3.1+LLD	BBR	Light RS	3–5 ms	High (up to 1440p/H.265)	AQM controls queue; LL flow bypasses bulk
Wi-Fi 5	GCC / SCReAM	Raptor (burst)	8–15 ms	Adaptive	Contention spikes need burst recovery + pacing
Wi-Fi 6	BBR	Moderate RS	5–8 ms	High	OFDMA reduces contention; legacy devices remain
Wi-Fi 6E	BBR	Light RS	3–5 ms	High	Clean spectrum; near Wi-Fi 7 without MLO
Wi-Fi 7	BBR	Light RS	Minimal (1–3 ms)	Max quality (up to 4K/AV1)	MLO provides link-level redundancy
5G NR	GCC	Raptor (periodic)	10–15 ms	Adaptive	Handover spikes need periodic burst FEC
FWA (5G)	Copa	Raptor (adaptive)	10–20 ms	Conservative (720p H.265/AV1)	Shared cell; weather-dependent; high variance
LEO Satellite	Copa / delay-tolerant	Heavy RS + interleaver	20–40 ms	Conservative (720p H.265/AV1)	High baseline RTT; orbital handovers

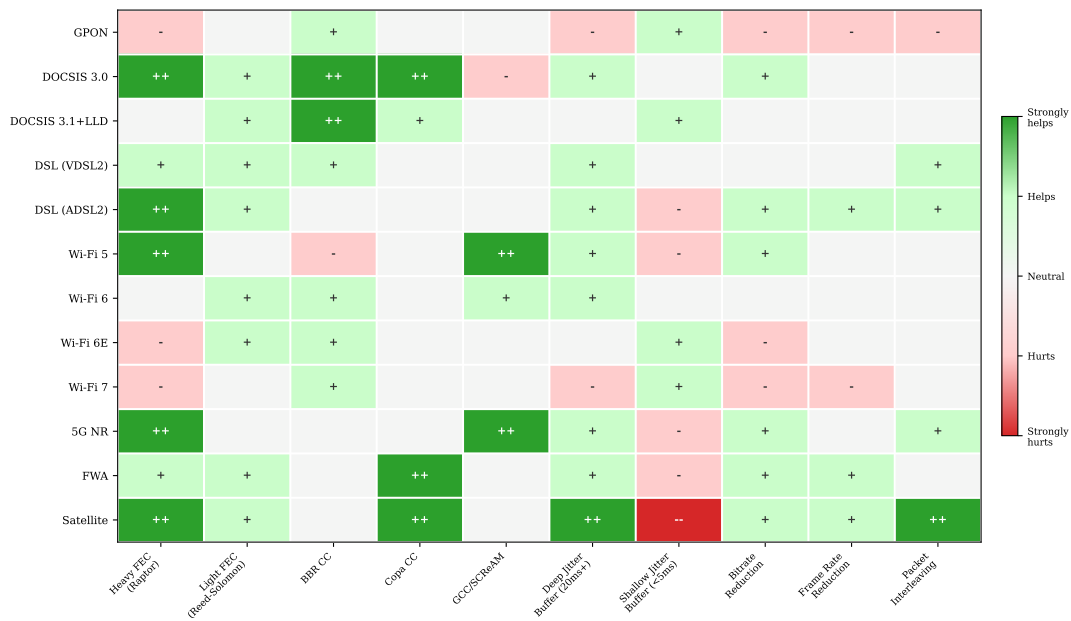


Figure 76. Mitigation effectiveness matrix. Each cell indicates whether a given mitigation technique helps (green) or hurts (red) for the corresponding access technology, based on the fingerprint characteristics and calibration analysis of this report.

10 Calibrating the Response: Under- vs. Over-Correction

The central challenge of adaptive mitigation is *calibration*: not only applying the right *degree* of correction, but selecting the right *type* of correction for the observed conditions. Because the inference engine operates on a 10–30 second decision cycle (section 4), applying the wrong mitigation strategy has consequences that persist for the duration of that window—and can actively compound the original impairment rather than alleviating it.

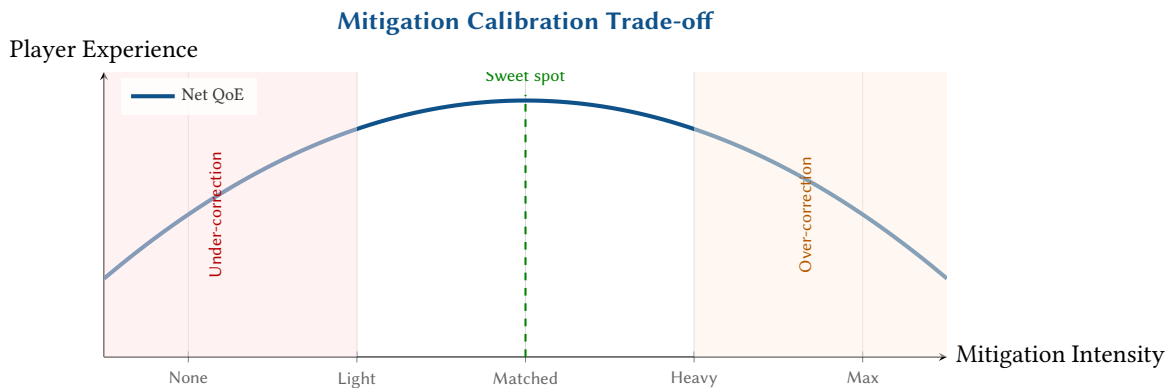


Figure 77. The mitigation calibration trade-off. Under-correction leaves impairments visible; over-correction unnecessarily reduces quality. The sweet spot depends on the inferred network profile.

Figure 78 illustrates the three failure modes side by side, showing how each affects the player’s perceived quality over time.

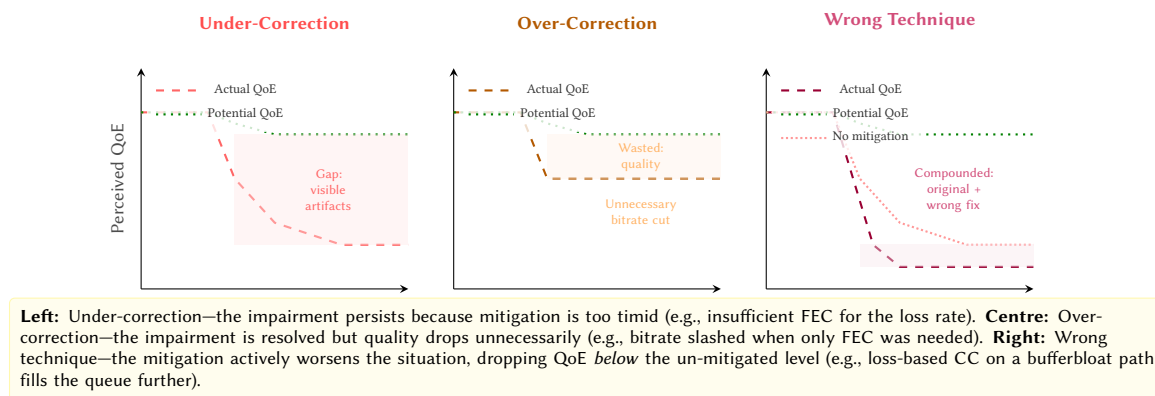


Figure 78. The three calibration failure modes. Under-correction (left) leaves impairments visible. Over-correction (centre) wastes quality unnecessarily. Wrong-technique application (right) is the most damaging: QoE drops below even the un-mitigated baseline, as the mitigation compounds the original impairment.

- **Under-correction:** Insufficient FEC for the observed loss rate; frame rate too high for the available throughput; compression too light for the bandwidth envelope. Result: visual artifacts, stuttering, missed inputs.
- **Over-correction:** Excessive FEC wasting bandwidth that could carry higher-quality video; unnecessary frame rate or resolution reduction on a path that could support more. Result: degraded visual quality despite adequate network conditions.
- **Wrong-technique application:** Perhaps the most damaging failure mode. Applying a mitigation designed for one channel characteristic to a fundamentally different impairment can actively *compound* the problem. For example, detecting Wi-Fi 5 MAC-layer contention spikes and responding by slashing video bitrate is doubly harmful: the underlying contention (a MAC-layer wait time) is unaffected by bitrate, but the user now experiences both the original stutter *and* degraded visual quality. Similarly, switching to a loss-based congestion control algorithm on a path exhibiting bufferbloat (where the problem is queuing delay, not loss) will fill the buffer further before eventually backing off. At the 10–30 second decision timescale, these compounding errors persist long enough to severely degrade the session.

10.1 Hysteresis and Strategy Switching

Beyond calibrating the *degree* of mitigation, the system must manage the *frequency* of strategy changes. Users are highly sensitive to perceptible quality transitions [0]. Rapid oscillation between strategies—even locally

correct ones—produces a “flickering” experience that users perceive as worse than a stable, slightly suboptimal configuration.

Design Rule 2: Hysteresis in Strategy Transitions

The inference engine should employ hysteresis at two levels. First, **parameter hysteresis**: degrade quickly in response to worsening conditions but wait for sustained improvement before upgrading (as discussed in section 9 for resolution scaling). Second, and more critically, **strategy hysteresis**: require a higher confidence threshold to *switch* between fundamentally different mitigation approaches (e.g., from a loss-oriented FEC profile to a delay-oriented congestion control adjustment) than to adjust parameters within the current strategy. A wrong strategy switch at the macro level is far more costly than a suboptimal parameter within the correct strategy, because the micro-level algorithms need time to converge after any strategy change.

Unnecessary strategy switching (convergence delay, compounded impairment, perceptual disruption) typically costs more than remaining on a slightly suboptimal but stable strategy for one additional observation window.

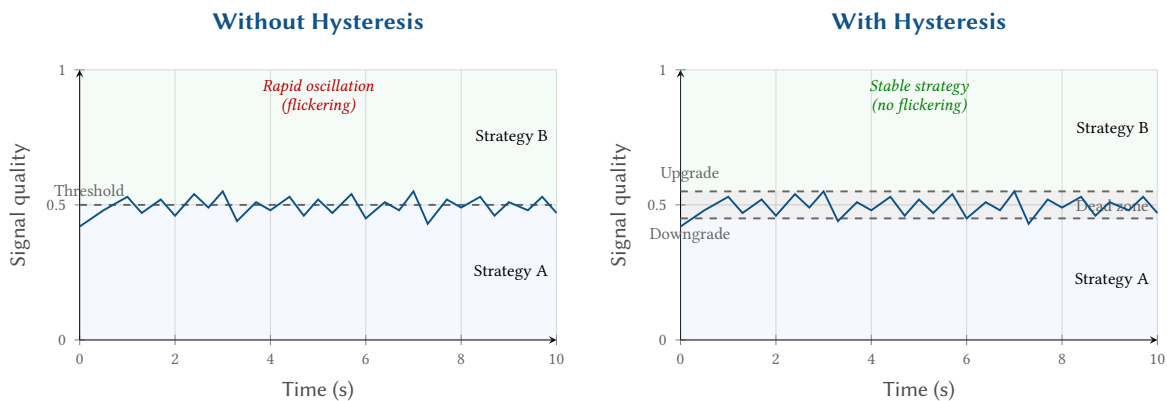


Figure 79. Hysteresis prevents strategy flickering. **Left:** A single threshold causes the noisy signal to trigger rapid oscillation between strategies. **Right:** Separate upgrade and downgrade thresholds create a dead zone; the system requires sustained improvement before upgrading, eliminating transient switching.

10.2 Game-Type Aware Calibration

The appropriate calibration point varies by game genre [0]:

Table 8. Game-type aware mitigation priorities.

Genre	Delay Tolerance	Priority	Preferred Mitigation
FPS / Racing	< 80 ms	Latency	Lower FEC, lower resolution, higher fps
MOBA / Sports	< 100 ms	Balance	Moderate FEC, 60 fps
Strategy / RPG	< 150 ms	Quality	Higher FEC, higher resolution, 30 fps OK
Puzzle / Casual	< 200 ms	Quality	Max FEC, best visual quality

10.3 Upstream Input Protection

For the upstream direction, the mitigation problem is different: input packets are tiny but critical. Strategies include:

- **Input redundancy:** Sending each input event 2–3 times to ensure at least one copy arrives. The bandwidth cost is negligible (< 1 kbit/s additional).
- **Input prediction:** Server-side prediction of likely player input (based on game state and input history) to mask the effect of late-arriving input packets. Lee et al. [0] demonstrated this with speculative frame rendering, predicting player input one full RTT ahead.
- **Input smoothing / rejection:** Filtering anomalous input samples (e.g. a key press that arrives with 200 ms delay may be stale and should be discarded rather than acted upon). This parallels techniques developed for robotics teleoperation, where delayed control signals must be evaluated for staleness before actuation.

Design Rule 3: Proportional Response

Every mitigation action has a cost. FEC consumes bandwidth. Frame rate reduction degrades smoothness. Resolution scaling reduces visual fidelity. The adaptation engine must treat these as a constrained optimisation problem: minimise perceived impairment subject to the inferred network capacity and characteristics, not apply blanket defensive corrections.

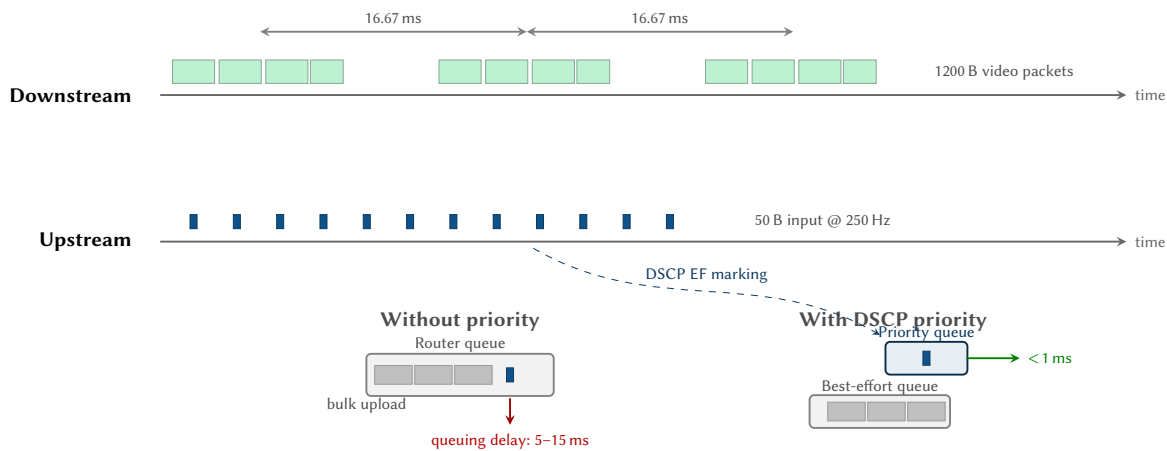


Figure 80. Upstream input packets are tiny but time-critical. Without priority marking, they queue behind bulk uploads; DSCP-based priority queuing ensures sub-millisecond forwarding.

10.4 Audio Stream Considerations

While this report focuses primarily on video frame delivery, cloud gaming also streams audio as a separate RTP stream (typically Opus codec at 48 kHz, 64–128 kbit/s). Audio has distinct sensitivity characteristics:

- **Jitter sensitivity:** Audio playout buffers are typically 20–60 ms deep. Jitter exceeding the buffer depth produces audible glitches (clicks, dropouts) that are perceptually more jarring than equivalent visual artifacts, as the human auditory system is exceptionally sensitive to temporal discontinuities.
- **Loss concealment:** Audio codecs employ Packet Loss Concealment (PLC) algorithms that synthesise missing audio segments from surrounding context. Opus PLC can mask isolated packet losses effectively, but consecutive losses (> 2–3 packets) produce audible artifacts.
- **FEC approach:** Audio FEC is typically handled independently from video—Opus supports in-band Forward Error Correction at the codec level, duplicating critical signal information across packets. This is lightweight (10–20% bandwidth overhead) and does not require the Raptor/Reed-Solomon machinery used for video.
- **Synchronisation:** Audio and video must remain lip-synchronised within ± 40 ms. Differential jitter between the audio and video streams (if they traverse different network paths or queues) can break synchronisation, requiring adaptive playout buffer management at the client.

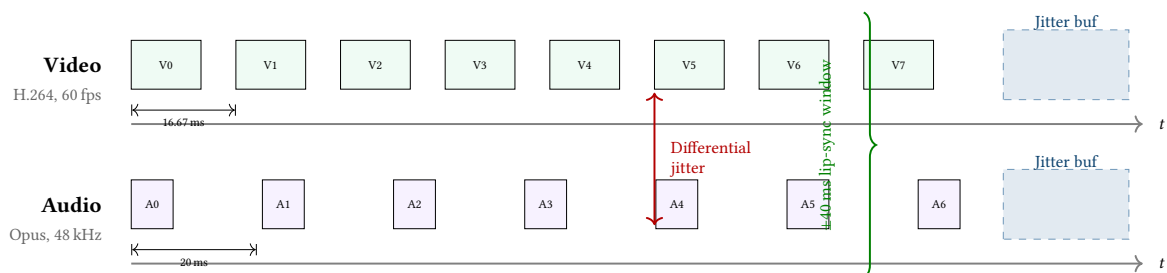


Figure 81. Audio–video synchronisation timeline. The video stream (H.264 at 60 fps) produces frames every 16.67 ms, while the Opus audio stream emits 20 ms frames. Differential jitter between the two streams causes progressive drift; the client must keep both streams within the perceptual ± 40 ms lip-sync window via adaptive playout buffer management.

The inference engine’s channel diagnosis applies equally to the audio stream, but the mitigation response differs: audio mitigation focuses on jitter buffer depth adaptation and codec-level FEC, rather than the bitrate/resolution/frame-rate trade-offs that dominate video adaptation.

DEPLOYMENT

11 Cloud Gaming Platform Architectures

Each cloud gaming platform makes distinct architectural choices (fig. 82) that affect both the feasibility of telemetry collection and the available mitigation toolkit.

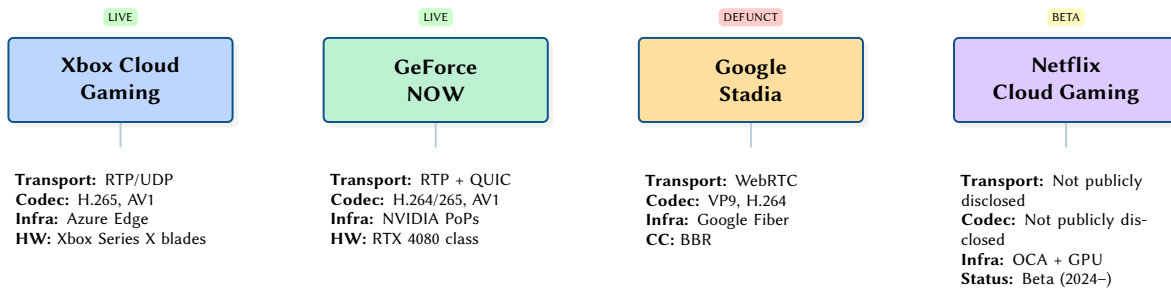


Figure 82. Comparative architecture of four major cloud gaming platforms.

Xbox Cloud Gaming deploys custom Xbox Series X blades across Azure’s edge regions (140+ countries), leveraging Azure’s private backbone [0]. The edge-first placement strategy reduces backbone latency for the majority of users.

NVIDIA GeForce NOW (GFN) uses dedicated datacenters with RTX-class GPUs and an RTP + QUIC protocol stack [0]. Sessions consume 5–45 Mbit/s with NVENC hardware encoding at 2–5 ms per frame.

Google Stadia (defunct January 2023) pioneered WebRTC-based cloud gaming with QUIC transport, BBR congestion control, and VP9 codec [0].

Netflix Cloud Gaming (beta, 2024–) is the most architecturally interesting case for last-mile inference: Netflix’s existing Open Connect Appliance (OCA) network of thousands of servers embedded within ISP networks provides an unmatched last-mile advantage for video—but gaming OCAs require GPU hardware, a fundamentally different and more expensive deployment. The GPU placement constraint is discussed further in section 12.

12 The GPU Placement Problem

A fundamental tension exists between the desire to place rendering servers as close to players as possible and the economic reality of GPU hardware deployment (fig. 83).

12.1 Video CDN vs. Gaming Edge

A conventional video Content Delivery Network (CDN) deploys commodity storage and networking at thousands of locations. These appliances are inexpensive, power-efficient, and require minimal maintenance. Netflix’s OCAs, for example, are essentially high-capacity SSDs with network interfaces—deployed at the cost of a few thousand dollars per unit.

Gaming edge servers require high-end GPUs capable of real-time rendering (RTX 4080-class or custom silicon like Xbox Series X), substantially more power, active cooling, and more frequent hardware refresh cycles. The per-unit cost is 10–50× that of a video CDN appliance, and the power/cooling requirements limit viable deployment locations.

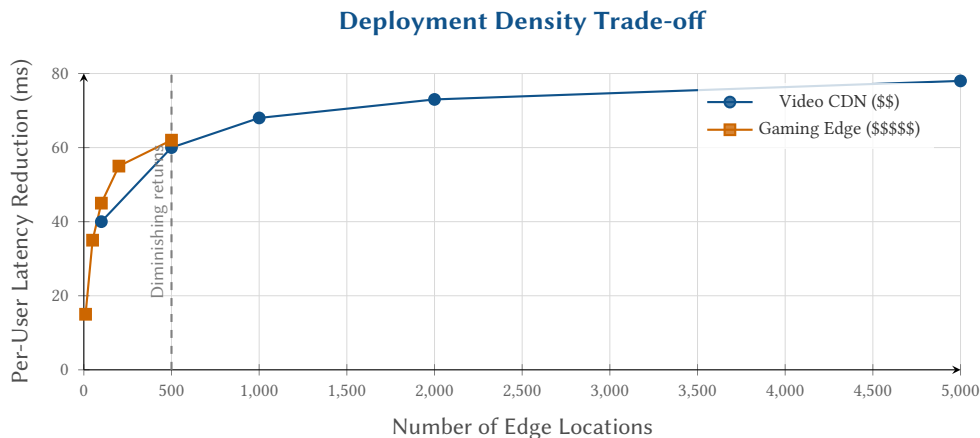


Figure 83. Video CDN appliances can be deployed at thousands of locations economically. Gaming edge servers hit cost barriers at much lower density, limiting the achievable latency reduction.

12.2 Implications for Last-Mile Inference

The alternative—disaggregated edge rendering, where rendering is split across edge and cloud tiers—is an active research area but introduces its own complexity and is beyond the scope of this report.

Insight:
Because GPU servers cannot be deployed everywhere, many users will always have a non-trivial backbone component in their latency. This makes last-mile inference and adaptive mitigation more important, not less—if you cannot eliminate the path, you must adapt to it.

13 Emerging Techniques

13.1 Speculative Frame Rendering

Lee et al. [0] (Outatime, MobiSys 2015 Best Paper) render speculative frames one full RTT ahead based on predicted input, masking up to 120 ms of network latency. This trades server-side GPU compute for perceived latency reduction.

13.2 VSync Coordination

Li et al. [0] (LoopTailor, NSDI 2025) identified VSync misalignment between cloud server and client display as a previously overlooked latency source, reducing interactive latency by $\approx 34\%$.

13.3 Edge-Based Packet Recovery

Meng et al. [0] (Hairpin, NSDI 2024) propose placing FEC recovery at edge nodes rather than requiring server-side retransmission, reducing the recovery round-trip from server-to-player to edge-to-player.

13.4 5G Network Slicing

Dedicated Ultra-Reliable Low-Latency Communication (URLLC) network slices for cloud gaming traffic [0] offer QoS-guaranteed paths, but require coordination between cloud gaming platforms, mobile operators, and device manufacturers.

13.5 Production-Ready Inference SDK: Design and Deployment

To bridge the gap between simulation and real-world deployment, we are developing a production-quality **Last-Mile Inference SDK**. This lightweight library (written in Rust with a C-FFI wrapper) is designed to be integrated directly into game engines (Unity, Unreal) or standalone cloud gaming clients.

- **ROCKET-Powered Classification:** The SDK implements the ROCKET transform for near-real-time technology classification. By using a pre-trained set of 10,000 random convolutional kernels, the engine can fingerprint the access technology within the first 10–30 seconds of a session with $> 94\%$ accuracy.
- **Privacy-Preserving Telemetry:** Following the "Prefix Cohort" model, the SDK anonymises user telemetry before it leaves the device. Using K-anonymity, telemetry is aggregated into technology-specific cohorts (e.g., "DOCSIS-3.1-USA-East"), ensuring that no individual user's network behaviour is identifiable.
- **Multi-Modal Signal Fusion:** The SDK provides built-in support for ingesting OS-level signals (RSSI, SINR) alongside network telemetry, enabling the "Multi-Modal Disambiguation" described in section 8.

DISCUSSION AND CONCLUSIONS

14 Discussion

14.1 The Inference Feedback Loop and Implicit QoE

Effective systems close the loop between inference and mitigation: infer $\rightarrow$ adapt $\rightarrow$ observe effect $\rightarrow$ refine inference (fig. 84). This loop operates at two timescales: per-frame telemetry collection (micro) and dampened 10–30 s strategy decisions (macro). The requirements:

1. Per-frame telemetry collection at the client and server (micro-level input).
2. Profile convergence over a dampened window with exponential decay (macro-level confidence building).
3. Constrained optimisation over the Layer 7 mitigation toolkit, selecting *which* algorithms to activate.
4. Game-type awareness to weight latency vs. quality.
5. Cohort-level signals to distinguish local from shared impairments.

Implicit Feedback via Session Behaviour: The user's behaviour provides an implicit QoE signal. An abnormally short session or rapid game switching indicates dissatisfaction.

ML models can establish baseline session lengths for a given game and user profile. An early termination, correlated with concurrent telemetry (severe loss, excessive jitter), may indicate that the *wrong* mitigation was applied—for example, dropping bitrate on a transient Wi-Fi 5 contention spike, degrading video quality without resolving the MAC-layer wait, compounding frustration. These behavioural signals let the platform flag specific strategies as harmful under certain channel conditions.

Insight:
A production inference engine must be lightweight enough to run on mobile hardware without impacting decode latency, yet statistically robust enough to handle the extreme non-stationarity of real-world last-mile links.

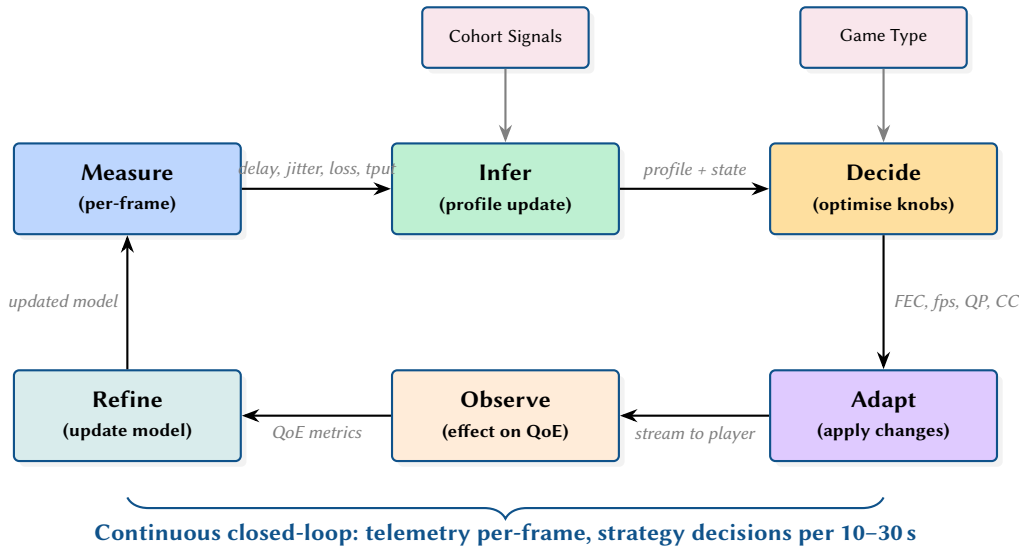


Figure 84. The inference feedback loop. Telemetry drives profile inference, which drives mitigation decisions, whose effects are observed and used to refine the model—continuously, at frame rate.

14.2 The Netflix OCA Question

Netflix operates one of the largest CDN footprints in the world through its Open Connect Appliance (OCA) programme, embedding caching servers directly inside ISP networks. If even a fraction of these appliances were equipped with GPUs capable of real-time game rendering, the backbone segment of the delivery path would vanish for those users. Latency would collapse to the local loop plus a single switch hop.

Under this architecture, last-mile inference becomes the primary differentiating capability. The backbone classifier in fig. 61 would output near-zero delay for OCA-served users, leaving the home network (Wi-Fi generation, station count, router AQM policy) as the dominant source of variance. The mitigation engine would shift its focus entirely to local-loop optimisation: adaptive jitter buffering for Wi-Fi contention, FEC tuning for wireless loss, and codec selection constrained by the user’s upstream capacity.

Two open questions arise. First, whether the GPU silicon required for cloud rendering at 4K/60 fps can be economically integrated into the OCA form factor, which today uses commodity storage-optimised hardware. Second, whether ISPs hosting OCAs would permit GPU workloads that increase power draw and rack density. Neither question has a public answer, but the strategic logic—eliminating backbone latency for the platform’s highest-value subscribers—makes OCA-based rendering a plausible medium-term trajectory that last-mile inference systems should anticipate.

14.3 Simulation-Based Hypothesis Evaluation

A companion report (LM-SIM-2026-01, *Last-Mile Monte Carlo Simulator*) presents a stochastic queuing-model simulator for evaluating the hypotheses identified here. Each access technology is abstracted as a parameterised queuing model whose service time distribution and loss process reproduce the fingerprints described in section 5. The simulator generates 60 fps video frame streams, passes them through composable queuing stages, and produces QoE distributions across 10^5 parallel Monte Carlo trials—enabling controlled evaluation of questions difficult to answer through live experimentation alone (e.g., quantifying the QoE benefit of Raptor over Reed-Solomon FEC under a DOCSIS bufferbloat profile, or measuring the cost of wrong-technique application during a 5G handover).

The simulator employs several variance-reduction techniques to improve statistical efficiency. *Common Random Numbers* (CRN) ensure that scenario-versus-scenario comparisons share identical random draws per trial via ChaCha8 stream separation, reducing the variance of paired differences [0]. *Antithetic variates* further halve variance for monotone functions by pairing each trial with a complemented-uniform counterpart and averaging the results. Confidence intervals are computed using bias-corrected and accelerated (BCa) bootstrap resampling [0], which adjusts for skewness in the percentile distributions of latency, jitter, and loss. Sensitivity analysis uses Partial Rank Correlation Coefficients (PRCC) with OLS residual regression over Latin Hypercube samples to identify which parameters—buffer depth, cross-traffic intensity, Wi-Fi station count—most influence QoE outcomes [0]. An automated convergence checker monitors the relative half-width of the P95 latency confidence interval after each batch of trials, early-stopping when $CI_{95}/|\bar{x}| < \epsilon$ (default $\epsilon = 0.02$), avoiding unnecessary computation while ensuring statistical reliability.

14.3.1 Headline Results

Table 9 summarises the baseline performance of each technology under idle/clean conditions across 10,000 Monte Carlo trials at 60 fps for 60 s sessions. All eleven technologies achieve “Acceptable” QoE under baseline conditions, but P95 latencies span an order of magnitude—from 7.17 ms (FWA clear-sky) to 53.39 ms (LEO satellite)—confirming that the access technology is the dominant source of last-mile variance.

Table 9. Simulated baseline performance per access technology (10,000 trials, 60 fps, 60 s sessions). CV denotes the coefficient of variation of P95 across trials.

Family	Technology	P95 (ms)	Loss (%)	CV (%)	Best fit	QoE
Wired	GPON	8.12	1.0	0.1	Normal	Acceptable
Wired	DOCSIS 3.0	9.17	1.3	0.0	Gamma	Acceptable
Wired	DOCSIS 3.1	9.08	1.2	0.0	Normal	Acceptable
Wired	DSL (VDSL2)	8.17	1.1	0.0	Gamma	Acceptable
Wi-Fi	Wi-Fi 5	8.97	1.1	0.3	Log-normal	Acceptable
Wi-Fi	Wi-Fi 6	17.60	1.1	0.1	Gamma	Acceptable
Wi-Fi	Wi-Fi 6E	17.60	1.1	0.1	Gamma	Acceptable
Wi-Fi	Wi-Fi 7	8.25	1.1	0.1	Normal	Acceptable
Cellular	5G NR	31.50	1.3	5.5	Log-normal	Acceptable
Fixed Wireless	FWA	7.17	1.1	5.1	Log-normal	Acceptable
Satellite	LEO Satellite	53.39	1.1	0.0	Normal	Acceptable

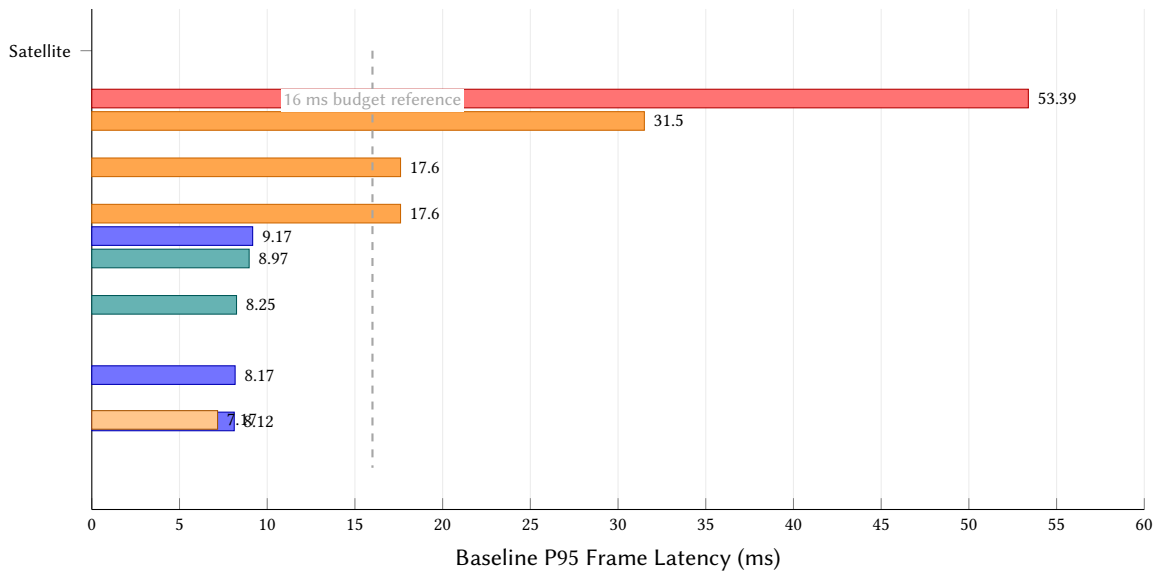


Figure 85. Comparative benchmark: baseline P95 frame latency ranked by access technology. The dashed line marks a 16 ms reference budget for the network share of a 60 fps interactive path.

Three distributional families emerge: wired technologies (GPON, DOCSIS, DSL) and Wi-Fi 7 are well-described by Normal or Gamma distributions with near-zero inter-run variability ($CV < 0.3\%$); Wi-Fi 5 and the wireless technologies (5G NR, FWA) follow Log-Normal distributions with higher variance (CV up to 5.5% for 5G NR); and LEO satellite follows a Normal distribution but at a much higher baseline (53.39 ms) dominated by propagation delay.

14.3.2 Technology Distinguishability

Pairwise Cohen’s d effect sizes confirm that the eleven technologies are statistically distinguishable (all pairs $|d| > 0.5$), with the largest separations between satellite and wired technologies ($d > 10$). This validates the premise of section 5: each access technology produces a distinct latency fingerprint that an inference engine can exploit for classification.

For an operational reader, the practical message is simpler than the raw statistics: these technologies fall into distinct latency bands, so a platform should not treat them as small variations around one generic network model.

14.3.3 Extreme Value Tail Bounds

Generalised Pareto Distribution (GPD) fits to the upper decile of each technology’s P95 distribution yield shape parameters $\xi < 0$ in all cases, indicating bounded (Weibull-type) tails. Table 10 reports the estimated return levels.

Table 10. EVT return levels from GPD fits to the upper 10% of P95 latency distributions. All shape parameters $\xi < 0$ indicate bounded tails—worst-case latencies are finite and predictable.

Technology	Threshold (ms)	ξ	RL ₁₀₀ (ms)	RL ₁₀₀₀ (ms)	RL ₁₀₀₀₀ (ms)
GPON	7.36	-0.184	7.37	7.38	7.38
DOCSIS 3.0	9.61	-0.130	9.62	9.63	9.64
Wi-Fi 5	9.21	-0.012	9.43	9.65	9.86
5G NR	16.22	-0.276	16.22	16.22	16.22
FWA	17.48	-0.160	17.57	17.63	17.67
Satellite	88.35	-0.149	89.84	90.91	91.66

The practical implication is that even at 1/10,000 exceedance probability, satellite latency is bounded below 92 ms and wired technologies remain below 10 ms. This means the QoE guarantees derived from median or P95 statistics hold reliably at the tail—a critical property for production SLA design.

14.3.4 Scenario Impact Analysis

Table 11 quantifies the P95 latency impact of adverse conditions relative to each technology’s baseline. The most dramatic degradation occurs on DSL under crosstalk (+850%), followed by satellite beam switching (+54%) and 5G NR during fast driving (+27%). DOCSIS 3.0 bufferbloat adds ~12% to P95, while Wi-Fi 7 MLO congestion adds only ~9%, demonstrating that MLO’s multi-band steering provides meaningful resilience under load.

Table 11. Scenario impact on P95 latency relative to baseline. Technologies are grouped by family; each row shows one adverse condition.

Family	Scenario	Baseline P95 (ms)	Scenario P95 (ms)	Δ (ms)	Δ (%)
GPON	Cross-Traffic Burst	8.12	8.12	0.00	0.0%
GPON	Cross-Traffic CBR	8.12	8.12	0.00	0.0%
DOCSIS 3.0	Bufferbloat Light	9.17	10.18	+1.01	+11.0%
DOCSIS 3.0	Bufferbloat Severe	9.17	10.27	+1.10	+12.0%
DOCSIS 3.1	AQM Load	9.08	9.37	+0.28	+3.1%
Wi-Fi 5	Congestion (16 Stations)	8.97	10.08	+1.12	+12.4%
Wi-Fi 6	OFDMA Congestion	17.60	17.60	+0.00	+0.0%
Wi-Fi 6E	Congestion 6 GHz	17.60	17.60	+0.00	+0.0%
Wi-Fi 7	Congestion with MLO	8.25	9.02	+0.76	+9.2%
5G NR	Driving Fast	31.50	39.99	+8.49	+27.0%
FWA	Heavy Rain Fade	7.17	8.07	+0.90	+12.5%
Satellite	Beam Switch	53.39	82.00	+28.61	+53.6%
DSL (ADSL2)	Crosstalk	14.89	141.41	+126.53	+850.0%

These deltas inform the mitigation calibration discussed in section 10: a technology with a +850% worst case (DSL ADSL2 crosstalk) requires aggressive intervention, while one with +9% (Wi-Fi 7 congestion) warrants only light adaptation.

14.3.5 Loss-Latency Joint Distribution

Figure 87 plots the joint distribution of P95 latency and frame loss rate across all baseline scenarios. The tight, non-overlapping clusters confirm that even a simple two-dimensional feature space (latency, loss) provides strong discriminative power for technology classification—supporting the fingerprint-based inference approach of section 5.

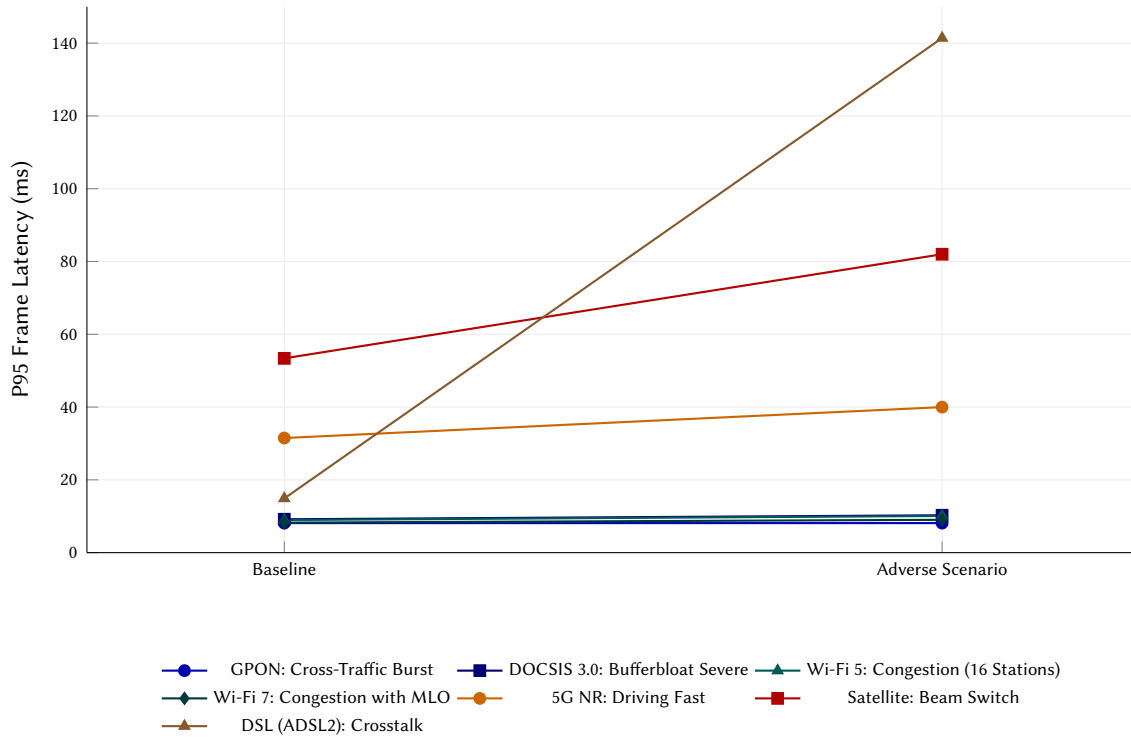


Figure 86. Comparative benchmark: baseline-to-adverse slope chart for representative stress scenarios. The steepest slopes identify the cases where the inference engine should move quickly from passive observation to active mitigation.

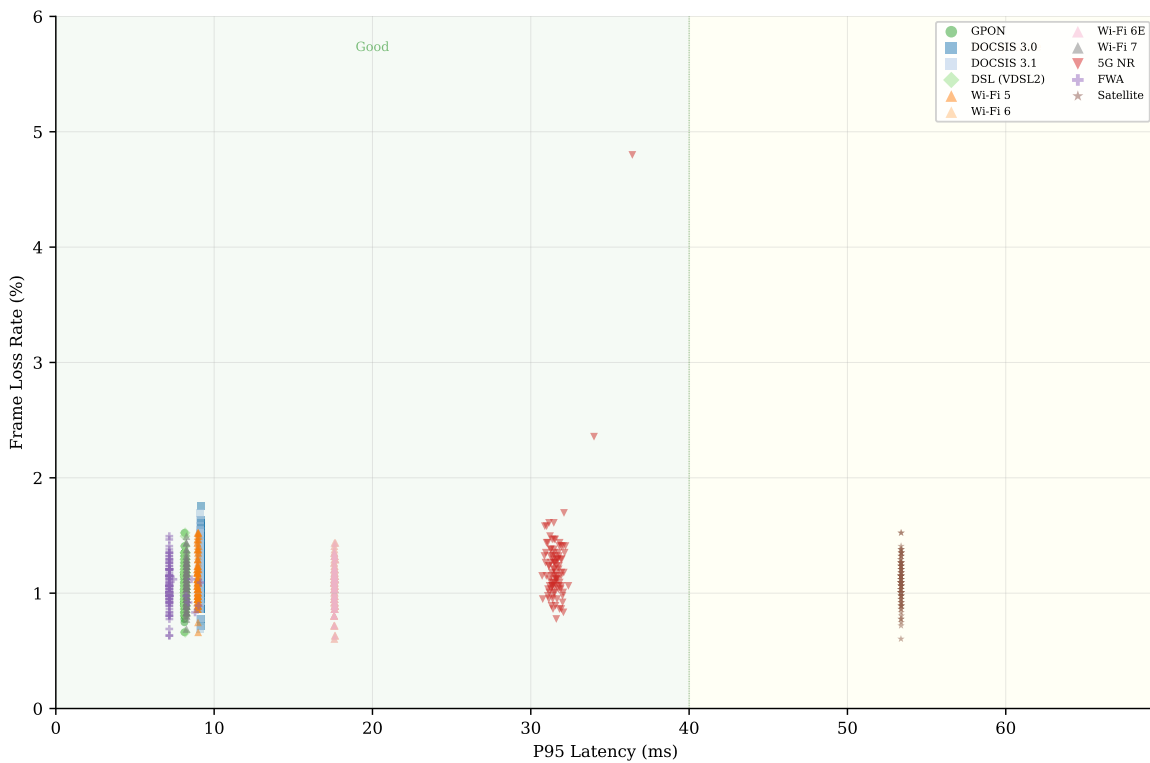


Figure 87. Joint distribution of P95 latency and frame loss rate across all technology baselines. Each point is one Monte Carlo trial. Technologies occupy distinct regions of the (latency, loss) space.

14.4 Privacy and Ethical Considerations

The cross-user correlation and feature store mechanisms described in section 5.5 raise important privacy considerations. Correlating telemetry across users on the same network prefix inherently involves observing and comparing individual users' network behaviour patterns. Several safeguards should be considered:

- **Aggregation thresholds:** Prefix cohort analysis should require a minimum cohort size (e.g., ≥ 5 users) before correlation signals are generated, preventing identification of individual users' behaviour patterns from small groups.
- **Feature anonymisation:** The feature store should store aggregated statistical features (mean shift magnitudes, motif class distributions) rather than raw per-user telemetry traces. Individual session traces should be discarded after feature extraction.
- **Data retention:** Exponential decay weighting (as described for profile evolution) serves a dual purpose: it maintains inference accuracy while automatically aging out historical data. A maximum retention window (e.g., 90 days) provides a hard boundary.
- **Consent and transparency:** Users should be informed that anonymised network telemetry is used to improve streaming quality, with opt-out mechanisms available. The diagnostic information surfaced to users (e.g., "your connection shows signs of bufferbloat") should be framed as helpful rather than surveillance.
- **Purpose limitation:** Telemetry data collected for inference and mitigation should not be repurposed for ISP performance reporting, advertising targeting, or other secondary uses without explicit consent.

14.5 Production Validation Methodology

Validating that the inference engine improves QoE in production requires a rigorous A/B testing framework. The recommended approach:

Experimental design: Users are randomly assigned to one of three groups: (1) *control*—static mitigation with fixed, conservative FEC and default congestion control; (2) *treatment-A*—adaptive mitigation with the full inference engine; (3) *treatment-B*—adaptive mitigation without cross-user correlation (to isolate the value of cohort signals). Assignment should be persistent per user (not per session) to avoid within-user confounding.

Primary metrics:

- **Session duration:** Mean and median session length, controlling for game type. Longer sessions indicate higher satisfaction.
- **Frame delivery rate:** Fraction of frames arriving before their playout deadline, measured at the client.
- **Quality switches:** Number of resolution, frame rate, or bitrate transitions per session. Fewer switches generally indicate better stability.
- **User retention:** 7-day and 30-day return rates, controlling for game catalogue and platform factors.

Statistical requirements: Given the high variance in network conditions across users, achieving statistical significance requires large sample sizes. Power analysis should target $>80\%$ power to detect a 5% improvement in frame delivery rate at $p < 0.01$. Stratification by access technology (using the inference engine's own classification) enables subgroup analysis—the benefit of adaptive mitigation may be large for DOCSIS 3.0 users but negligible for FTTH users.

14.6 Convergence with Robotics Teleoperation

The cloud gaming latency challenge shares deep structural similarity with robotics teleoperation: both involve a real-time control loop over an imperfect network, where delayed or lost control signals must be evaluated for staleness, and visual feedback must arrive within strict timing constraints. Groshev et al. [0] demonstrated FoReCo at SIGCOMM '22, a forecast-based recovery mechanism for real-time remote control of robotic manipulators that uses ML prediction to recover lost control commands over lossy wireless links—reducing trajectory error by over 34%. The parallel to cloud gaming input recovery is direct: a lost or stale controller input can be inferred from recent input history and game state, much as a lost robot command can be inferred from the intended trajectory. Techniques developed for one domain—input smoothing, predictive control, adaptive FEC, staleness rejection—transfer directly to the other.

14.7 Limitations

This framework rests on several assumptions that constrain its applicability and should be acknowledged explicitly:

- **Telemetry availability:** The framework assumes that per-frame telemetry (delay, jitter, loss, throughput) can be collected at the server at 30–120 Hz. Some platform architectures—particularly those relying on third-party WebRTC stacks—may expose only aggregated statistics (e.g., 1-second RTCP reports), limiting the temporal resolution available for motif detection.
- **Single-server perspective:** The inference engine observes the end-to-end path but cannot directly attribute impairments to specific hops. The multi-hop decomposition (section 5.9) is inherently approximate: without in-network measurement points, separating Wi-Fi jitter from DOCSIS bufferbloat relies on frequency-domain heuristics rather than ground truth.

- **Absence of ground-truth labels:** The fingerprint corpus is built from inferred, not confirmed, access technology labels. Without ISP cooperation to provide subscriber line types, the system cannot validate its classifications directly. Misclassification rates remain unknown until labelled datasets become available.
- **Cross-user correlation scalability:** Prefix cohort analysis requires concurrent sessions on the same prefix. In regions with low platform adoption or sparse ISP prefixes, the cohort may be too small to produce statistically meaningful correlation signals. The minimum cohort size for reliable inference is an open empirical question.
- **Illustrative, not empirical:** All telemetry plots and numerical examples in this report are illustrative. The characteristic shapes and scales are drawn from the measurement literature, but the specific values have not been validated against a production deployment. The companion simulation report (LM-SIM-2026-01) provides Monte Carlo validation with statistically rigorous confidence intervals (BCa bootstrap) and sensitivity analysis (PRCC), but live A/B testing against real player sessions remains necessary for production deployment.
- **Stationarity assumption for motifs:** The framework assumes that access technology fingerprints are sufficiently stable to be learned and reused across sessions. Infrastructure upgrades, firmware updates, and changing usage patterns can shift fingerprints over time—the exponential decay weighting partially addresses this, but abrupt fleet-wide changes (e.g., an ISP enabling LLD overnight) may temporarily confuse the classifier.

14.8 Future Work

Several directions merit further investigation:

- **ML-based fingerprint classification:** The motif-matching and statistical indicator approach described in this report is inherently rule-based. A supervised or semi-supervised model trained on labelled telemetry traces—where ground-truth access technology is known from ISP metadata or user surveys—could learn higher-dimensional decision boundaries that capture compound fingerprints more accurately than hand-crafted rules.
- **Standardised telemetry APIs:** Current inference relies on application-layer measurements (RTP timestamps, RTCP reports, WebRTC stats). Emerging standards such as the W3C Network Information API and L4S ECN markings could expose lower-layer signals directly to the application, reducing the need for indirect inference.
- **ISP cooperation and ground-truth labels:** The feature store would benefit enormously from ground-truth access technology labels. Partnerships with ISPs to correlate anonymised subscriber line types with observed telemetry would enable supervised model training and validation—though privacy constraints (section 14) make this non-trivial.
- **Wi-Fi 7 MLO in practice:** Multi-Link Operation fundamentally changes the inference problem by introducing link-level redundancy that masks impairments before they reach the application. As Wi-Fi 7 deployment matures, the inference engine must learn to distinguish between a genuinely clean path and one where MLO is hiding underlying issues that may resurface under load.
- **Network slicing integration:** 5G network slicing promises QoS-guaranteed paths for latency-sensitive traffic. If the inference engine can request a URLLC slice when it detects degradation on the default bearer, the mitigation toolkit expands from purely application-layer adaptations to cross-layer coordination—though this requires business agreements between cloud gaming platforms and mobile operators.
- **Predictive pre-adaptation:** Rather than reacting to observed impairments, the feature store’s temporal patterns (e.g., “this user’s DOCSIS path degrades every weekday at 19:00”) could drive *predictive* strategy pre-positioning—applying the bufferbloat mitigation *before* the ramp begins, eliminating the detection latency entirely.

15 Conclusion

Inferring last-mile network characteristics for cloud gaming is a systems design challenge, not merely a measurement problem. It operates at two timescales: per-frame telemetry feeds micro-level algorithms that react autonomously, while macro-level inference builds confidence over 10–30 s windows before selecting strategies. The mitigation must be calibrated, proportional, game-aware, and stable enough that strategy transitions themselves do not degrade the player’s experience.

The key findings of this survey:

1. **Buffering is not an option.** Cloud gaming’s interactive loop demands real-time frame delivery, making last-mile inference a first-order problem.
2. **Four signals at frame granularity.** Delay, jitter, loss, and throughput measured at 30–120 Hz capture the transient impairments that dominate Quality of Experience (QoE) degradation.

3. **Each access technology leaves a distinct fingerprint.** The eleven technologies surveyed each produce characteristic telemetry signatures distinguishable through shape motifs and statistical indicators. Multi-hop paths (Wi-Fi → powerline → DOCSIS) produce compound fingerprints requiring frequency decomposition to disentangle.
4. **Priors and correlation accelerate classification.** Regional metadata halves the hypothesis space before the first frame arrives; cross-user correlation across prefix cohorts separates shared upstream impairments from in-home events. The accumulated fingerprint corpus improves classification over time.
5. **Upstream, downstream, and audio require different strategies.** Input protection (redundancy, prediction, staleness rejection), video adaptation (FEC, compression, frame rate), and audio management (jitter buffering, PLC, lip-sync) each demand distinct mitigation approaches.
6. **Wrong-technique application is the most dangerous failure mode.** Applying a mitigation designed for one channel to a different impairment compounds the problem, dropping QoE below even the unmitigated baseline.
7. **Device variability constrains the network budget.** A weak client decoder consumes more of the MTP budget, leaving less headroom for last-mile transit and requiring earlier intervention.
8. **Game type drives calibration.** A strategy optimal for a turn-based RPG is wrong for a competitive FPS, and vice versa. GPU placement further constrains deployment: gaming edge servers are expensive, making adaptive mitigation essential for users outside the edge footprint.
9. **Privacy and validation are first-order concerns.** Cross-user correlation requires careful anonymisation and aggregation thresholds; production validation demands A/B testing stratified by access technology.
10. **The session lifecycle structures inference.** A state machine (probe → classify → steady state → re-evaluate) with feature store priors enables progressively faster classification across sessions, transforming cold-start latency into a diminishing cost.
11. **Simulation validates distinguishability.** Monte Carlo simulation across 10,000 trials confirms that all eleven access technologies produce statistically distinct P95 latency distributions (pairwise Cohen's $d > 0.5$), with bounded Weibull-type tails ($\xi < 0$) ensuring that worst-case latencies are finite and predictable (tables 9 and 10).

References

- [0] T. Kämäräinen, M. Siekkinen, A. Ylä-Jääski, W. Zhang, and P. Hui, “A measurement study on achieving imperceptible latency in mobile cloud gaming,” in *Proc. 8th ACM Multimedia Systems Conference*, ser. MMSys '17, Taipei, Taiwan, 2017, pp. 88–99. DOI: [10.1145/3083187.3083191](https://doi.org/10.1145/3083187.3083191)
- [0] S. S. Sabet, S. Schmidt, S. Zadtootaghaj, C. Griwodz, and S. Möller, “Delay sensitivity classification of cloud gaming content,” in *Proc. 12th ACM International Workshop on Immersive Mixed and Virtual Environment Systems*, ser. IMVS '20, 2020. DOI: [10.1145/3386293.3397116](https://doi.org/10.1145/3386293.3397116)
- [0] K.-T. Chen, Y.-C. Chang, P.-H. Tseng, C.-Y. Huang, and C.-L. Lei, “Measuring the latency of cloud gaming systems,” in *Proc. 19th ACM International Conference on Multimedia*, ser. MM '11, Scottsdale, AZ, USA, 2011, pp. 1269–1272. DOI: [10.1145/2072298.2071991](https://doi.org/10.1145/2072298.2071991)
- [0] U. Lampe, M. Weiß, R. Steinmetz, R. Hans, and D. Burgstahler, “Assessing latency in cloud gaming,” in *Cloud Systems and Digital Experiences*, ser. Lecture Notes in Computer Science, Springer, 2014. DOI: [10.1007/978-3-319-11561-0_4](https://doi.org/10.1007/978-3-319-11561-0_4)
- [0] F. Diomede, L. De Cicco, and S. Mascolo, “A network analysis on cloud gaming: Stadia, GeForce Now and PSNow,” *Games*, vol. 1, no. 3, p. 15, 2020. DOI: [10.3390/games1030015](https://doi.org/10.3390/games1030015)
- [0] M. Carrascosa and B. Bellalta, “Cloud-gaming: Analysis of Google Stadia traffic,” *Computer Communications*, vol. 188, pp. 99–116, 2022. DOI: [10.1016/j.comcom.2022.03.006](https://doi.org/10.1016/j.comcom.2022.03.006)
- [0] M. Lyu, S. Madanapalli, A. Vishwanath, and V. Sivaraman, “Network anatomy and real-time measurement of NVIDIA GeForce NOW cloud gaming,” in *Passive and Active Measurement (PAM 2024)*, ser. Lecture Notes in Computer Science, Springer, 2024, pp. 61–91. DOI: [10.1007/978-3-031-56249-5_3](https://doi.org/10.1007/978-3-031-56249-5_3)
- [0] M. Lyu and V. Sivaraman, “A large-scale network measurement study of NVIDIA GeForce NOW cloud gaming in the wild,” *IEEE Transactions on Networking*, vol. 33, pp. 1–16, 2025.
- [0] N. Barman, S. Schmidt, S. Zadtootaghaj, M. G. Martini, and S. Möller, “An analysis of cloud gaming platforms behaviour under synthetic network constraints and real cellular networks conditions,” *Journal of Network and Systems Management*, vol. 31, no. 2, p. 39, 2023. DOI: [10.1007/s10922-023-09720-9](https://doi.org/10.1007/s10922-023-09720-9)
- [0] F. Metzger et al., “An introduction to online video game QoS and QoE influencing factors,” *IEEE Communications Surveys and Tutorials*, vol. 24, no. 3, pp. 1894–1925, 2022. DOI: [10.1109/COMST.2022.3177251](https://doi.org/10.1109/COMST.2022.3177251)
- [0] Y. Wang, M. Lyu, and V. Sivaraman, “Games are not equal: Classifying cloud gaming contexts for effective user experience measurement,” in *Proc. ACM Internet Measurement Conference*, ser. IMC '25, Madison, WI, USA, 2025. DOI: [10.1145/3730567.3764455](https://doi.org/10.1145/3730567.3764455)

- [0] J. Wu, C. Yuen, N.-M. Cheung, J. Chen, and C. W. Chen, "Streaming mobile cloud gaming video over TCP with adaptive source-FEC coding," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 27, no. 1, pp. 32–48, 2017. DOI: [10.1109/TCSVT.2016.2527398](https://doi.org/10.1109/TCSVT.2016.2527398)
- [0] C. An et al., "Tooth: Toward optimal balance of video QoE and redundancy cost by fine-grained FEC in cloud gaming streaming," in *22nd USENIX Symposium on Networked Systems Design and Implementation*, ser. NSDI '25, Philadelphia, PA, 2025, pp. 635–651.
- [0] K. Chen, H. Wang, S. Fang, X. Li, M. Ye, and H. J. Chao, "RL-AFEC: Adaptive forward error correction for real-time video communication based on reinforcement learning," in *Proc. 13th ACM Multimedia Systems Conference*, ser. MMSys '22, 2022. DOI: [10.1145/3524273.3528184](https://doi.org/10.1145/3524273.3528184)
- [0] Z. Meng et al., "Hairpin: Rethinking packet loss recovery in edge-based interactive video streaming," in *21st USENIX Symposium on Networked Systems Design and Implementation*, ser. NSDI '24, Santa Clara, CA, 2024, pp. 907–926.
- [0] A. Shokrollahi, "Raptor codes," *IEEE Transactions on Information Theory*, vol. 52, no. 6, pp. 2551–2567, 2006. DOI: [10.1109/TIT.2006.874390](https://doi.org/10.1109/TIT.2006.874390)
- [0] M. Luby, A. Shokrollahi, M. Watson, T. Stockhammer, and L. Minder, *RaptorQ forward error correction scheme for object delivery*, RFC 6330, 2011. DOI: [10.17487/RFC6330](https://doi.org/10.17487/RFC6330) [Online]. Available: <https://www.rfc-editor.org/info/rfc6330>
- [0] M. Luby, *Adaptive receiver-side scheduling for smooth interactive delivery*, 2025. arXiv: [2511.16902](https://arxiv.org/abs/2511.16902) [cs.NI].
- [0] V. Arun and H. Balakrishnan, "Copa: Practical delay-based congestion control for the internet," in *15th USENIX Symposium on Networked Systems Design and Implementation*, ser. NSDI '18, Renton, WA, 2018, pp. 329–342.
- [0] G. Carlucci, L. De Cicco, S. Holmer, and S. Mascolo, "Analysis and design of the google congestion control for web real-time communication (WebRTC)," in *Proc. 7th Int. Conf. Multimedia Systems*, ser. MMSys '16, 2016. DOI: [10.1145/2910017.2910605](https://doi.org/10.1145/2910017.2910605)
- [0] I. Johansson and Z. Sarker, *Self-clocked rate adaptation for multimedia*, RFC 8298, 2017. DOI: [10.17487/RFC8298](https://doi.org/10.17487/RFC8298)
- [0] X. Xu and M. Claypool, "Measurement of cloud-based game streaming system response to competing TCP cubic or TCP BBR flows," in *Proc. 22nd ACM Internet Measurement Conference*, ser. IMC '22, Nice, France, 2022. DOI: [10.1145/3517745.3561464](https://doi.org/10.1145/3517745.3561464)
- [0] K. Nichols and V. Jacobson, "Controlling queue delay," *ACM Queue*, vol. 10, no. 5, 2012. DOI: [10.1145/2209249.2209264](https://doi.org/10.1145/2209249.2209264)
- [0] R. Pan et al., "Proportional integral controller enhanced (PIE): A lightweight control scheme to address the bufferbloat problem," RFC 8033, 2017.
- [0] B. Briscoe, K. D. Schepper, M. Bagnulo, and G. White, "Low latency, low loss, and scalable throughput (l4s) internet service: Architecture," RFC Editor, RFC 9330, Jan. 2023.
- [0] K. D. Schepper, B. Briscoe, and G. White, "Dual-queue coupled active queue management (AQM) for low latency, low loss, and scalable throughput (L4S)," RFC 9332, 2023.
- [0] J. Jiang, V. Sekar, I. Stoica, and H. Zhang, "Shedding light on the structure of internet video quality problems in the wild," in *Proc. 9th ACM Conference on Emerging Networking Experiments and Technologies*, ser. CoNEXT '13, Santa Barbara, CA, USA, 2013, pp. 357–368. DOI: [10.1145/2535372.2535394](https://doi.org/10.1145/2535372.2535394)
- [0] C.-C. M. Yeh, Y. Zhu, L. Ulanova, N. Nurjaha, G. E. Batista, and E. Keogh, "Matrix profile i: All pairs similarity joins for time series: A unifying view that includes motifs, discords and shapelets," in *2016 IEEE 16th International Conference on Data Mining (ICDM)*, IEEE, 2016, pp. 1317–1322.
- [0] A. Dempster, F. Petitjean, and G. I. Webb, "ROCKET: Exceptionally fast and accurate time series classification using random convolutional kernels," *Data Mining and Knowledge Discovery*, vol. 34, no. 5, pp. 1454–1495, 2020.
- [0] A. Dempster, D. F. Schmidt, and G. I. Webb, "MINIROCKET: A very fast (almost) deterministic transform for time series classification," in *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, 2021, pp. 248–257.
- [0] Federal Communications Commission, "Measuring broadband america: Thirteenth report," FCC Office of Engineering and Technology, Tech. Rep., 2023.
- [0] Ofcom, "UK home broadband performance: Technical report," Office of Communications, Tech. Rep., 2024.
- [0] R. N. Staff, "RIPE atlas: A global internet measurement network," in *IEEE Internet Computing*, vol. 19, 2015, pp. 48–56.
- [0] S. Sundaresan, W. de Donato, N. Feamster, R. Teixeira, S. Crawford, and A. Pescapé, "Broadband internet performance: A view from the gateway," in *Proceedings of ACM SIGCOMM*, 2011, pp. 134–145.
- [0] C. Baena, O. S. Peñaherrera-Pulla, R. Barco, and S. Fortes, "Measuring and estimating key quality indicators in cloud gaming services," *Computer Networks*, vol. 231, p. 109 808, 2023. DOI: [10.1016/j.comnet.2023.109808](https://doi.org/10.1016/j.comnet.2023.109808)

- [0] H. Wan and K. Jamieson, *Evolving mobile cloud gaming with 5G standalone network telemetry*, 2024. arXiv: [2402.04454](https://arxiv.org/abs/2402.04454) [cs.NI].
- [0] IEEE, *IEEE standard for information technology—wireless LAN medium access control (MAC) and physical layer (PHY) specifications—amendment 1: Enhancements for high-efficiency WLAN*, 2021. DOI: [10.1109/IEEESTD.2021.9442429](https://doi.org/10.1109/IEEESTD.2021.9442429)
- [0] IEEE, *IEEE standard for information technology—telecommunications and information exchange between systems—local and metropolitan area networks—specific requirements—part 11: Wireless LAN medium access control (MAC) and physical layer (PHY) specifications*, 2021. DOI: [10.1109/IEEESTD.2021.9363693](https://doi.org/10.1109/IEEESTD.2021.9363693)
- [0] G. Bianchi, “Performance analysis of the IEEE 802.11 distributed coordination function,” *IEEE Journal on Selected Areas in Communications*, vol. 18, no. 3, pp. 535–547, 2000. DOI: [10.1109/49.840210](https://doi.org/10.1109/49.840210)
- [0] IEEE, *IEEE standard for information technology—amendment: Enhancements for extremely high throughput (EHT)*, 2024.
- [0] ITU-T, “Asymmetric digital subscriber line 2 transceivers (ADSL2) — extended bandwidth ADSL2 (ADSL2plus),” International Telecommunication Union, Tech. Rep. G.992.1, 2003.
- [0] ITU-T, “Very high speed digital subscriber line transceivers 2 (VDSL2),” International Telecommunication Union, Tech. Rep. G.993.1, 2006.
- [0] J. Maes, M. v. Helvoort, and P. Spruyt, “Crosstalk cancellation in VDSL2: System and algorithm design,” *IEEE Transactions on Signal Processing*, vol. 57, no. 12, pp. 4786–4797, 2009.
- [0] CableLabs, “Data-over-cable service interface specifications (DOCSIS) 3.1: MAC and upper layer protocols interface specification,” CableLabs, Tech. Rep. CM-SP-MULPIv3.1, 2017.
- [0] J. Gettys and K. Nichols, “Bufferbloat: Dark buffers in the internet,” *Communications of the ACM*, vol. 55, no. 1, pp. 57–65, 2012. DOI: [10.1145/2063176.2063196](https://doi.org/10.1145/2063176.2063196)
- [0] T. Høiland-Jørgensen, P. McKenney, D. Taht, J. Gettys, and E. Dumazet, “The flow queue CoDel packet scheduler and active queue management algorithm,” RFC Editor, RFC 8290, Jan. 2018.
- [0] CableLabs. “Low latency DOCSIS: Technology overview.” [Online]. Available: <https://www.cablelabs.com/technologies/low-latency-docsis>
- [0] G. White and R. Pan, “Active queue management (AQM) based on proportional integral controller enhanced (PIE) for Data-Over-Cable Service Interface Specifications (DOCSIS) cable modems,” RFC 8034, 2017.
- [0] Y. Ni, Z. Meng, et al., “CellFusion: Multipath vehicle-to-cloud video streaming with network coding in the wild,” in *Proc. ACM SIGCOMM 2023*, 2023. DOI: [10.1145/3603269.3604832](https://doi.org/10.1145/3603269.3604832)
- [0] C.-Y. Huang, K.-T. Chen, D.-Y. Chen, H.-J. Hsu, and C.-H. Hsu, “GamingAnywhere: The first open source cloud gaming system,” *ACM Transactions on Multimedia Computing, Communications, and Applications*, vol. 10, no. 1s, 10:1–10:25, 2014. DOI: [10.1145/2537855](https://doi.org/10.1145/2537855)
- [0] F. Dossena and A. Trentini, “OpenLDAT: A system for the measurement of display latency metrics,” *Journal of the Society for Information Display*, vol. 30, 2022. DOI: [10.1002/jsid.1104](https://doi.org/10.1002/jsid.1104)
- [0] C. Bachhuber and E. Steinbach, “A system for high precision glass-to-glass delay measurements in video communication,” in *2016 IEEE Int. Conf. Image Processing*, ser. ICIP 2016, 2016, pp. 2132–2136. DOI: [10.1109/ICIP.2016.7532735](https://doi.org/10.1109/ICIP.2016.7532735)
- [0] A. Shokrollahi and M. Luby, “Raptor codes,” *Foundations and Trends in Communications and Information Theory*, vol. 6, no. 3–4, pp. 213–322, 2011. DOI: [10.1561/01000000060](https://doi.org/10.1561/01000000060)
- [0] K. Lee et al., “Outatime: Using speculation to enable low-latency continuous interaction for mobile cloud gaming,” in *Proc. 13th ACM Int. Conf. Mobile Systems, Applications, and Services*, ser. MobiSys ’15, 2015. DOI: [10.1145/2742647.2742656](https://doi.org/10.1145/2742647.2742656)
- [0] Y. Li et al., “Dissecting and streamlining the interactive loop of mobile cloud gaming,” in *Proc. 22nd USENIX Symposium on Networked Systems Design and Implementation*, ser. NSDI ’25, Philadelphia, PA, USA, 2025.
- [0] C. Baena, S. Fortes, O. Peñaherrera-Pulla, E. Baena, and R. Barco, “Gaming in the cloud: 5G as the pillar for future gaming approaches,” *IEEE Communications Magazine*, 2024. DOI: [10.1109/MCOM.005.2300256](https://doi.org/10.1109/MCOM.005.2300256)
- [0] A. M. Law, *Simulation Modeling and Analysis*, 5th. McGraw-Hill Education, 2014, ISBN: 978-0-07-340132-4.
- [0] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. Chapman & Hall/CRC, 1993, ISBN: 978-0-412-04231-7.
- [0] A. Saltelli, P. Annoni, I. Azzini, F. Campolongo, M. Ratto, and S. Tarantola, “Variance based sensitivity analysis of model output: Design and estimator for the total sensitivity index,” *Computer Physics Communications*, vol. 181, no. 2, pp. 259–270, 2010. DOI: [10.1016/j.cpc.2009.09.018](https://doi.org/10.1016/j.cpc.2009.09.018)
- [0] M. Groshev, J. Sacido, and J. Martín-Pérez, “Demo: FoReCo—a forecast-based recovery mechanism for real-time remote control of robotic manipulators,” in *Proc. ACM SIGCOMM ’22 Poster and Demo Sessions*, Amsterdam, The Netherlands, 2022, pp. 7–9. DOI: [10.1145/3546037.3546047](https://doi.org/10.1145/3546037.3546047)

Revision History

Version	Date	Changes
0.2.0	March 2026	Reframed around network inference and adaptive mitigation
0.1.0	March 2026	Initial conceptual report

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